

Exploring Statistics with R

An R workbook to accompany Exploring Statistics: Tales of Distributions

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Instructor Guide

Exploring Statistics with R, prepared by Erick Axxe, is an R companion to *Exploring Statistics: Tales of Distributions*. It follows the textbook's research questions and statistical sequence while replacing menu-based software instructions with guided R and RStudio activities.

The workbook is written for introductory statistics students, including students who have never programmed. It teaches the R skills needed for the analyses, but it is not intended to be a complete programming course.

Download The Complete Instructor Materials

[Download the complete instructor materials \(ZIP, approximately 11 MB\)](#)

The archive includes the rendered website, editable Quarto source, Word chapters, the combined PDF, student starter files, data, documentation, scripts, assets, and a README.

For student use, distribute the smaller [student starter files](#). Students need those files regardless of whether they read the workbook online, in Word, or as a PDF.

The [combined PDF](#) is also available as a separate download.

Choose A Workbook Format

The three editions contain the same instructional substance:

- **HTML** is the most convenient teaching edition. It includes sidebar navigation, search, code-copy controls, and links to the codebook and reference pages.
- **PDF** combines the complete workbook into one fixed-layout document with a table of contents and PDF bookmarks. It is useful for printing, archiving, or distributing one file.
- **Word** provides one editable file for each start page, chapter, reference page, and the instructor guide. Instructors may add local directions, adjust pacing, or upload selected chapters to a learning-management system.

Internal website links are removed from the Word edition because a link tied to a website folder may fail after an individual document is downloaded or moved. External links, including the starter-file download, remain active.

What Students Need

Students need:

1. a computer on which they can install or access R and RStudio;
2. the complete, unzipped student starter folder;
3. the workbook in HTML, PDF, or Word; and
4. permission to install the required R packages if those packages are not already available.

The Set Up RStudio page provides installation, download, and unzip directions. Students should begin every work session by opening `exploring-statistics-r.Rproj`, not by opening a script alone. Opening the project gives R a consistent starting location for the data and scripts.

Use The Student Starter Files

The student archive contains:

- `exploring-statistics-r.Rproj`, which opens the project;
- `data/EAMMI2-REDUCED.csv`, the reduced teaching dataset;
- `scripts/_setup.R`, which prepares packages, data, graph styling, and helper functions;
- `scripts/00-open-and-check-data.R`, the first-run check;
- one annotated R script for each workbook chapter; and
- `resources/EAMMI2_Survey_withcodes.pdf`, the survey and coding documentation.

Ask students to extract the entire ZIP archive and keep its folders together. They should then:

1. open `exploring-statistics-r.Rproj`;
2. open `scripts/00-open-and-check-data.R` from the **Files** tab in RStudio;
3. run that script one section at a time; and
4. open the script named in each assigned chapter.

The first run may take several minutes while R installs any missing packages. Later sessions should start more quickly.

Follow The Chapter Structure

Most chapters begin with one or more research questions from the textbook. A worked question typically moves through preparation, analysis, investigation of the output, and interpretation. Practice questions gradually ask students to reuse familiar code with less guidance.

Every chapter ends with **Check Your Work**. Instructors can ask students to delay that section until they have run the analysis and written their own interpretation. Sample responses demonstrate an acceptable answer, but another answer may also be defensible when a prompt asks students to describe a graph or explain a data decision.

The chapters deliberately reduce support over time. Early chapters explain file navigation and each major part of a command. Later chapters expect students to copy a familiar pattern, change object or variable names, check the resulting output, and describe the statistics in context. The R Command Index shows where each command is first explained.

R Packages And Supplied Helpers

The workbook uses `tidyverse` for data preparation, summaries, and graphs. Chapter 15 also uses `PMCMRplus` for Dwass-Steel-Critchlow-Fligner post hoc comparisons after a Kruskal-Wallis test. The setup script installs missing packages and loads them; students do not need separate package-management instructions.

The setup script also supplies small helper functions so students can focus on the statistical idea without repeatedly reconstructing a long calculation. These helpers:

- calculate the statistical mode and sample skewness;
- calculate Cohen's *d* for one-sample, independent-group, repeated, and Tukey comparisons;
- calculate eta-squared measures for analysis of variance;

- calculate rank-biserial correlations and epsilon squared for nonparametric tests;
- calculate Cramer's V ; and
- organize selected repeated-measures and two-way ANOVA results.

The R Command Index labels supplied helpers and identifies the chapter in which each is introduced. Instructors may show the definitions in `scripts/_setup.R` when a class would benefit from seeing how a calculation is assembled.

Important Data And Analysis Decisions

The R edition uses the included reduced EAMMi2 data as its canonical source. Most results reproduce the jamovi and SPSS modules closely. The following decisions are worth knowing.

Chapter 2. A Satisfaction With Life Scale value of 0 is retained and discussed as a reason to inspect a frequency table before interpreting a distribution.

Chapters 3 and 4. One IMPcareer value of 180 falls outside the stated 0–100 scale. Students identify it, set that observation aside for the chapter analysis, and report the decision.

Chapter 6. The special-someone support correlation is reported as positive because that is the sign produced by the included data; some source prose gives the opposite sign.

Chapter 10. Political-group counts and belonging results are calculated from the included data. The source modules contain inconsistent values, so the R output controls where the printed statistics cannot all be reconciled.

Chapter 11. Bartlett's test checks equality of variances across the three groups. The earlier two-group F test is not reused because it compares only two variances.

Chapter 12. The repeated-measures analysis and Tukey-adjusted comparisons are calculated from the included data with the supplied R helpers.

Chapter 13. The interaction is interpreted as nonsignificant because that is the result produced by the included data. The significant interaction described in a source module is not supported by the analysis.

Chapter 14. Political identity categories and counts follow the included data and its codebook. Missing PoliticsDichotomous and PartyDichotomous values represent participants outside the two retained categories and are excluded only from analyses that require those dichotomous variables.

Chapter 15. PMCMRplus supplies the DSCF post hoc procedure. The workbook reports the independent-samples Wilcoxon statistic as U and the paired statistic as V , matching the conventions taught in the chapter.

The analyses are instructional demonstrations. Any data-quality decision should be described in a student report rather than treated as an invisible correction.

Adapt The Materials

Word chapters may be edited directly for local use. Instructors can remove unassigned research questions, add course-specific directions, or move Check Your Work to a separate answer document. Keep the student starter scripts synchronized with any changed variable names or analysis steps.

The editable source pages are Quarto `.qmd` files. After editing:

- render the HTML site by selecting **Render Website** in RStudio or running `quarto render`;
- render the Word collection by running `python tools/render_word.py`; and
- rebuild the combined PDF by running `python tools/build_combined_pdf.py`.

Open `exploring-statistics-r.Rproj` before rendering. The complete instructor archive includes the required configuration, filters, styles, assets, data, and helper tools.

Troubleshoot Common Problems

When a student reports an error, ask for the script name, the exact section run, and the first complete error message. Common fixes include:

- Open `exploring-statistics-r.Rproj` and keep the starter folders together.
- Run `source("scripts/_setup.R")` before chapter code.
- If the Console shows `+`, press `Esc` and look for a missing parenthesis or quotation mark.
- If an object is missing, rerun the section that creates it before running later code.
- If output differs from the workbook, restart R and run the chapter script from the beginning in order.
- On institution-managed computers, ask technical support to preinstall `tidyverse` and `PMCMRplus`.

The student-facing Troubleshooting page provides a longer plain-language guide.

Data And Documentation

The EAMMi2 project is available through the [Open Science Framework study page](#), and the [survey with variable codes](#) is included in the student starter files. The project record identifies the study materials with a CC0 1.0 license. The workbook uses a reduced teaching copy containing only the variables needed for its analyses.

Choose Your Path

You can use this workbook even if you have never written code. Choose the path below that best describes your experience.

If R Is New To You

Complete the Start Here pages in order:

1. Set Up RStudio.
2. Run Your First Script.
3. Understanding EAMMi2.
4. Begin Chapter 1.

The first chapters provide complete code and explain new commands when they first matter. Focus on understanding the purpose of each step. Memorization can come later through practice.

If You Have Used R Before

You may begin with Chapter 1 after you:

1. download and unzip the starter files;
2. open `exploring-statistics-r.Rproj` in RStudio; and
3. run `scripts/00-open-and-check-data.R` to confirm that the data open correctly.

Review Run Your First Script if the project, script, or Console workflow differs from the way you usually work.

What Everyone Needs

Download the [starter files](#). The folder includes:

- `exploring-statistics-r.Rproj`, which opens the workbook project;
- `data/EAMMi2-REDUCED.csv`, which contains the teaching data;
- `scripts/_setup.R`, which prepares the R tools and data used by the chapter scripts;
- `scripts/00-open-and-check-data.R`, which checks the starter setup;
- one R script for each workbook chapter; and
- survey documentation in the `resources` folder.



Keep the folder together

After unzipping the starter files, do not move the project file, data folder, or scripts folder away from one another. The chapter scripts use their locations inside this folder to find the data.

A Useful Learning Routine

When you reach a code example:

1. Read the sentence above the code.
2. Predict what the code should produce.
3. Run the code in the chapter script.
4. Find the result in RStudio.
5. Check whether the result is reasonable.
6. Explain what the result means for the research question.

Small mistakes are a normal part of learning R. The Troubleshooting page explains common messages and how to recover from them.

Set Up RStudio

R performs the statistical calculations. RStudio gives you a place to write code, run scripts, inspect data, and view graphs. You need both programs.

Install R And RStudio

Install the programs in this order:

1. Download and install [R](#).
2. Download and install [RStudio Desktop](#).
3. Open RStudio once both installations are complete.

For a more complete tour, including current download links, see Posit's [Get Started guide](#).

Download And Unzip The Starter Files

Download the [starter files](#).

The file arrives as a compressed .zip folder. Extract its contents before opening the project:

- On Windows, follow Microsoft's [zip and unzip instructions](#).
- On a Mac, follow Apple's [zip and unzip instructions](#).

Move the unzipped folder somewhere easy to find, such as your Documents folder. Keep all files and folders inside it together.

Open The Project

❗ Open the project before a script

Begin each work session by opening `exploring-statistics-r.Rproj`. Opening the project first helps R find the data and scripts used in the workbook.

1. Open RStudio.
2. Choose **File > Open Project....**
3. Select `exploring-statistics-r.Rproj` inside the unzipped starter folder.
4. In the Output pane, select the **Files** tab.
5. Confirm that you can see folders named `data`, `resources`, and `scripts`.

Find Your Way Around RStudio

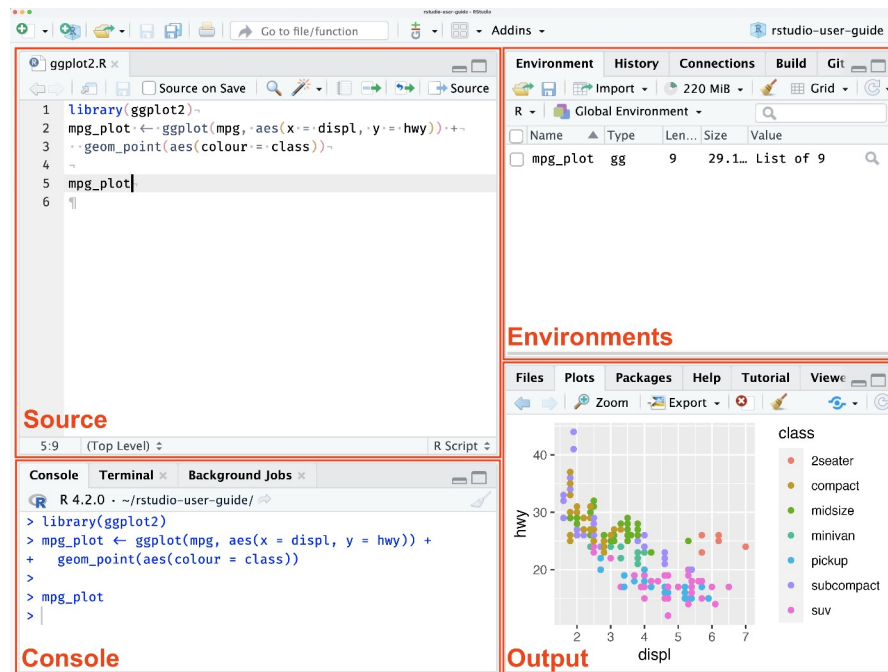


Figure 1: RStudio interface with labels for the Source, Console, Environment, and Output panes.

Pane	What you do there
Source	Read, edit, and run .R scripts.
Console	See commands run, messages, and printed results.
Environment	See the objects currently available in R.
Output (Files, Plots, Packages, Help, Viewer)	Browse files, view graphs, review packages, read help, and preview output.

You do not need to remember every tab. Most workbook activities use the Source pane, Console, Environment, and the **Files** and **Plots** tabs in the Output pane.

Check Your Setup

Before continuing, confirm that:

- RStudio is open;
- the project name appears in the RStudio window;
- the **Files** tab shows the data and scripts folders; and
- you have not opened a second copy of the starter folder from inside the .zip file.

Next, use Run Your First Script to check the data and learn how RStudio responds when you run code.

Run Your First Script

An R script is a saved set of instructions. You can run the instructions again, correct them, or adapt them for another analysis. This makes your work reproducible: you and another reader can see how the result was produced.

Open The Starter Script

1. Open `exploring-statistics-r.Rproj` in RStudio.
2. In the Output pane, select the **Files** tab.
3. Open the `scripts` folder.
4. Select `00-open-and-check-data.R`.

The file opens in the Source pane. Read the comments at the top before running the code.

Run Code One Step At A Time

Place your cursor anywhere on a line of code and select **Run**. RStudio sends that line to the Console. You can also select several complete lines and run them together.

Run the starter script from top to bottom. Pause after each section and look for the result described by its comments.



Watch the Console prompt

The `>` prompt means R is ready for another command. If the prompt has not returned, R may still be calculating.

A `+` prompt means R thinks the command is unfinished. Press `Esc`, check for a missing parenthesis or quotation mark, and run the complete command again.

After the setup section runs:

- the Environment pane should contain an object named `eammi`; and
- the Console should show that the data contain 2,073 observations and 39 variables.

The first run may take a few minutes because the setup file checks for and prepares the R tools used throughout the workbook, including `tidyverse`.

The rest of this page introduces the central ideas you need to navigate R: objects, functions, data frames, the pipe, and comments. Examples of these features also appear at the end of `00-open-and-check-data.R`, so you can practice with the script or review the ideas on this page by themselves.

Objects: Names R Remembers

An object is a named result stored in R's memory. The assignment operator `<-` gives a result a name.

```
favorite_number <- 7  
favorite_number
```

Read the first line as “store the value 7 in an object named `favorite_number`.” Running the object name prints the stored value.

You will see the object `eammi` throughout this workbook. It represents the data we are working with.

Functions: Actions R Performs

A function tells R to perform an action. Its name is followed by parentheses.

```
round(21.428, digits = 2)
```

Here, `round()` performs the action, 21.428 is the value, and `digits = 2` provides an instruction about the result.

When you meet a new function, ask three questions:

1. What object or variable does it use?
2. What action does it perform?
3. What result should it return?

Data Frames: Observations And Variables

A data frame is a rectangular dataset.

- Each observation usually represents one participant or case.
- Each variable records one characteristic or measurement.
- Variables may contain numbers, categories, text, or missing responses.

The setup file reads the EAMMi2 teaching data and stores the resulting data frame as `eammi`.

These commands ask basic questions about that data frame:

```
nrow(eammi)
ncol(eammi)
names(eammi)
```

`nrow()` counts observations, `ncol()` counts variables, and `names()` lists the variable names.

The Pipe: Work From Left To Right

The native pipe `|>` passes the result of one step into the next step.

```
eammi |>
  count(Gender)
```

Read this as: “begin with `eammi`, and then count how often each value of `Gender` appears.”

Older R materials may use `%>%`. This workbook was developed with R 4.6.0 and RStudio 2026.07.0+139. It uses `|>`, which is built into R and is the default pipe inserted by this version of RStudio. The chapter code also uses functions from `tidyverse`; the setup file prepares those functions for you.

Comments: Notes For Readers

R ignores anything after # on a line. These comments explain the purpose of the code to you and anyone else who reads the script.

```
# Count participants in each gender category.  
eammi |>  
  count(Gender)
```

When you copy or adapt code, update its comments too. A short, accurate comment will help you understand the analysis when you return to it later.

Practice The Routine

Use the final examples in 00-open-and-check-data.R to practice:

1. Read the comment above an example.
2. Predict what will appear.
3. Run the complete example.
4. Find the result in the Console or Environment pane.
5. Describe the result in one sentence.

You can now continue to Understanding EAMMi2 or use Troubleshooting if your result differs from the description above.

Understanding EAMMi2

Emerging Adulthood Measured at Multiple Institutions-2

Throughout this workbook, you will use the EAMMi2 data set to learn about statistical analyses and practice using R. The information that follows provides background about the EAMMi2 study and a codebook to help you navigate and understand the data file.



Important note

As you progress through this workbook, remember that the analyses and resulting conclusions are for demonstration purposes only.

What Is The EAMMi2 Study?

The EAMMi2 data set includes responses from undergraduate students at multiple institutions. Participants answered questions across several variables, including basic demographic questions, questions about personal experiences, and questions about well-being. The overarching goal of the study was to learn more about the psychological experiences of 18- to 29-year-olds, the age range associated with the transition from adolescence to adulthood.

The final sample size of respondents between the ages of 18 and 29 was 2,073. You can learn more about the [EAMMi2 study](#) and view the [survey-with-codes PDF](#).

Our Data File

This workbook uses a reduced version of the original EAMMi2 data. The data file includes a subset of variables selected to allow you to conduct the types of analyses covered in *Exploring Statistics*.

The name of the data file used throughout this workbook is `EAMMi2-REDUCED.csv`. It is included in the starter files.

You are encouraged to save the starter folder somewhere on your computer where you can easily find it throughout your time in this course.

If you are new to R, begin with Choose Your Path and run `scripts/00-open-and-check-data.R` before working through the chapters.

Download the starter files here:

[Download starter files](#)

In the next sections, you will find a codebook for the data set. The codebook describes the variables used in the reduced data set and tells you how each variable is represented in the data file.

Opening The Data In R

The starter files include the data set and the R scripts. Open `exploring-statistics-r.Rproj`, then run the setup line at the top of a chapter script:

```
source("scripts/_setup.R")
```

The setup line creates a data frame named `eammi`. Most chapter scripts begin with this same line.

The data frame contains 39 variables.

Reading Coded Categories

Some variables use numbers to stand for categories. For example, the survey codes `Adult` as 1 = Yes, 2 = Maybe, and 3 = No; `Gender` as 1 = Male, 2 = Female, and 3 = Other; and `Disability` as 1 = Yes and 2 = No.

The chapter scripts will usually turn these numeric codes into readable labels before making tables or graphs. You do not need to memorize all of the codes. When a chapter uses a coded variable, the chapter will explain the codes needed for that analysis.

Missing Values

Missing values appear as `NA` in R. `NA` means that a value is not available for that person on that variable.

You can count missing values for a variable with:

```
sum(is.na(eammi$Stress))
```

Each chapter will point out missing-value checks when they matter for the analysis.

Survey Documentation

For more detail about the original survey questions and response options, see the [EAMMi2 survey-with-codes PDF](#).

EAMMi2 Codebook

Variable	Brief description	Response options
Duration	How many minutes it took each participant to finish the survey	Duration automatically recorded from survey website
Finished	Whether or not the participant finished the survey	0 = no; 1 = yes
FinanciallyInd	Extent to which participant reported being financially independent	1 = no; 2 = somewhat; 3 = yes
NoLongerHome	Extent to which participant has moved out of parents' home	1 = no; 2 = somewhat; 3 = yes
MOA_ACH	Extent to which participant has achieved markers of adulthood. Sample item: the extent to which	20 items were rated 1 = no; 2 = somewhat; 3 = yes. The final score is the total across all provided

Variable	Brief description	Response options
	the participant has settled into a long-term career.	markers of adulthood, with a possible range of 20-60.
Adult	Answer to “Are you an adult?”	1 = yes; 2 = maybe; 3 = no
Exploration	Responses to the Inventory of Dimensions of Emerging Adulthood. This 2-item subscale measured the extent to which this period of life was a time for exploration.	Each item was rated on a scale of 1 = strongly disagree to 4 = strongly agree. The subscale score represents the average rating across two items.
Negativity	Responses to the Inventory of Dimensions of Emerging Adulthood. This 2-item subscale measured the extent to which this period of life was a time for negativity.	Each item was rated on a scale of 1 = strongly disagree to 4 = strongly agree. The subscale score represents the average rating across two items.
Identity	Responses to the Inventory of Dimensions of Emerging Adulthood. This 2-item subscale measured the extent to which this period of life was a time for identity formation.	Each item was rated on a scale of 1 = strongly disagree to 4 = strongly agree. The subscale score represents the average rating across two items.
BetweenStages	Responses to the Inventory of Dimensions of Emerging Adulthood. This 2-item subscale measured the extent to which this period of life was a time for feeling between stages of life.	Each item was rated on a scale of 1 = strongly disagree to 4 = strongly agree. The subscale score represents the average rating across two items.
Politics	Self-reported political views	1 = extremely liberal to 7 = extremely conservative
PoliticsDichotomous	Self-reported political views, dichotomized into liberal and conservative categories	1 = liberal; 2 = conservative
Party	Self-reported political party affiliation	1 = extremely democrat to 7 = extremely republican
PartyDichotomous	Self-reported political party affiliation, dichotomized into democrat and republican categories	1 = democrat; 2 = republican
SWLS	Answers to the 5-item Satisfaction with Life Scale. Sample item: “In most ways my life is close to my ideal.”	Each item was rated on a scale of 1 = strongly disagree to 7 = strongly agree. The final SWLS score represents the sum of all 5 items, with a possible range of 5-35.
Mindfulness	Answers to the 15-item Mindful Attention Awareness Scale. Sample	The final Mindfulness score is an average rating across items. Higher

Variable	Brief description	Response options
	item: "I do jobs or tasks automatically, without being aware of what I'm doing."	scores indicate greater mindfulness.
Belonging	Agreement with statement "I feel like I belong"	1 = not at all to 5 = very much
SelfEfficacy	Answers to the 10-item Generalized Self-Efficacy Scale. Sample item: "I can always manage to solve difficult problems if I try hard enough."	Each item was rated on a scale of 1 = not at all true to 4 = exactly true. The final SelfEfficacy score is the sum across all 10 items, with a possible range of 10-40.
SupportFriends	Answers to the Perceived Social Support Scale. This subscale measured perceived support from friends. Sample item: "I can count on my friends when things go wrong."	Items were rated on a scale of 1 = very strongly disagree to 7 = very strongly agree. This subscale is an average across relevant items.
SupportFamily	Answers to the Perceived Social Support Scale. This subscale measured perceived support from family.	Items were rated on a scale of 1 = very strongly disagree to 7 = very strongly agree. This subscale is an average across relevant items.
SupportSpecial	Answers to the Perceived Social Support Scale. This subscale measured perceived support from a "special someone."	Items were rated on a scale of 1 = very strongly disagree to 7 = very strongly agree. This subscale is an average across relevant items.
SocialSupport	Answers to the Perceived Social Support Scale across friends, family, and a "special someone."	SocialSupport represents the sum of all perceived social support items, with a possible range of 12-84.
SMmaintain	Items that assessed the extent to which participants reported using social media to maintain existing social connections.	Items were rated on a scale of 1 = never to 5 = a lot. This subscale is an average across relevant items.
SMnewconnect	Items that assessed the extent to which participants reported using social media to make new social connections.	Items were rated on a scale of 1 = never to 5 = a lot. This subscale is an average across relevant items.
SMinformation	Items that assessed the extent to which participants reported using social media to get or share information.	Items were rated on a scale of 1 = never to 5 = a lot. This subscale is an average across relevant items.
SMtotal	Overall use of social media across	Items were rated on a scale of 1 =

Variable	Brief description	Response options
	11 items.	never to 5 = a lot. SMtotal represents the sum of all items, with a possible range of 11-55.
USdream1	Answer to "Do you believe in the importance of achieving the American Dream?"	1 = definitely not to 5 = yes, strongly
USdream2	Answer to "Do you believe you will be able to achieve the American Dream?"	1 = definitely not to 5 = yes, strongly
Conflict	Answer to "In the past two weeks, how often has someone gotten upset with you in response to something that you said or did?"	1 = not at all to 7 = constantly
Symptoms	13 items that assessed the extent to which participants were bothered by a variety of physical symptoms, such as stomach pain, back pain, and difficulty sleeping, during the past 4 weeks.	Each item was rated on a scale of 1 = not bothered at all to 3 = bothered a lot. This variable represents the sum of all 13 items, with a possible range of 13-39.
Stress	Answers to the 10-item Perceived Stress Scale. Sample item: "In the last month, how often have you felt confident about your ability to handle your personal problems?"	Each item was rated on a scale of 1 = never to 5 = very often. The final Stress score represents the sum of all 10 items, with a possible range of 10-50.
IMPmarriage	Importance placed on marriage as an eventual or current aspect of life.	Participants assigned a percentage from 0-100. They were instructed to answer IMPmarriage, IMPparenting, IMPcareer, and IMPhobbies so the total across all four aspects equaled 100%.
IMPparenting	Importance placed on parenting as an eventual or current aspect of life.	Participants assigned a percentage from 0-100. They were instructed to answer IMPmarriage, IMPparenting, IMPcareer, and IMPhobbies so the total across all four aspects equaled 100%.
IMPCareer	Importance placed on career as an eventual or current aspect of life.	Participants assigned a percentage from 0-100. They were instructed to answer IMPmarriage, IMPparenting, IMPcareer, and IMPhobbies so the total across all four aspects equaled 100%.

Variable	Brief description	Response options
IMP hobbies	Importance placed on hobbies as an eventual or current aspect of life.	Participants assigned a percentage from 0-100. They were instructed to answer IMP marriage, IMP parenting, IMP career, and IMP hobbies so the total across all four aspects equaled 100%.
Gender	Self-reported gender identity	1 = male; 2 = female; 3 = another identity
Age	Age of the participant, in years	Responses span ages 18-29
Race	Self-reported racial identity	1 = White; 2 = African American; 3 = Hispanic/Latinx; 4 = Asian/Pacific Islander; 5 = Native American; 6 = Another racial identity; 7 = Multiracial (2 selected); 8 = Multiracial (3 or more selected)
Disability	Self-identified disability status	1 = yes; 2 = no

Chapter 1: Introduction

Chapter 1 of *Exploring Statistics* introduces the field and the language used to describe data. This chapter adds a practical question: **How can R help us examine the people and variables in the EAMMi2 dataset?**

If this is your first time using R, begin with Choose Your Path. The Start Here pages explain how to install the software, download the starter files, open the RStudio project, and run your first script.

Learning Goals

By the end of this chapter, you should be able to:

- inspect observations, variables, variable names, and variable types;
- distinguish statistical variable types from the way R represents them;
- give numeric category codes readable labels;
- create a frequency table and bar chart; and
- describe what the table and graph show.



Learning tip

Focus on reading and running code. You do not need to memorize every function. Your goal is to understand that an analysis is a sequence of instructions you can save, check, and run again.

Research Question

What does the reduced EAMMi2 dataset contain, and what does it tell us about participants' self-reported gender?

Get Ready

This chapter uses `scripts/01-introduction.R` in the starter files.

Open the Chapter 1 script

1. Open `exploring-statistics-r.Rproj`.
2. In the Output pane, select the **Files** tab.
3. Open the `scripts` folder.
4. Open `01-introduction.R`.
5. Place your cursor on the first line of code and select **Run**. Continue running one line or complete expression at a time as you follow this chapter.

The script begins by running the shared setup file. This opens the EAMMi2 dataset and loads the R tools used in the workbook.

```
source("scripts/_setup.R")
```

After setup runs, `eammi` refers to the EAMMi2 data frame. You will see this object throughout the workbook.

Predict

Before running the next section of the script, answer these questions:

1. What does one observation in this dataset represent?
2. What does one variable represent?
3. Do you expect the dataset to contain more observations or more variables?

You are not expected to know the exact numbers yet. A prediction gives you something to compare with the output.

Run The Analysis

Inspect The Dataset

The functions below reveal the dataset's dimensions and variable names.

```
dim(eammi)
[1] 2073  39

names(eammi)
[1] "Duration"          "Finished"          "FinanciallyInd"
[4] "NoLongerHome"      "MOA_ACH"           "Adult"
[7] "Exploration"       "Negativity"        "Identity"
[10] "BetweenStages"     "Politics"          "PoliticsDichotomous"
[13] "Party"             "PartyDichotomous"  "SWLS"
[16] "Mindfulness"       "Belonging"         "SelfEfficacy"
[19] "SupportFriends"    "SupportFamily"     "SupportSpecial"
[22] "SocialSupport"     "SMmaintain"        "SMnewconnect"
[25] "SMinformation"     "SMtotal"           "USdream1"
[28] "USdream2"          "Conflict"          "Symptoms"
[31] "Stress"            "IMPmarriage"       "IMPparenting"
[34] "IMPcareer"         "IMP hobbies"       "Gender"
[37] "Age"              "Race"              "Disability"
```

`dim()` reports observations first and variables second. In this dataset, each observation represents a participant and each variable represents something measured about that participant. `names(eammi)` lists the names of all variables in the data frame.

The next two functions request the same dimensions separately.

```
nrow(eammi)
[1] 2073

ncol(eammi)
[1] 39
```

Investigate The Code

Each line follows the same basic pattern: the function name comes first, and the object the function works on appears inside parentheses.

Code	Question it asks R
<code>dim(eammi)</code>	What are the dimensions of this data frame?
<code>names(eammi)</code>	What are the variable names?
<code>nrow(eammi)</code>	How many observations are there?
<code>ncol(eammi)</code>	How many variables are there?

The output should agree across the two approaches. If `dim(eammi)` begins with 2,073, then `nrow(eammi)` should also return 2,073.

Understand Variable Types

The textbook distinguishes categorical and quantitative variables, then describes their scales of measurement. R also records how each variable is represented. These ideas overlap, but they are not identical. A variable can be stored as numbers even when those numbers represent categories.

Statistical variable types and common R representations

Statistical variable type	Common R representation	Example
Quantitative	numeric or integer	Age, Stress, SWLS
Nominal categorical	factor	Gender, Race, Disability
Ordinal categorical	ordered factor	Adult, NoLongerHome

R can store labeled categories as factors:

- a factor represents categories without a required progression; and
- an ordered factor represents categories with a meaningful order, such as No, Maybe, Yes.

Gender begins as the numeric codes 1, 2, and 3. The next step gives those codes readable category names. We retrieved the meaning of each code from the Gender entry in the EAMMi2 codebook.

```
eammi <- eammi |>
  mutate(
    Gender = factor(
      Gender,
      levels = c(3, 2, 1),
      labels = c("Another identity", "Female", "Male")
    )
  )
```

```
levels(eammi$Gender)
```

```
[1] "Another identity" "Female"          "Male"
```

Investigate The Code

- `mutate()` changes or creates a variable inside a data frame.
- `factor()` tells R that Gender should be treated as categorical.
- `levels` gives the original numeric codes.
- `labels` gives the category names that should be displayed.

- `$` selects one variable from a data frame; `eammi$Gender` means the Gender variable inside `eammi`.
- `levels(eammi$Gender)` shows the labels R now uses for the variable.

The pipe `|>` passes the result on its left into the function on its right. Read it as “and then.” The code starts with `eammi`, and then changes Gender into a labeled factor.

The order of factor levels controls the order R uses in tables and graphs. The order in which we write the levels is important. When a variable is ordinal, use a meaningful order. When a variable is nominal and has no meaningful order, alphabetical order is a useful default because it reduces the chance that presentation order will suggest an unsupported ranking.

Create A Frequency Table

Before running the code, predict which gender category will have the highest frequency in this sample.

```
gender_summary <- eammi |>
  count(Gender, name = "Frequency") |>
  mutate(Percent = round(100 * Frequency / sum(Frequency), 1))
```

gender_summary

A tibble: 3 × 3

	Gender <fct>	Frequency <int>	Percent <dbl>
1	Another identity	30	1.4
2	Female	1524	73.5
3	Male	519	25

`count()` creates a frequency table. R then works like a flexible calculator: it divides each frequency by the total frequency, multiplies by 100, and rounds to one decimal place. `mutate()` creates the new Percent variable inside the summary table.

Create A Bar Chart

The same information can be represented visually.

```
ggplot(gender_summary, aes(x = Gender, y = Frequency)) +
  geom_col() +
  labs(
    x = "Self-reported gender",
    y = "Number of participants"
  )
```

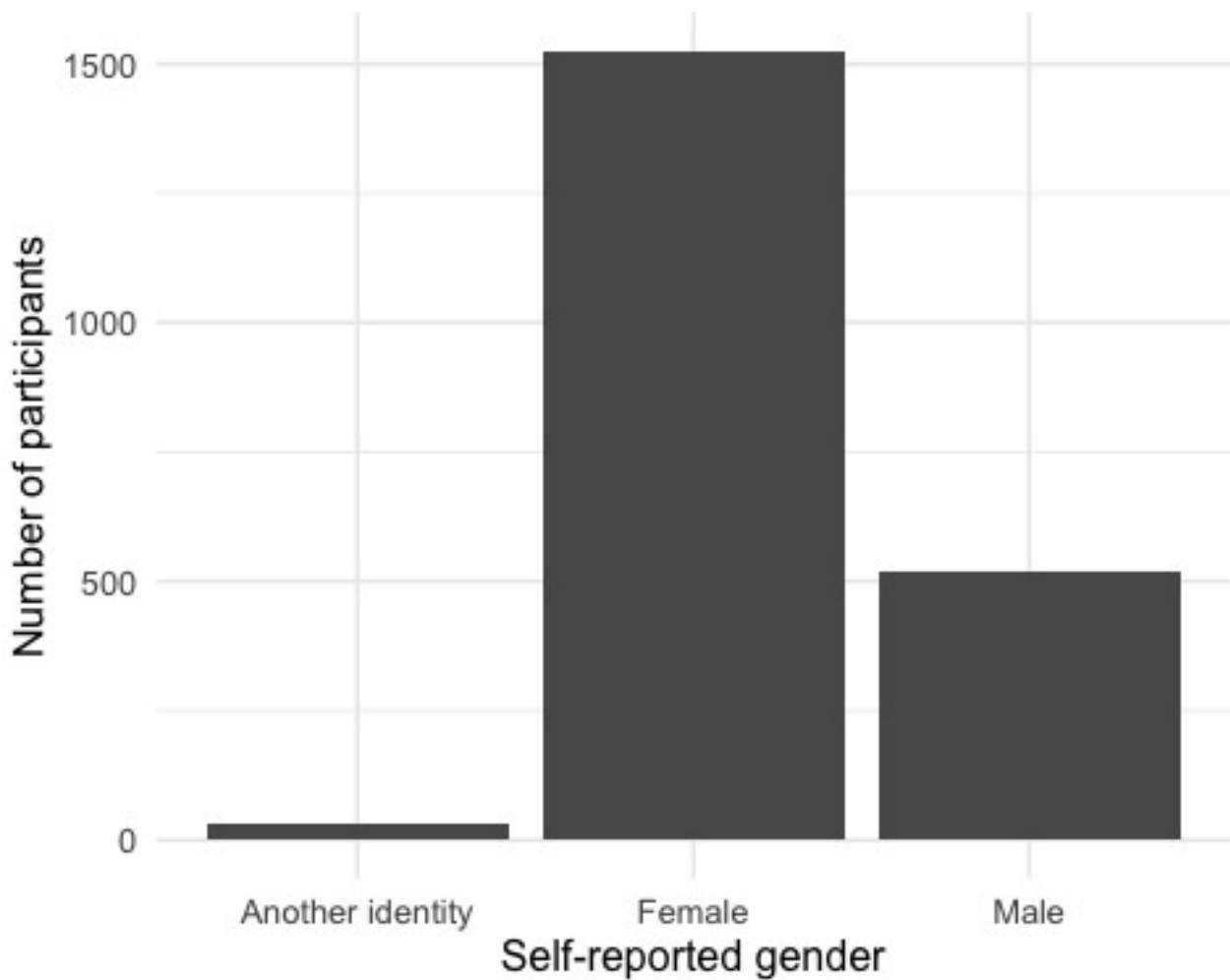


Figure 2: Number of EAMMi2 participants in each self-reported gender category.

Investigate The Code

Every graph in `ggplot2` follows a repeated structure:

1. `ggplot(gender_summary, ...)` names the data or summary object.
2. `aes(x = Gender, y = Frequency)` maps variables to positions on the graph.
3. `geom_col()` requests bars whose heights come from the frequency table.
4. `labs()` adds readable axis labels.

The table supplies exact values; the graph makes the distribution easy to compare. Neither display replaces an interpretation in words.

Produce Your Interpretation

Write two or three sentences that answer the research question. Include:

- the total number of participants and variables in the reduced dataset;
- the most common gender category; and

- at least one frequency and one percentage from gender_summary.

Practice

Exercise 1: Classify Variables

Classify each variable as primarily quantitative, nominal categorical, or ordinal categorical: Duration, NoLongerHome, Adult, PoliticsDichotomous, SWLS, Belonging, Gender, and Age.

Exercise 2: Compare Dataset Views

Run the following code:

```
glimpse(eammi)

Rows: 2,073
Columns: 39
$ Duration      <dbl> 30.65000, 24.45000, 36.41667, 20.48333, 34.46667, ...
$ Finished      <dbl> 1, 1, 0, 1, 1, 0, 0, 0, 1, 1, 1, 1, 1, 0, 1, 1,...
$ FinanciallyInd <dbl> 2, 1, 2, 2, 1, 1, 2, 1, 1, 1, 1, 1, 2, 1, 1, 2, 2,...
$ NoLongerHome  <dbl> 1, 1, 1, 1, 1, 1, 2, 1, 2, 1, 2, 1, 1, 1, 1, 1, 1,...
$ MOA_ACH       <dbl> 38, 33, 33, 47, 27, 34, 39, 34, 31, 29, 36, 33, 37,...
$ Adult         <dbl> 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 1, 2, 1,...
$ Exploration   <dbl> 3.5, 4.0, 4.0, 4.0, 3.5, 3.5, 3.5, 3.5, 4.0, 2.0, ...
$ Negativity    <dbl> 3.5, 4.0, 4.0, 3.5, 3.0, 4.0, 3.5, 3.0, 4.0, 4.0, ...
$ Identity      <dbl> 4.0, 3.5, 4.0, 3.0, 4.0, 3.5, 2.5, 4.0, 4.0, 2.0, ...
$ BetweenStages <dbl> 4.0, 4.0, 3.0, 3.5, 2.5, 3.0, 2.5, 4.0, 3.0, 3.0, ...
$ Politics      <dbl> 2, 1, 2, 1, 8, 4, 2, 3, 5, 1, 8, 2, 8, 6, 1, 3, 2,...
$ PoliticsDichotomous <dbl> 1, 1, 1, 1, NA, NA, 1, 1, 2, 1, NA, 1, NA, 2, 1, 1,...
$ Party        <dbl> 3, 4, 8, 8, 8, 4, 1, 8, 2, 4, 8, 4, 8, 2, 2, 3, 2,...
$ PartyDichotomous <dbl> 1, NA, NA, NA, NA, NA, 1, NA, 1, NA, NA, NA, NA, 1,...
$ SWLS         <dbl> 23, 21, 9, 17, 20, 18, 27, 25, 12, 32, 23, 14, 29,...
$ Mindfulness  <dbl> 2.400000, 1.800000, 2.200000, 3.200000, 3.400000, ...
$ Belonging    <dbl> 4, 4, 2, 4, 3, 4, 4, 4, 5, 1, 3, 5, 4, 5, 2, 4, 5,...
$ SelfEfficacy <dbl> 34, 34, 22, 30, 24, 23, 30, 33, 30, 31, 24, 36, 29,...
$ SupportFriends <dbl> 6.75, 6.50, 5.50, 6.25, 1.75, 4.50, 7.00, 5.75, 6,...
$ SupportFamily <dbl> 6.0, 6.6, 4.0, 5.8, 4.2, 3.6, 5.2, 6.2, 5.8, 3.8, ...
$ SupportSpecial <dbl> 5.25, 7.00, 6.25, 6.00, 7.00, 7.00, 6.50, 5.75, 5,...
$ SocialSupport <dbl> 72, 81, 62, 72, 54, 60, 73, 72, 72, 32, 72, 47, 54,...
$ SMmaintain    <dbl> 3.8, 2.4, 3.0, 3.0, 1.2, 2.8, 4.4, 3.8, 2.6, 3.0, ...
$ SMnewconnect <dbl> 4.75, 1.25, 3.25, 3.50, 1.25, 3.25, 3.25, 3.25, 2,...
$ SMinformation <dbl> 4.5, 3.0, 3.0, 4.0, 1.0, 5.0, 4.0, 1.5, 2.0, 5.0, ...
$ SMtotal      <dbl> 47, 23, 34, 37, 13, 37, 43, 35, 27, 37, 40, 39, 26,...
$ USdream1     <dbl> 4, 4, 2, 3, 1, 4, 2, 1, 4, 5, 1, 2, 3, 2, 3, 1, 3,...
$ USdream2     <dbl> 4, 4, 2, 4, 1, 3, 4, 2, 4, 4, 1, 3, 4, 2, 2, 1, 3,...
$ Conflict     <dbl> 3, 4, 6, 4, 2, 1, 5, 2, 2, 5, 2, 3, 3, 2, 2, 2, 2,...
$ Symptoms     <dbl> 24, 24, 27, 16, 24, 25, 20, 25, 19, 22, 16, 18, 35,...
$ Stress       <dbl> 33, 36, 33, 35, 29, 32, 30, 35, 33, 38, 29, 32, 38,...
$ IMPmarriage  <dbl> 10, 10, 1, 25, 13, 70, 15, 10, 25, 0, 2, 13, 0, 40,...
$ IMPparenting <dbl> 25, 25, 1, 25, 33, 70, 25, 20, 25, 0, 28, 60, 0, 1,...
$ IMPcareer    <dbl> 30, 35, 59, 25, 21, 90, 40, 20, 25, 70, 40, 13, 50,...
$ IMPhobbies   <dbl> 35, 30, 39, 25, 33, 90, 20, 50, 25, 30, 30, 13, 50,...
$ Gender       <fct> Female, Male, Male, Male, Female, Female, Female, ...
$ Age          <dbl> 20, 23, 23, 18, 23, 25, 22, 21, 25, 22, 23, 19, 25,...
$ Race         <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,...
$ Disability   <dbl> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2,...
```

```
summary(eammi$Age)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
18.00	19.00	20.00	20.24	21.00	29.00

In one or two sentences, explain the difference between the information returned by these functions. Remember that the abbreviations in `glimpse()` describe how R stores each variable; they do not determine its statistical role.

Challenge: Create An Ordered Factor

Begin by modifying the frequency-table analysis:

1. Copy the code that creates `gender_summary`.
2. Give the new object a descriptive name, such as `adult_summary`.
3. Replace `Gender` with `Adult`.
4. Run the code and examine the numeric category codes.

`Adult` still contains numeric category codes. Use the `Adult` entry in Understanding EAMMi2 to identify what those codes mean.

Next, adapt the `Gender` factor code to convert `Adult` into an ordered factor. Then create a new frequency table and explain why the order of the factor levels matters. Try completing these steps before checking the code below.

Check Your Work

Dataset Dimensions

`nrow(eammi)` returns 2,073, the number of participants. `ncol(eammi)` returns 39, the number of variables retained in the reduced teaching dataset.

Sample Interpretation

The reduced EAMMi2 dataset contains 2,073 participants and 39 variables. A majority of participants identified as female (73.5%, or $n = 1,524$), followed by 519 participants who identified as male (25.0%) and 30 participants who identified as another gender (1.4%).

Exercise 1

- Quantitative: `Duration`, `SWLS`, `Belonging`, `Age`
- Nominal categorical: `PoliticsDichotomous`, `Gender`
- Ordinal categorical: `NoLongerHome`, `Adult`

Reasonable disagreements about treating bounded rating-scale variables as ordinal or quantitative are worth discussing. In this workbook, scale scores such as `SWLS` and `Belonging` are generally analyzed as quantitative.

Exercise 2

`glimpse()` provides a compact overview of the entire data frame, including each column's R representation and example values. `summary(eammi$Age)` focuses on one variable and returns its minimum, quartiles, median, mean, maximum, and missing-value count when applicable.

Challenge

```
eammi <- eammi |>
  mutate(
    Adult = factor(
      Adult,
      levels = c(3, 2, 1),
      labels = c("No", "Maybe", "Yes"),
      ordered = TRUE
    )
  )

count(eammi, Adult)
```

The order tells R how the categories progress conceptually. It also controls their default order in tables and graphs.

Chapter Takeaway

A reproducible R analysis begins with a sequence you can save and rerun: load the data, inspect the variables, make their statistical roles explicit, summarize, graph, and interpret. Every later chapter builds on this sequence.

In this chapter, we focused on these R commands and code patterns:

- `source()` runs the shared setup script.
- `dim()`, `nrow()`, and `ncol()` report dataset dimensions.
- `names()` lists variable names.
- `factor()` gives categorical codes readable labels and an order.
- `levels()` shows the labels for a factor.
- `count()` creates a frequency table.
- `mutate()` changes or creates variables.
- `ggplot()` starts a graph and names its data.
 - `aes()` maps variables to visual positions.
 - `geom_col()` creates bars from values already stored in a table.
 - `labs()` adds readable labels.

Chapter 2: Exploring Data

Chapter 2 of *Exploring Statistics* introduces frequency distributions and graphs as tools for organizing data. For starters, we will use R to learn who participated in EAMMi2 and what their scores look like. Tables give us the exact values; graphs help us see the larger pattern.

Learning Goals

By the end of this chapter, you should be able to:

- create frequency tables for categorical and quantitative variables;
- calculate and interpret percentages and cumulative percentages;
- produce bar charts for categorical variables;
- produce a histogram for a quantitative variable;
- describe the shape of a quantitative distribution; and
- adapt a demonstrated analysis for new variables.

Research Questions

This chapter is organized around two research questions:

1. Who are our EAMMi2 participants?
2. Where do our participants stand on adulthood status?

Research Question 1 provides a worked analysis. Research Question 2 asks you to modify the same code patterns for two new variables.

Research Question 1: Who Are Our EAMMi2 Participants?

For starters, let's examine two variables: gender identity and racial identity. Then we will create a frequency table and graph for a quantitative variable: Satisfaction With Life.

Get Ready

This chapter uses `scripts/02-frequency-distributions.R` in the starter files.

Open the Chapter 2 script

1. Open `exploring-statistics-r.Rproj`.
2. In the Output pane, select the **Files** tab.
3. Open the `scripts` folder.
4. Open `02-frequency-distributions.R`.
5. Run one complete expression at a time as you work through this chapter.

Begin by running the shared setup file.

```
source("scripts/_setup.R")
```

The first research question uses three variables:

Research Question 1 variables

Variable	What it measures	Statistical variable type
Gender	Self-reported gender identity	categorical
Race	Self-reported racial identity	categorical
SWLS	Satisfaction With Life Scale score	quantitative

Use Understanding EAMMi2 when you need to check a variable's category codes or possible score range.

Predict

Before preparing the variables, record your predictions:

1. Which gender category and racial identity category do you expect to appear most often?
2. Will a bar chart or histogram be appropriate for Gender and Race?
3. Will a bar chart or histogram be appropriate for SWLS?

Run The Analysis**Prepare Gender**

Gender is stored as the numeric codes 1, 2, and 3. The next code gives those codes readable labels.

```
eammi <- eammi |>
  mutate(
    Gender = factor(
      Gender,
      levels = c(3, 2, 1),
      labels = c("Another identity", "Female", "Male")
    )
  )
```

This is the `mutate()` and `factor()` pattern introduced in Chapter 1. The categories are displayed alphabetically because Gender is nominal and does not have a meaningful low-to-high order.

Prepare Race

The next block uses the same pattern for Race. Before running it, identify the three parts that changed from the Gender code: the variable name, the original codes, and the displayed labels.

```
eammi <- eammi |>
  mutate(
    Race = factor(
      Race,
      levels = c(2, 6, 4, 3, 7, 8, 5, 1),
      labels = c(
        "African American",
        "Another racial identity",
        "Asian/Pacific Islander",
        "Hispanic/Latinx",
        "Multiracial (2 selected)",
        "Multiracial (3+ selected)"
      )
    )
  )
```

```

    "Native American",
    "White"
  )
)
)

```

Create Frequency Tables

This is the frequency-table code introduced in Chapter 1, with one addition: `CumulativePercent` adds percentages as you move down the table.

```

gender_table <- eammi |>
  count(Gender, name = "Frequency") |>
  mutate(
    Percent = round(100 * Frequency / sum(Frequency), 1),
    CumulativePercent = round(100 * cumsum(Frequency) / sum(Frequency), 1)
  )

```

```

gender_table
# A tibble: 3 × 4
  Gender      Frequency Percent CumulativePercent
<fct>      <int>    <dbl>          <dbl>
1 Another identity     30     1.4            1.4
2 Female             1524    73.5            75
3 Male                519    25            100

```

Investigate The Code

- `Frequency` gives the number of participants in each category.
- `Percent` gives each category's share of the responses.
- `cumsum(Frequency)` creates a running total.
- `CumulativePercent` gives a running total through the displayed rows. For an ordered variable, it can answer questions about the percentage at or below a value. For nominal variables such as `Gender` and `Race`, it is only a display total because the categories do not have a low-to-high order.

Now modify the same pattern for `Race`.

```

race_table <- eammi |>
  drop_na(Race) |>
  count(Race, name = "Frequency") |>
  mutate(
    Percent = round(100 * Frequency / sum(Frequency), 1),
    CumulativePercent = round(100 * cumsum(Frequency) / sum(Frequency), 1)
  )

```

```

race_table
# A tibble: 8 × 4
  Race                        Frequency Percent CumulativePercent
<fct>      <int>    <dbl>          <dbl>
1 African American           172     8.3            8.3
2 Another racial identity      76     3.7            12
3 Asian/Pacific Islander     141     6.8           18.8
4 Hispanic/Latinx           222    10.7           29.5
5 Multiracial (2 selected)    173     8.3           37.8
6 Multiracial (3+ selected)   28     1.4           39.2

```

7 Native American	8	0.4	39.6
8 White	1252	60.4	100

Race has one missing response. `drop_na(Race)` leaves out that response so the percentages describe participants who provided a racial identity response.

Create Bar Charts

Both Gender and Race are categorical, so each bar represents a category.

```
ggplot(gender_table, aes(x = Gender, y = Frequency)) +
  geom_col() +
  labs(
    x = "Self-reported gender",
    y = "Frequency"
  )
```

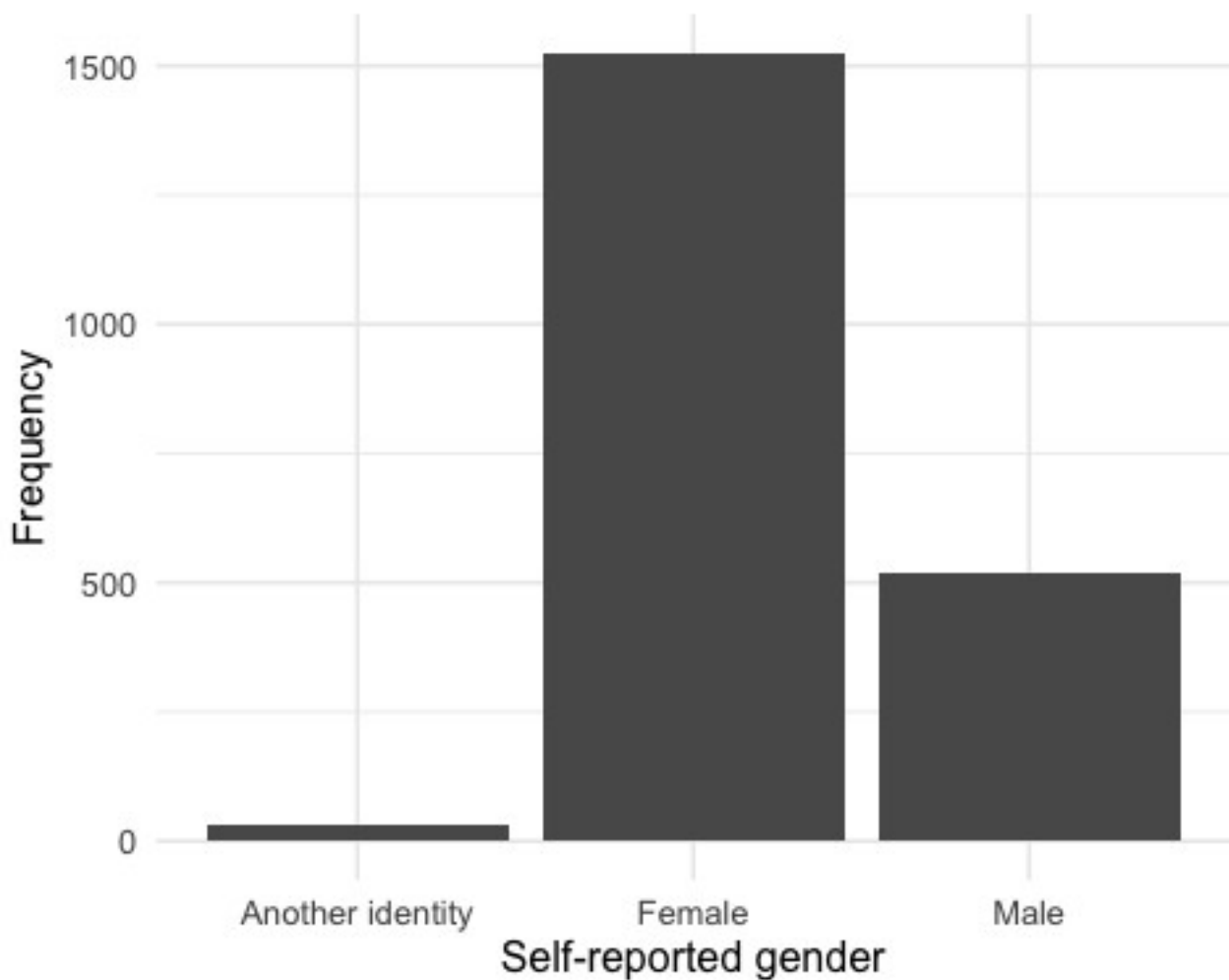


Figure 3: Number of EAMMi2 participants in each self-reported gender category.

This repeats the Chapter 1 pattern: name the table in `ggplot()`, map the category and frequency to the axes in `aes()`, request bars with `geom_col()`, and add readable labels with `labs()`.

Now apply the same pattern to Race. `coord_flip()` turns the graph sideways so the longer labels are easier to read.

```
ggplot(race_table, aes(x = Race, y = Frequency)) +
  geom_col() +
  coord_flip() +
  labs(
    x = "Self-reported racial identity",
    y = "Frequency"
  )
```

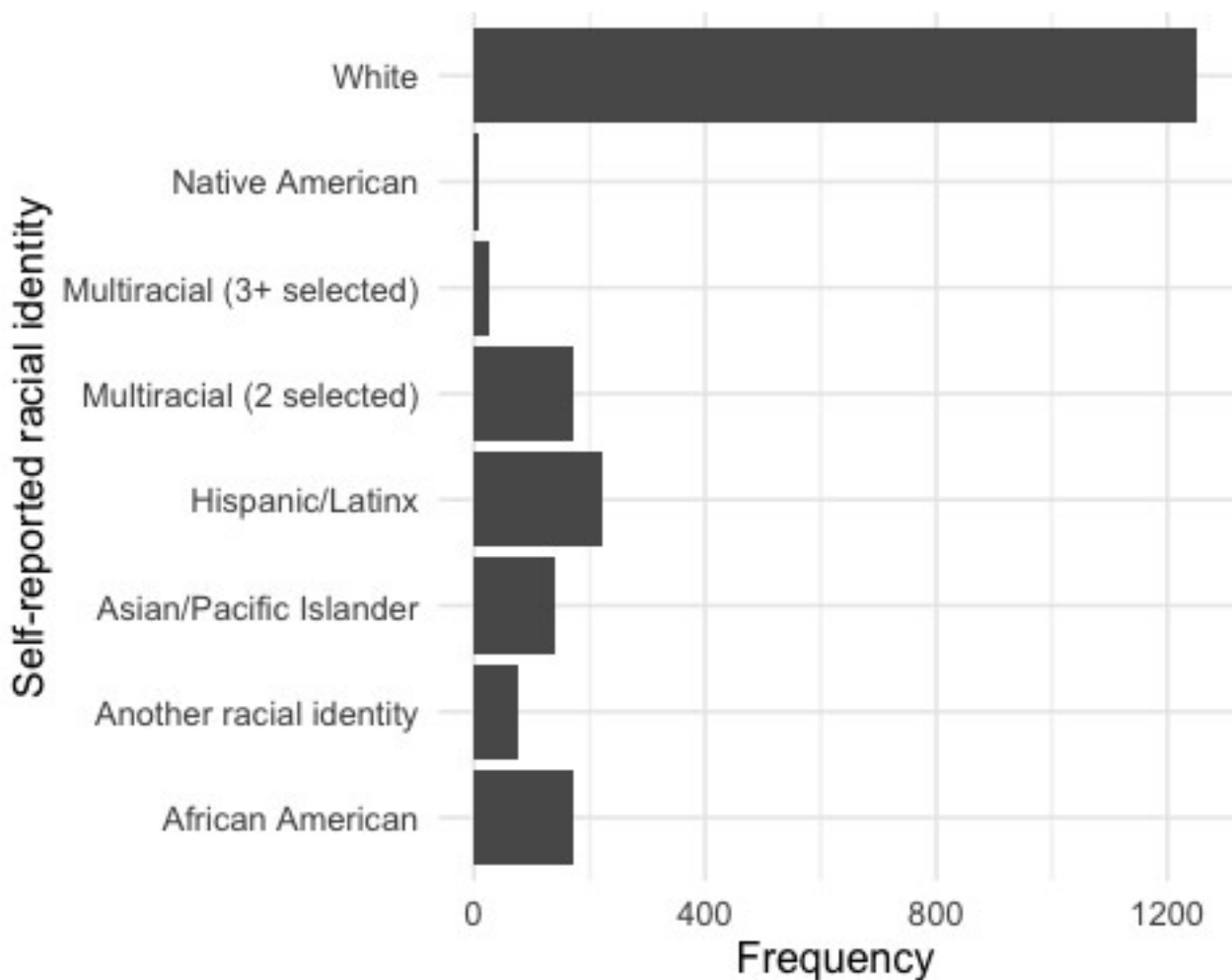


Figure 4: Number of EAMMi2 participants in each self-reported racial identity category.

Examine Satisfaction With Life

SWLS contains scores on the Satisfaction With Life Scale. The scale contains five questions, each answered from 1 to 7, so possible total scores range from 5 to 35.

You might remember from Chapter 2 of your textbook that this is the variable used to introduce frequency distributions and frequency polygons or histograms. If not, now is a good time to reread that section.

Use `class()` to check how R currently represents the variable.

```
class(eammi$SWLS)
[1] "numeric"
```

R represents SWLS as numeric. The code below creates a frequency table with the score, frequency, percent, and cumulative percent.

```
swls_table <- eammi |>
  count(SWLS, name = "Frequency") |>
  mutate(
    Percent = round(100 * Frequency / sum(Frequency), 1),
    CumulativePercent = round(100 * cumsum(Frequency) / sum(Frequency), 1)
  )
```

```
print(swls_table, n = 32)
```

```
# A tibble: 32 × 4
  SWLS Frequency Percent CumulativePercent
<dbl>      <int>   <dbl>      <dbl>
1      0         1      0.0         0.0
2      5        18      0.9         0.9
3      6        14      0.7         1.6
4      7        12      0.6         2.2
5      8        24      1.2         3.3
6      9        31      1.5         4.8
7     10        39      1.9         6.7
8     11        34      1.6         8.3
9     12        43      2.1        10.4
10    13        46      2.2        12.6
11    14        53      2.6        15.2
12    15        50      2.4        17.6
13    16        70      3.4        21.0
14    17        64      3.1        24.1
15    18        91      4.4        28.5
16    19        87      4.2        32.7
17    20        94      4.5        37.2
18    21       105      5.1        42.3
19    22        94      4.5        46.8
20    23       118      5.7        52.5
21    24       115      5.5        58.0
22    25       111      5.4        63.4
23    26       112      5.4        68.8
24    27       129      6.2        75.0
25    28       102      4.9        79.9
26    29        92      4.4        84.4
27    30       107      5.2        89.5
28    31        69      3.3        92.9
29    32        54      2.6        95.5
30    33        35      1.7        97.2
31    34        24      1.2        98.3
32    35        35      1.7       100.0
```

Pause And Inspect

Before reading further, scan the first and last rows of the table. Do all observed scores fall between 5 and 35?

One response has an observed SWLS value of 0, outside the possible range. This is not an error in R. Real datasets sometimes contain coding errors, impossible values, or unfinished responses. Examining variables before interpreting them helps us identify these issues.

The cumulative percent column answers questions such as, “What percent of the sample scored below 20?” The row for SWLS = 19 gives the cumulative percentage for scores below 20.

A frequency table gives exact counts. A histogram makes the shape of a quantitative distribution easier to see.

```
ggplot(eammi, aes(x = SWLS)) +  
  geom_histogram(binwidth = 1, boundary = 0) +  
  labs(  
    x = "Satisfaction With Life Scale score",  
    y = "Frequency"  
  )
```

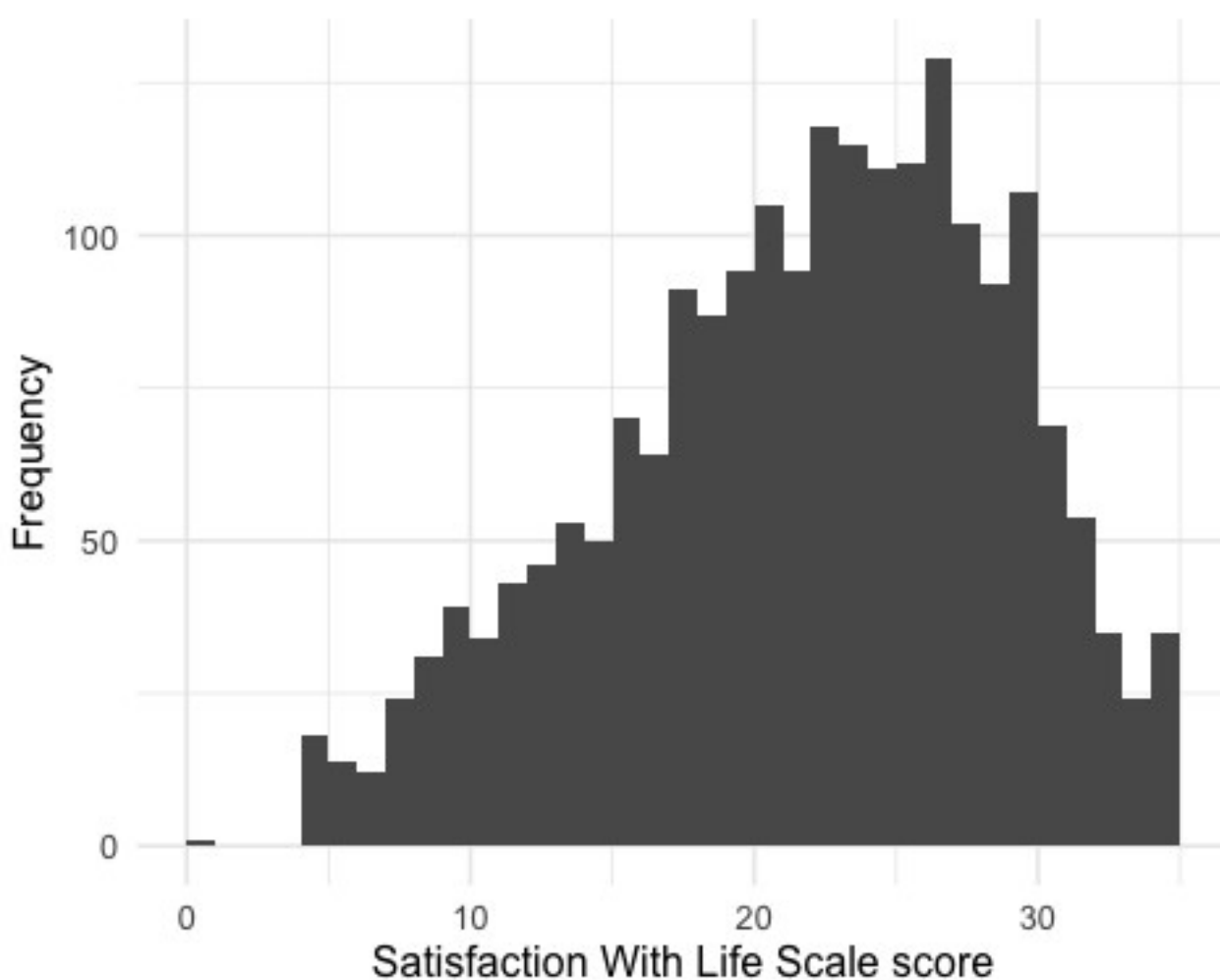


Figure 5: Distribution of Satisfaction With Life Scale scores in the reduced EAMMi2 sample.

Investigate The Graph

This graph starts with the original `eammi` data frame rather than a frequency table. Because SWLS is quantitative, it uses `geom_histogram()` instead of `geom_col()`.

- `geom_col()` draws bars from values already calculated in a table.
- `geom_histogram()` groups quantitative scores into intervals and counts the observations in each interval.

Read The Output

The reduced EAMMi2 sample was comprised of 2,073 respondents between the ages of 18 and 29 years. Of these respondents, a majority identified as female (73.5%, or $n = 1,524$), with 519 who identified as male (25.0%) and 30 participants who identified as another gender (1.4%). Furthermore, a majority of respondents identified as White (60.4%, or $n = 1,252$), followed by Hispanic/Latinx (10.7%, or $n = 222$) and Multiracial (2 selected) (8.3%, or $n = 173$). Frequencies for all other racial identities are presented in the table.

Neither the SWLS frequency table nor the histogram gives us all of the information we need by itself. Together, they provide a richer understanding: the table gives exact counts and percentages, while the histogram makes the overall shape easier to see.

Produce Your Interpretation

Write a paragraph summarizing SWLS. State the shape of the distribution, the percent of participants who scored below 20, and the most common score. Then compare the distribution with the one presented in Chapter 2 of your textbook.

Reporting sample sizes

In APA style, use capital N for the entire sample and lower-case n for a subgroup.

Research Question 2: Where Do Our Participants Stand On Adulthood Status?

As a reminder, EAMMi2 stands for the Emerging Adulthood Measured at Multiple Institutions study, second wave. Where do our participants stand on adulthood status?

Get Ready

Research Question 2 variables

Variable	What it measures	Statistical variable type
Adult	Whether the participant identifies as an adult	ordinal categorical
NoLongerHome	Whether the participant has moved out of their parents' home	ordinal categorical

These variables are ordinal, so place their levels in a meaningful order.

```
eammi <- eammi |>
  mutate(
    Adult = factor(
      Adult,
      levels = c(3, 2, 1),
```

```

    labels = c("No", "Maybe", "Yes"),
    ordered = TRUE
  ),
  NoLongerHome = factor(
    NoLongerHome,
    levels = c(1, 2, 3),
    labels = c("No", "Somewhat", "Yes"),
    ordered = TRUE
  )
)

```

Predict

For each variable, predict the most common response. Then decide which graph is appropriate.

Complete The Analysis

Create a frequency table and graph for each variable. Use Research Question 1 as your model.

1. Copy the frequency-table code.
2. Create new object names rather than overwriting `gender_table` or `race_table`.
3. Replace the variable name everywhere it appears.
4. Decide whether missing responses should be omitted from the table.
5. Copy and modify the bar-chart code.
6. Write a short interpretation naming the most common response and reporting counts and percentages.

Attempt the analysis before using the completed code in Check Your Work.

Check Your Work

Research Question 1

In the reduced EAMMi2 sample, there was a slight negative skew to Satisfaction With Life Scale scores, with 32.7% of the sample scoring below the midpoint of the scale, or below 20. Furthermore, the most common score was 27 out of a possible 35. This mirrors the distribution presented in Chapter 2 of *Exploring Statistics*, where 18% of the sample scored below the midpoint and 27 was again the most common score. In short, satisfaction with life appears to be relatively high in both samples.

Research Question 2 Code

```

adult_table <- eammi |>
  drop_na(Adult) |>
  count(Adult, name = "Frequency") |>
  mutate(
    Percent = round(100 * Frequency / sum(Frequency), 1),
    CumulativePercent = round(100 * cumsum(Frequency) / sum(Frequency), 1)
  )

```

```

adult_table
# A tibble: 3 × 4
  Adult Frequency Percent CumulativePercent
<ord>    <int>    <dbl>         <dbl>
1 No      147      7.1          7.1

```

```

2 Maybe      518    25      32.1
3 Yes       1407   67.9     100

```

```

home_table <- eammi |>
  drop_na(NoLongerHome) |>
  count(NoLongerHome, name = "Frequency") |>
  mutate(
    Percent = round(100 * Frequency / sum(Frequency), 1),
    CumulativePercent = round(100 * cumsum(Frequency) / sum(Frequency), 1)
  )

```

```

home_table
# A tibble: 3 × 4
  NoLongerHome Frequency Percent CumulativePercent
<ord>         <int>    <dbl>          <dbl>
1 No           831     40.5           40.5
2 Somewhat    656     32            72.5
3 Yes         565     27.5           100

```

Adult has one missing response, and NoLongerHome has 21 missing responses. `drop_na()` leaves those responses out of the tables.

```

ggplot(adult_table, aes(x = Adult, y = Frequency)) +
  geom_col() +
  labs(
    x = "Response to 'Are you an adult?'",
    y = "Frequency"
  )

```

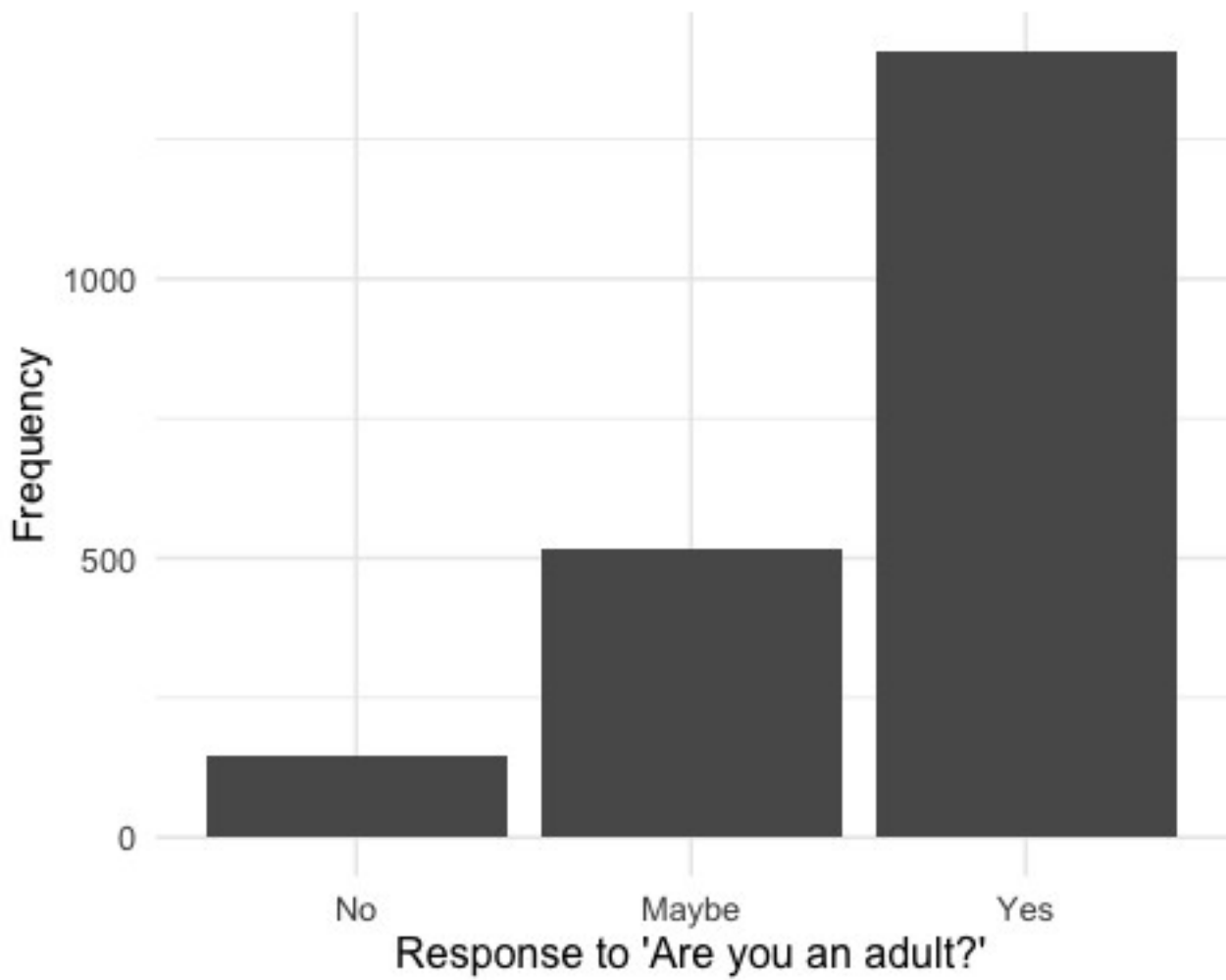


Figure 6: Number of EAMMi2 participants by self-reported adult status.

```
ggplot(home_table, aes(x = NoLongerHome, y = Frequency)) +  
  geom_col() +  
  labs(  
    x = "Moved out of parents' home",  
    y = "Frequency"  
  )
```

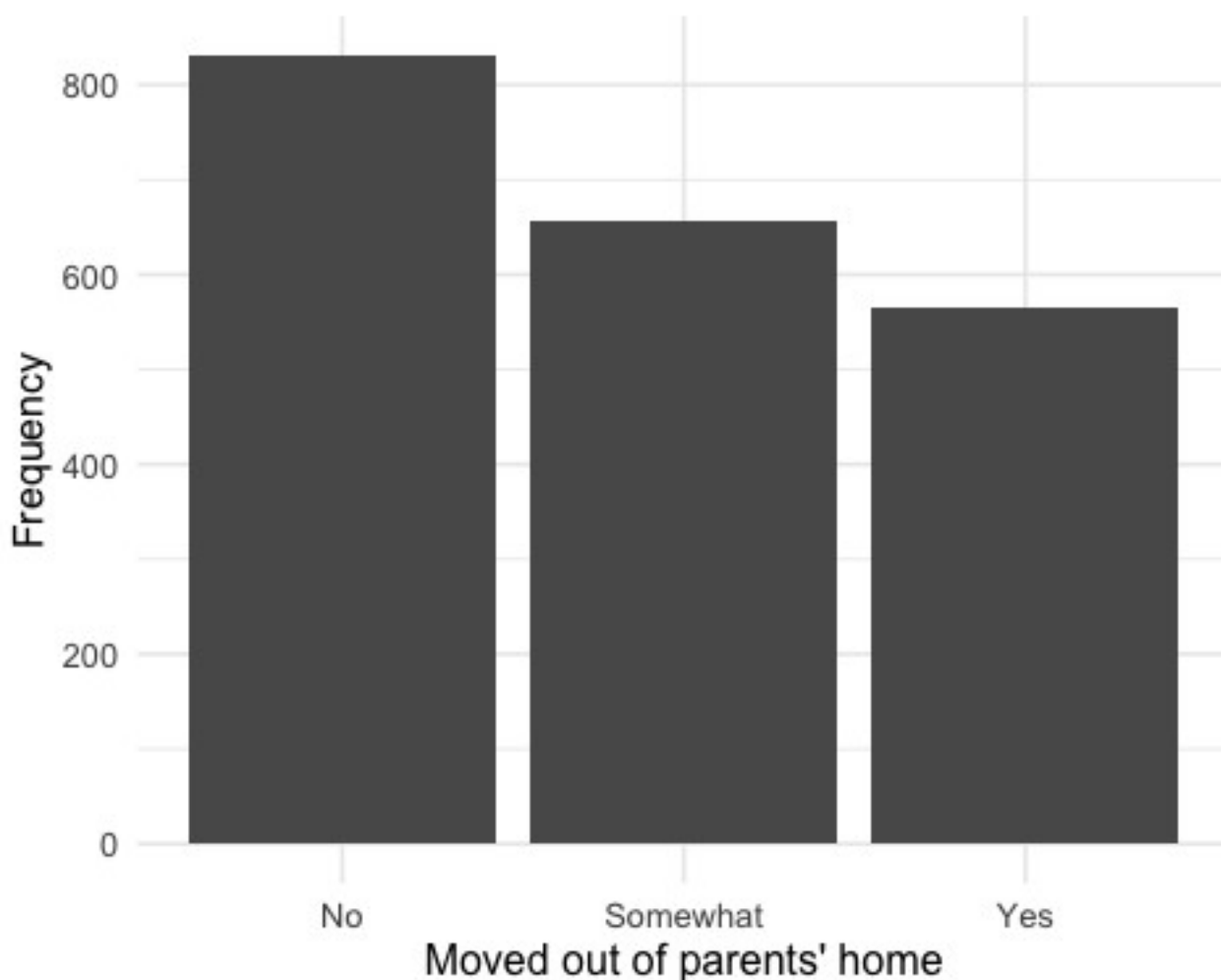


Figure 7: Number of EAMMi2 participants by whether they have moved out of their parents' home.

Research Question 2 Interpretation

The reduced EAMMi2 sample was comprised of 2,073 respondents between the ages of 18 and 29 years. Of these respondents, a majority identified as an adult (67.9%, or $n = 1,407$), with 518 who said they were “maybe” an adult (25.0%) and 147 who said they were not an adult (7.1%). Respondents were more evenly distributed in their responses about moving out of their parents' home: 831 had not moved out (40.5%), 656 had “somewhat” moved out (32.0%), and 565 had moved out (27.5%).

Chapter Takeaway

Frequency tables and graphs help us organize a distribution before interpreting it. For categorical variables, frequency tables and bar charts show how many people fall into each category. For quantitative variables, a frequency table provides exact counts while a histogram shows the distribution's shape.

In this chapter, we focused on these new R commands:

- `cumsum()` creates cumulative totals used for cumulative percentages.
- `drop_na()` omits missing responses from a specific analysis.

- `class()` shows how R represents a variable.
- `geom_histogram()` creates a histogram for a quantitative variable.
- `coord_flip()` turns a graph sideways when category labels are long.

Chapters 3 and 4: Central Tendency and Variability

Chapters 3 and 4 of *Exploring Statistics* introduce descriptive statistics that summarize quantitative variables. A single average never tells the whole story, so you will pair measures of central tendency with variability and distribution shape as you compare groups.

Learning Goals

By the end of this chapter, you should be able to:

- calculate and interpret the mean, median, mode, and standard deviation;
- use minimum and maximum values to check whether scores fit the intended scale;
- identify and set aside an impossible value;
- compare descriptive statistics across groups;
- use faceted histograms to compare distribution shape; and
- adapt a descriptive-statistics pattern for a variable you choose.

Research Questions

This chapter is organized around two research questions:

1. How important do emerging adults consider their career to be part of their lives? Do ratings of career importance, rated from 0-100%, vary as a function of whether or not the individual is still living at home?
2. What can descriptive statistics tell us about variables you choose from the EAMMi2 codebook?

Research Question 1 provides a worked grouped analysis. Research Question 2 asks you to produce a similar analysis with less guidance.

Research Question 1: Career Importance And Living At Home

How important do emerging adults consider their career to be part of their lives? Do ratings of career importance, rated from 0-100%, vary as a function of whether or not the individual is still living at home? Generate descriptive statistics for ratings of career importance for those still living at home, those “somewhat” living at home, and those no longer living at home.

Get Ready

This chapter uses `scripts/03-04-central-tendency-variability.R` in the starter files.

Open the Chapter 3/4 script

1. Open `exploring-statistics-r.Rproj`.
2. In the Output pane, select the **Files** tab.
3. Open the `scripts` folder.
4. Open `03-04-central-tendency-variability.R`.
5. Run one complete expression at a time as you work through this chapter.

```
source("scripts/_setup.R")
```

The first research question uses two variables:

Research Question 1 variables

Variable	What it measures	Role in the analysis
IMPcareer	Importance placed on career as an eventual or current aspect of life	quantitative outcome
NoLongerHome	Whether the participant has moved out of their parents' home	ordinal grouping variable

Participants were instructed to assign IMPcareer a percentage from 0 to 100. Give NoLongerHome readable labels in a meaningful order.

```
eammi <- eammi |>
  mutate(
    NoLongerHome = factor(
      NoLongerHome,
      levels = c(1, 2, 3),
      labels = c("No", "Somewhat", "Yes"),
      ordered = TRUE
    )
  )
```

Predict

Before calculating the statistics, record your expectations:

1. Do you expect the mean career-importance rating to be similar across the three groups?
2. What are the smallest and largest valid values the analysis should report?
3. If the mean is larger than the median, which direction of skew would you expect?

Run The Analysis

Inspect The Observed Range

Start with a complete descriptive-statistics table. `group_by(NoLongerHome)` tells R to calculate later results separately for each group. `summarise()` creates a smaller table containing the requested statistics.

```
eammi |>
  drop_na(NoLongerHome) |>
  group_by(NoLongerHome) |>
  summarise(
    N = n(),
    Missing = sum(is.na(IMPcareer)),
    Mean = mean(IMPcareer, na.rm = TRUE),
    Median = median(IMPcareer, na.rm = TRUE),
    Mode = statistical_mode(IMPcareer),
    SD = sd(IMPcareer, na.rm = TRUE),
    Minimum = min(IMPcareer, na.rm = TRUE),
    Maximum = max(IMPcareer, na.rm = TRUE)
  )

# A tibble: 3 × 9
  NoLongerHome      N Missing  Mean Median  Mode    SD Minimum Maximum
  <ord>         <int>   <int> <dbl> <dbl> <int> <dbl>   <dbl>   <dbl>
1 No             831     0  37.2    30    30  21.5     0    180
```

2	Somewhat	656	2	35.1	30	20	18.8	0	100
3	Yes	565	0	37.7	30	25	23.3	0	100

Investigate The Code

Read the summary from the outside inward:

1. start with `eammi`;
2. omit observations missing the grouping variable;
3. create groups using `NoLongerHome`; and
4. calculate the same set of statistics within each group.

`summarise()` is useful because it reduces many observations to a few descriptive values. Each named line creates one column in the result.

R's built-in `mode()` function does not calculate the statistical mode. The workbook setup file therefore supplies `statistical_mode()`, which returns the most common observed value.

Take a minute to scan the entire output. Which statistics look similar across the groups, and which one first catches your attention?

Check The Data

Compare the reported maximum with the intended 0-100 range. The maximum is 180, so at least one observation is impossible for this measure. This is not an error in R. It is a data-quality issue.

Use `filter()` to locate the problem.

```
eammi |>
  filter(IMPcareer > 100) |>
  count(NoLongerHome, IMPcareer, name = "Frequency")

# A tibble: 1 × 3
  NoLongerHome IMPcareer Frequency
  <ord>         <dbl>      <int>
1 No           180         1
```

`filter()` keeps observations that meet a condition. Here, `IMPcareer > 100` keeps only values above the intended maximum.

Because 180 is impossible on a 0-100 scale, set it aside before calculating the final statistics.

Recalculate The Summary

The next block repeats the descriptive-statistics pattern but adds a filter that keeps only valid career-importance ratings. Before running it, locate the new line and explain what each condition retains.

```
career_summary <- eammi |>
  filter(IMPcareer >= 0, IMPcareer <= 100) |>
  drop_na(NoLongerHome) |>
  group_by(NoLongerHome) |>
  summarise(
    N = n(),
    Mean = mean(IMPcareer),
    Median = median(IMPcareer),
```

```

    Mode = statistical_mode(IMPcareer),
    SD = sd(IMPcareer),
    Minimum = min(IMPcareer),
    Maximum = max(IMPcareer)
  )

```

```
career_summary
```

```
# A tibble: 3 × 8
```

	NoLongerHome	N	Mean	Median	Mode	SD	Minimum	Maximum
	<ord>	<int>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>
1	No	830	37.0	30	30	20.9	0	100
2	Somewhat	654	35.1	30	20	18.8	0	100
3	Yes	565	37.7	30	25	23.3	0	100

Compare Distribution Shapes

Histograms show the shape of a distribution. `facet_wrap()` creates one panel for each level of `NoLongerHome`.

```

eammi |>
  filter(IMPcareer >= 0, IMPcareer <= 100) |>
  drop_na(NoLongerHome) |>
  ggplot(aes(x = IMPcareer)) +
  geom_histogram(binwidth = 5, boundary = 0) +
  facet_wrap(~ NoLongerHome, ncol = 1) +
  labs(
    x = "Career importance rating",
    y = "Frequency"
  )

```

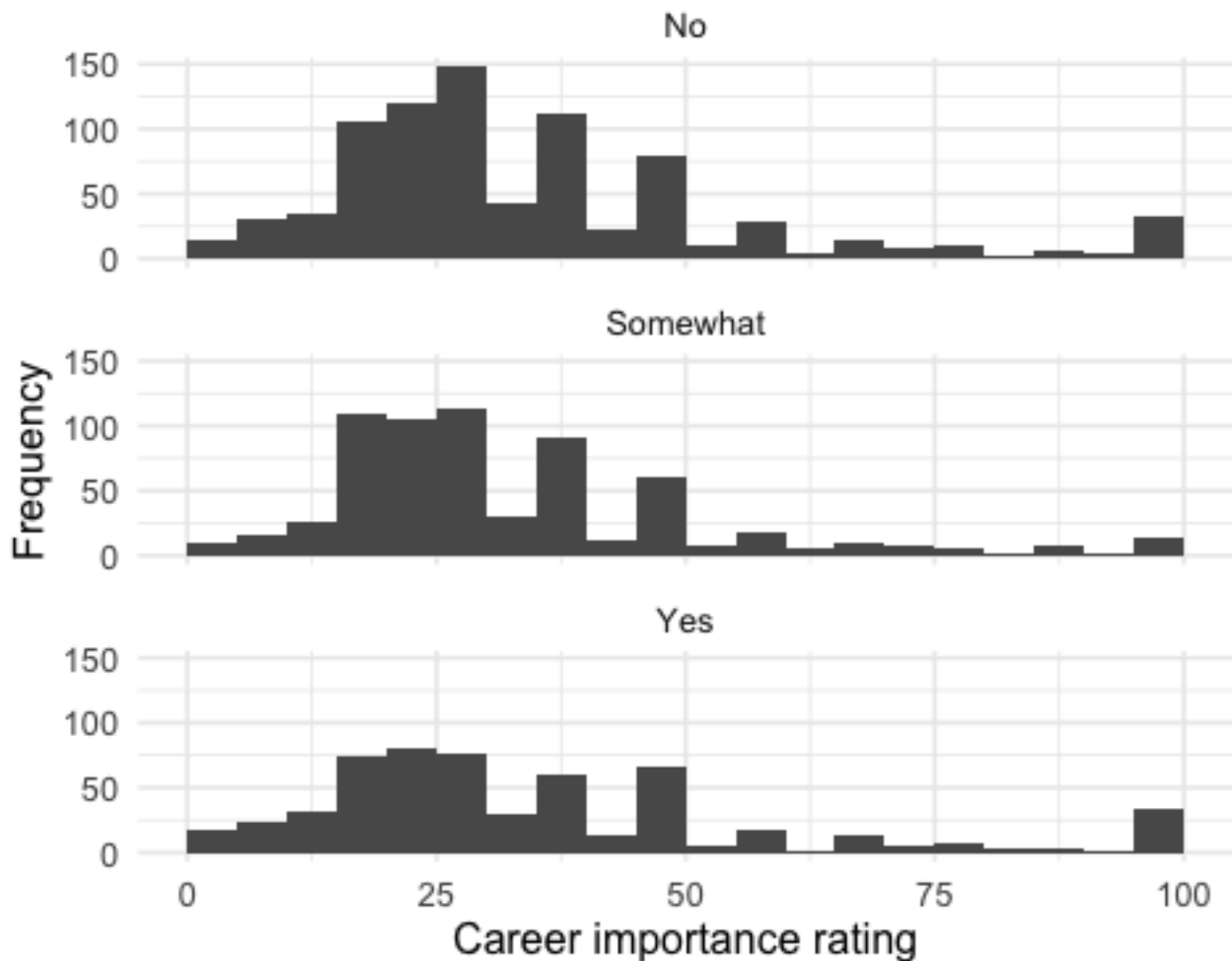


Figure 8: Career-importance ratings by whether participants have moved out of their parents' home.

Investigate The Graph

The graph reuses the same data-quality filter as the final table. The part after `ggplot()` adds two layers:

- `geom_histogram()` groups IMPcareer ratings into intervals; and
- `facet_wrap(~ NoLongerHome, ncol = 1)` repeats the histogram once for each living-at-home group and places the panels in one column.

The tilde in `~ NoLongerHome` can be read as “separate the graph by NoLongerHome.”

Read The Output

All three distributions are positively skewed. The mean is fairly consistent across the three groups, and the medians are identical. Notice, however, that the mode varies from group to group. The mode will usually be the measure of central tendency that varies the most from sample to sample, so this should not be too surprising.

There is also some consistency in the amount of variability present in each group. Always remember that what constitutes a large or small standard deviation depends on how the variable is measured. Here, scores ranged from 0-100. The standard deviations show that scores deviate, on average, about 19 to 23 points from their group

means. That variability represents about one-fifth of the possible score range, so it might be considered moderate or even large.

The Minimum and Maximum columns confirm that the final descriptive table is based on values from 0 to 100. Checking the range before interpreting the results helped us identify and set aside one impossible value.

Produce Your Interpretation

Write a paragraph or two summarizing the output, referencing each of the three groups. Include:

- the data-quality decision;
- the mean, median, and standard deviation for each group;
- the general shape of the distributions; and
- a conclusion about whether career importance differs meaningfully across groups.

Research Question 2: Practice With Descriptive Statistics

Review Understanding EAMMi2 and choose one or two quantitative variables that interest you. Learn about each variable by producing a descriptive-statistics analysis, then write a short paragraph summarizing your findings.

Use Research Question 1 as your model. For each variable:

1. check its intended range;
2. predict its distribution shape;
3. inspect its observed minimum and maximum;
4. decide whether any observed value is impossible;
5. calculate descriptive statistics;
6. create a histogram; and
7. interpret the center, variability, and shape.

If you compare groups, reuse the `group_by()` and `facet_wrap()` patterns. Copying a working block and carefully replacing the variable and object names is an appropriate way to begin.

Check Your Work

Research Question 1

In our reduced EAMMi2 sample, we examined participants' ratings of the importance of their career to their lives across three groups: those who still lived at home, those who "somewhat" lived at home, and those who no longer lived at home. One observed value, `IMPcareer = 180`, fell outside the possible 0-100 range and was set aside before calculating the final descriptive statistics.

After setting aside that impossible value, responses ranged from 0-100, the full range of possible values, for each group. Both the mean importance rating and the standard deviation were relatively stable across those still living at home ($M = 37.00$, $SD = 20.90$), those "somewhat" living at home ($M = 35.08$, $SD = 18.75$), and those no longer living at home ($M = 37.74$, $SD = 23.32$). Furthermore, the median rating was 30.00 across all three groups. There was a moderate amount of response variability in all three groups, and all three distributions showed a positive skew.

Research Question 2

Use these questions to check your analysis:

- Did you select a quantitative variable?
- Did you check whether observed values fit its intended range?
- If an impossible value appeared, did you explain what you did with it?
- Did you report measures of central tendency and variability?
- Did you describe the distribution's shape?
- If you compared groups, did you name and compare each group clearly?
- Did your written interpretation answer the statistical question rather than describe the code?

Chapter Takeaway

The mean, median, and mode describe central tendency, while the standard deviation, minimum, maximum, and histogram help us understand variability and shape. Before interpreting any descriptive statistic, check whether the observed values make sense for the way the variable was measured.

In this chapter, we focused on these new R commands:

- `filter()` keeps observations that meet a condition.
- `group_by()` tells R to calculate later results separately for each group.
- `summarise()` creates a smaller table of summary values.
- `n()` counts observations inside a summary.
- `sum()` adds values.
- `is.na()` checks whether values are missing.
- `mean()` calculates the arithmetic average.
- `median()` finds the middle value.
- `sd()` calculates the standard deviation.
- `min()` and `max()` find the observed endpoints.
- `facet_wrap()` creates separate graph panels for groups.
- `statistical_mode()` is supplied by the setup file and returns the most common observed value.

Chapter 5: Other Descriptive Statistics

Chapter 5 of *Exploring Statistics* introduces additional tools for describing distributions. This time, we will look beyond the average: z scores show where individual scores sit, boxplots flag values worth investigating, and effect sizes help us judge whether a group difference is substantial.

Learning Goals

By the end of this chapter, you should be able to:

- calculate and interpret standard deviation, range, quartiles, and IQR;
- create a new variable in a data frame;
- calculate z scores from the z-score formula;
- use z scores and IQR boundaries to identify potential outliers;
- create and interpret boxplots;
- explain why outlier decisions require judgment; and
- compare descriptive statistics across two groups.

Research Questions

1. How is perceived stress distributed in emerging adults? Are there any participants in our EAMMi2 data file who are particularly high or low on stress? To what extent is perceived stress distributed differently for people with and without a disability?
2. Pick another variable in our data set that you suspect might not be normally distributed. Use the IQR method to identify potential outliers. Filter out the outliers and then write a paragraph summarizing your final descriptive-statistics report. Then, compare two groups on this variable and write another descriptive-statistics report.

Research Question 1 provides a complete worked analysis. Research Question 2 asks you to apply the same process to a variable you select.

Research Question 1: Perceived Stress

Get Ready

Open `scripts/05-other-descriptive-statistics.R` in the starter project and run one complete expression at a time.

```
source("scripts/_setup.R")
```

Research Question 1 variables

Variable	What it measures	Role in the analysis
Stress	Perceived stress over the last month	quantitative outcome
Disability	Whether the participant identified as having a disability	categorical grouping variable

Stress is the sum of 10 questions rated from 1 to 5, so possible scores range from 10 to 50. Give Disability readable labels.

```
eammi <- eammi |>
  mutate(
    Disability = factor(
      Disability,
      levels = c(1, 2),
      labels = c("Yes", "No")
    )
  )
```

Run The Analysis

Describe Stress

```
stress_summary <- eammi |>
  drop_na(Stress) |>
  summarise(
    N = n(),
    Mean = mean(Stress),
    Median = median(Stress),
    Mode = statistical_mode(Stress),
    SD = sd(Stress),
    Minimum = min(Stress),
    Maximum = max(Stress),
    Range = Maximum - Minimum,
    Q1 = quantile(Stress, 0.25),
    Q3 = quantile(Stress, 0.75),
    IQR = IQR(Stress)
  )
```

```
stress_summary
```

```
# A tibble: 1 × 11
```

	N	Mean	Median	Mode	SD	Minimum	Maximum	Range	Q1	Q3	IQR
	<int>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	2071	32.7	33	33	4.05	10	50	40	30	35	5

Investigate The Code

Most of this summary repeats the pattern from Chapters 3 and 4. The new lines are:

- `quantile(Stress, 0.25)` for Q1;
- `quantile(Stress, 0.75)` for Q3; and
- `IQR(Stress)` for the distance between Q1 and Q3.

```
eammi |>
  drop_na(Stress) |>
  ggplot(aes(x = Stress)) +
  geom_histogram(binwidth = 1, boundary = 0) +
  labs(
    x = "Perceived stress score",
    y = "Frequency"
  )
```

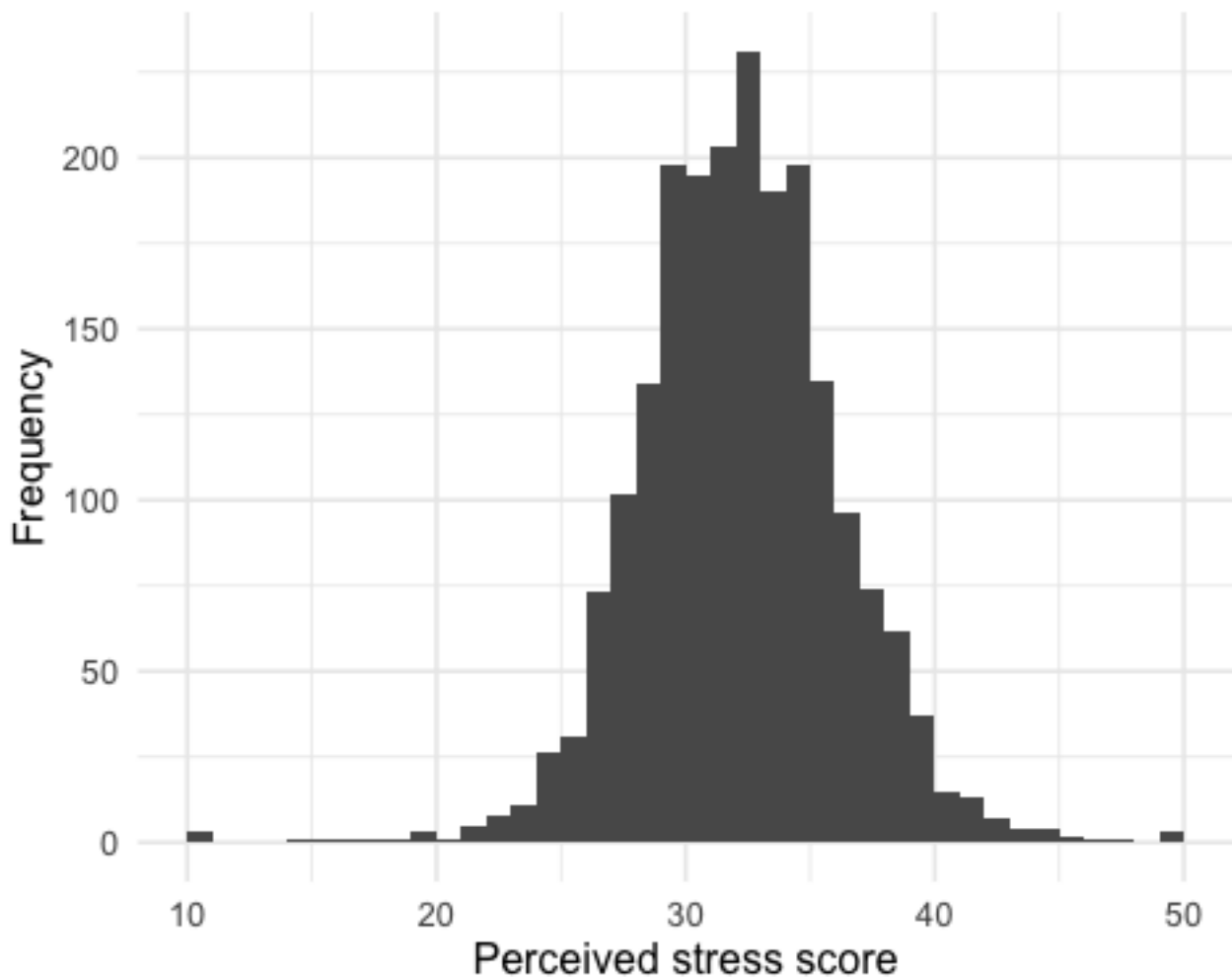


Figure 9: Distribution of perceived stress scores in the reduced EAMMi2 sample.

Looking at this output, you might say that this distribution has a slight negative skew. The mean is similar to the median, but the histogram has a tail pointing toward lower scores.

Create Z Scores

The formula for a z score is:

$$z = \frac{X - M}{SD}$$

In words, subtract the mean from a person's score and divide by the standard deviation.

Use `head()` to display the first two observed stress scores.

```
head(eammi$Stress, n = 2)
```

```
[1] 33 36
```

Using what you learned in the textbook, calculate the z score for each of these observations by hand. Use the mean and standard deviation for `Stress` from the summary output above.

If you use the rounded values shown in the output, you should get approximately 0.074 and 0.815. Keep those answers nearby while R creates the new variable.

```
eammi <- eammi |>
  mutate(
    Stress_z = (Stress - mean(Stress, na.rm = TRUE)) / sd(Stress, na.rm = TRUE)
  )

head(eammi$Stress_z, n = 2)
[1] 0.07741713 0.81854603
```

This `mutate()` step creates `Stress_z` inside `eammi`. The formula in the code follows the statistical formula in the same order. R reports approximately 0.077 and 0.819 for the first two observations. These results differ slightly from the hand calculations because R uses the complete, unrounded mean and standard deviation stored in the data.

To check your work creating the z scores, ask R for the mean and standard deviation of the new variable. Before running the code, decide what these numbers should be for a distribution of z scores.

```
eammi |>
  drop_na(Stress_z) |>
  summarise(
    Mean = mean(Stress_z),
    SD = sd(Stress_z),
    Minimum = min(Stress_z),
    Maximum = max(Stress_z)
  )

# A tibble: 1 × 4
   Mean      SD Minimum Maximum
  <dbl> <dbl>   <dbl>   <dbl>
1 3.77e-16 1.000    -5.60     4.28
```

The mean is displayed in scientific notation as a number extremely close to 0, and the standard deviation is 1. This is the expected result for a distribution of z scores.

Next, call back to our discussion in the textbook about how to use z scores to identify outliers in a distribution of scores. Use the general rule that z scores greater than an absolute value of 3 may be outliers. Compare that rule with the descriptive statistics for `Stress_z` to identify whether we have potential outliers before continuing.

Using this approach, we do have potential outliers: the minimum z score is -5.60 and the maximum is 4.28.

Identify Potential Outliers With IQR

Another way to identify potential outliers relies on the interquartile range. The IQR boundaries are:

$$\text{Lower outlier boundary} = Q1 - (1.5 \times IQR)$$

$$\text{Upper outlier boundary} = Q3 + (1.5 \times IQR)$$

```
stress_iqr <- eammi |>
  drop_na(Stress) |>
  summarise(
    Q1 = quantile(Stress, 0.25),
    Q3 = quantile(Stress, 0.75),
```

```

    IQR = IQR(Stress),
    LowerBoundary = Q1 - (1.5 * IQR),
    UpperBoundary = Q3 + (1.5 * IQR)
  )

```

```
stress_iqr
```

```
# A tibble: 1 × 5
```

	Q1	Q3	IQR	LowerBoundary	UpperBoundary
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	30	35	5	22.5	42.5

The lower boundary is 22.5 and the upper boundary is 42.5. Scores below 22.5 or above 42.5 are flagged as potential outliers. With this approach, many more scores will be considered outliers compared with the z-score approach.

```

eammi |>
  drop_na(Stress) |>
  ggplot(aes(y = Stress)) +
  geom_boxplot() +
  labs(
    x = NULL,
    y = "Perceived stress score"
  )

```

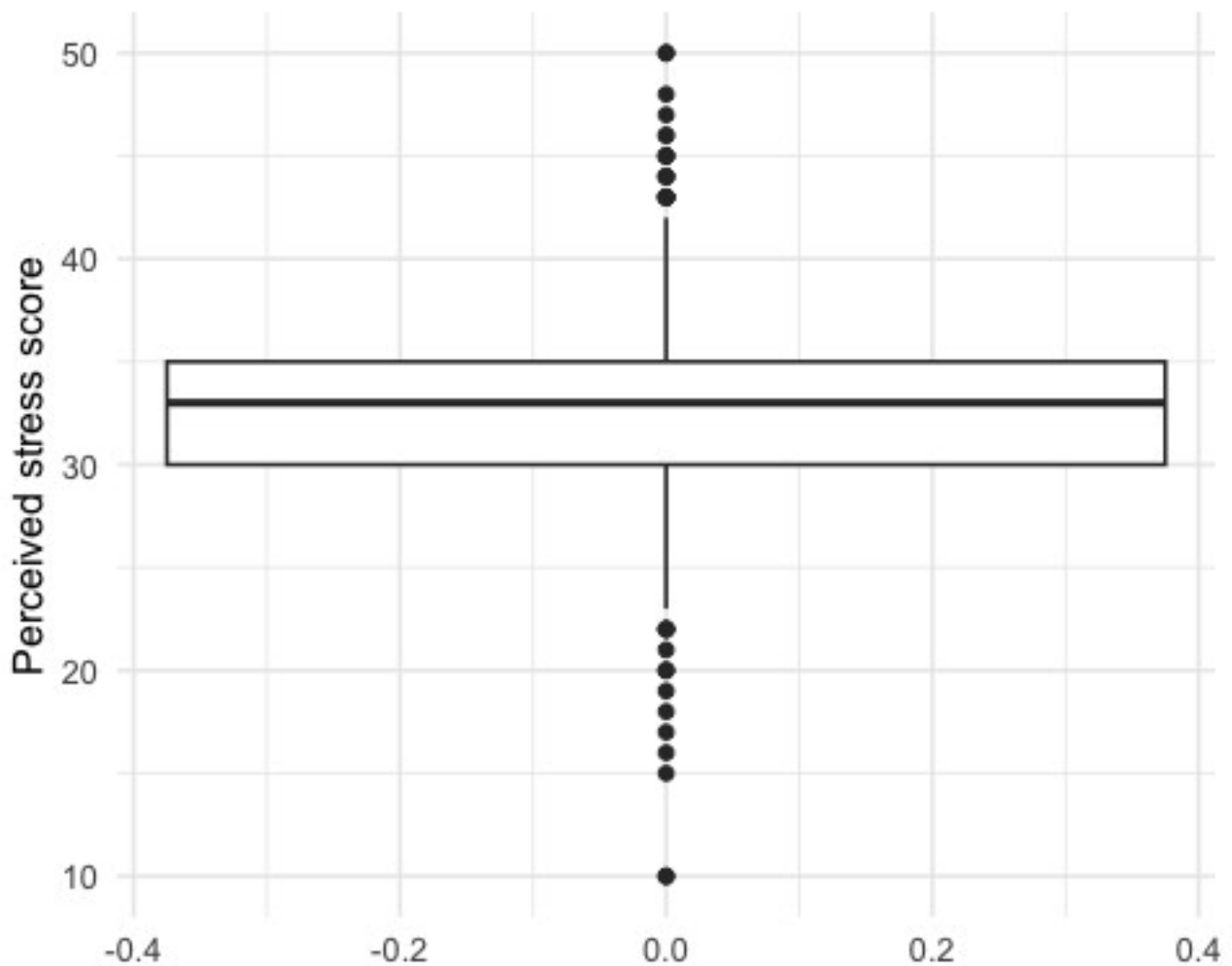


Figure 10: Boxplot of perceived stress scores in the reduced EAMMi2 sample.

The IQR approach is what R uses to identify potential outliers in boxplots. You can see that the dots beyond the whiskers begin around 22.5 and 42.5, respectively.

Filter The Potential Outliers

The next code creates a new data frame that contains only stress scores between the two IQR boundaries.

```
stress_filtered <- eammi |>
  filter(Stress > 22.5, Stress < 42.5)
```

The new object is named `stress_filtered`. The original `eammi` data frame remains unchanged. Keeping the original data untouched makes it possible to check the decision, compare results, or choose a different approach later.

Now repeat the descriptive statistics and graphs with the filtered data.

```
stress_filtered_summary <- stress_filtered |>
  summarise(
    N = n(),
    Mean = mean(Stress),
    Median = median(Stress),
```

```

    Mode = statistical_mode(Stress),
    SD = sd(Stress),
    Minimum = min(Stress),
    Maximum = max(Stress),
    Range = Maximum - Minimum,
    Q1 = quantile(Stress, 0.25),
    Q3 = quantile(Stress, 0.75),
    IQR = IQR(Stress)
  )

stress_filtered_summary
# A tibble: 1 × 11
      N Mean Median Mode   SD Minimum Maximum Range   Q1   Q3   IQR
<int> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1  2032  32.7    33    33  3.60    23    42    19    30    35     5

stress_filtered |>
  ggplot(aes(x = Stress)) +
  geom_histogram(binwidth = 1, boundary = 0) +
  labs(
    x = "Perceived stress score",
    y = "Frequency"
  )

```

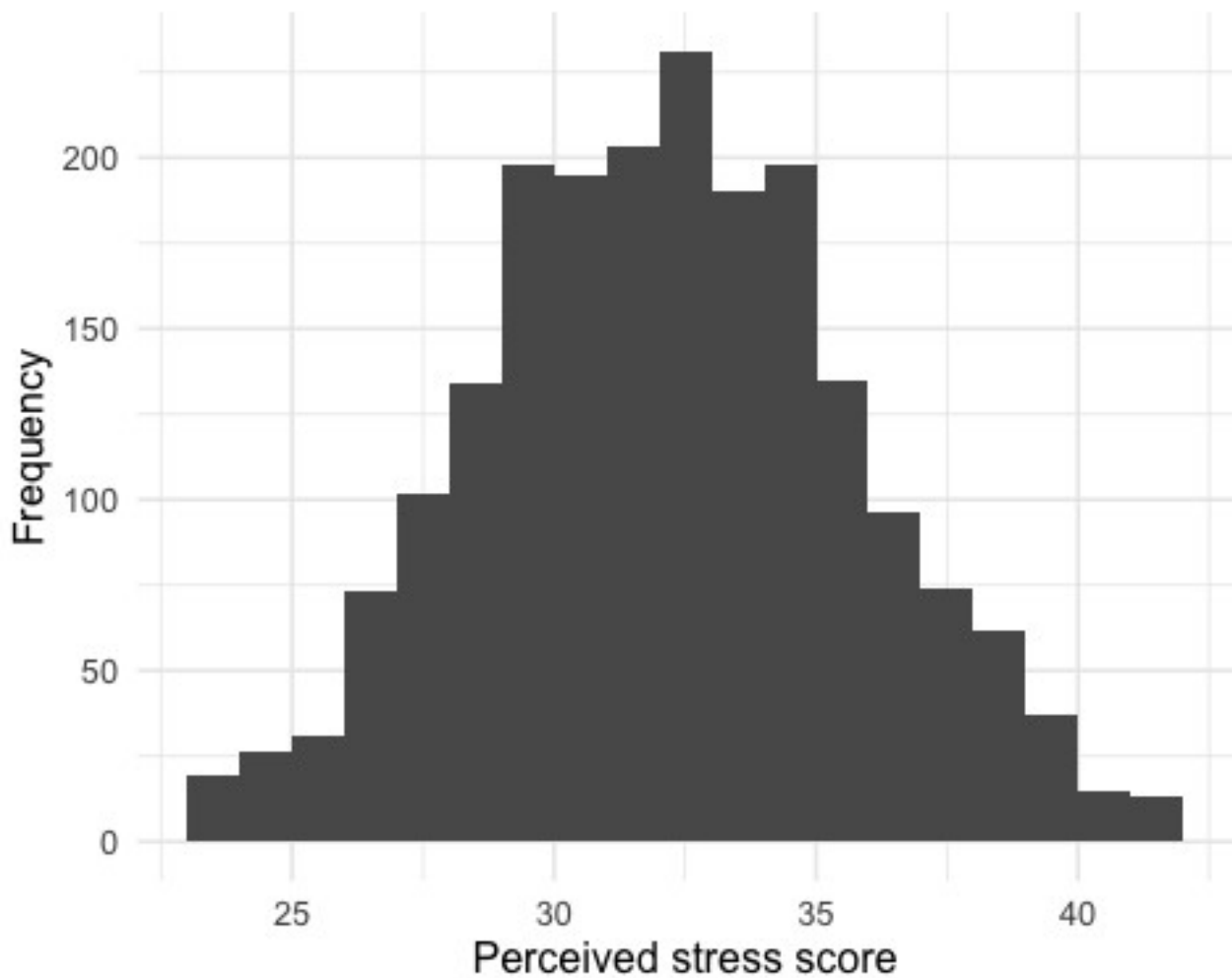


Figure 11: Distribution of perceived stress after scores outside the original IQR boundaries were filtered.

```
stress_filtered |>
  ggplot(aes(y = Stress)) +
  geom_boxplot() +
  labs(
    x = NULL,
    y = "Perceived stress score"
  )
```

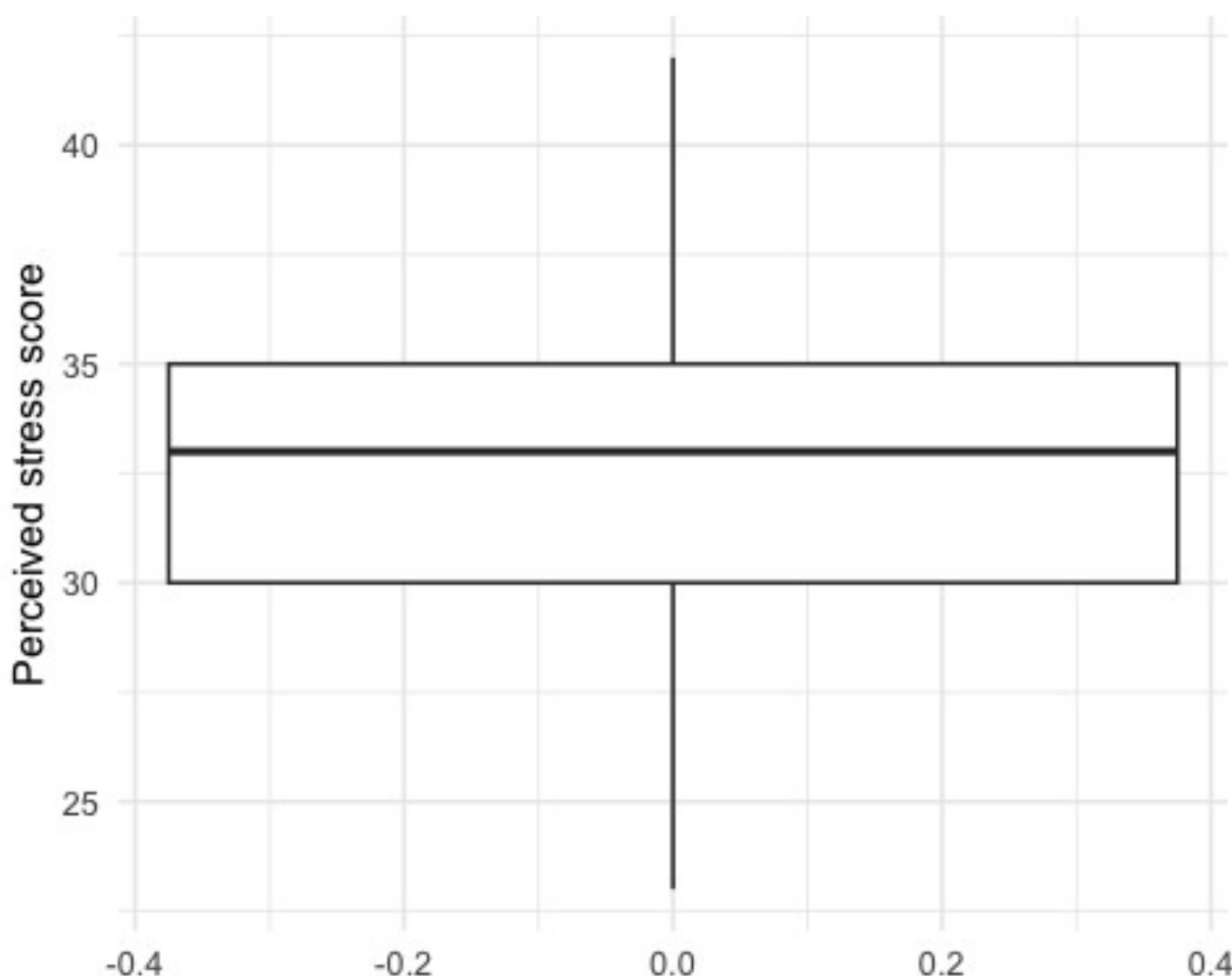


Figure 12: Boxplot of perceived stress after scores outside the original IQR boundaries were filtered.

Because the original version of our `Stress` variable was not appreciably skewed by the potential outliers we filtered, not much has changed in the descriptive-statistics output. You may notice that the standard deviation has gotten smaller. Think through why this would be. You should also see that the boxplot is no longer flagging any potential outliers.

Before moving on, write a paragraph summarizing the filtered version of the `Stress` distribution that also tells your audience about our approach to handling potential outliers. You can check your work at the end of this module.

As a reminder, in real research, your decision to keep or set aside potential outliers is a judgment that you need to justify to your audience. In this case, the original variable was very close to a normal distribution, so we suggest erring on the side of inclusion by keeping all of the data. We filtered the potential outliers here for demonstration purposes.

Compare Stress By Disability Status

As a final step, let's compare how `Stress` is distributed between individuals who do and do not identify as having a disability.

The next summary reuses `group_by()` to compare the two groups. Before running it, identify the grouping variable and the quantitative outcome.

```
stress_disability_summary <- stress_filtered |>
  drop_na(Disability) |>
  group_by(Disability) |>
  summarise(
    N = n(),
    Mean = mean(Stress),
    Median = median(Stress),
    Mode = statistical_mode(Stress),
    SD = sd(Stress),
    Minimum = min(Stress),
    Maximum = max(Stress),
    Range = Maximum - Minimum,
    Q1 = quantile(Stress, 0.25),
    Q3 = quantile(Stress, 0.75),
    IQR = IQR(Stress)
  )
```

```
stress_disability_summary
```

```
# A tibble: 2 × 12
```

	Disability	N	Mean	Median	Mode	SD	Minimum	Maximum	Range	Q1	Q3
	<fct>	<int>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Yes	171	33.4	34	34	3.23	24	42	18	31	35
2	No	1861	32.6	33	33	3.63	23	42	19	30	35

```
# i 1 more variable: IQR <dbl>
```

```
stress_filtered |>
  drop_na(Disability) |>
  ggplot(aes(x = Disability, y = Stress)) +
  geom_boxplot() +
  labs(
    x = "Identifies as having a disability",
    y = "Perceived stress score"
  )
```

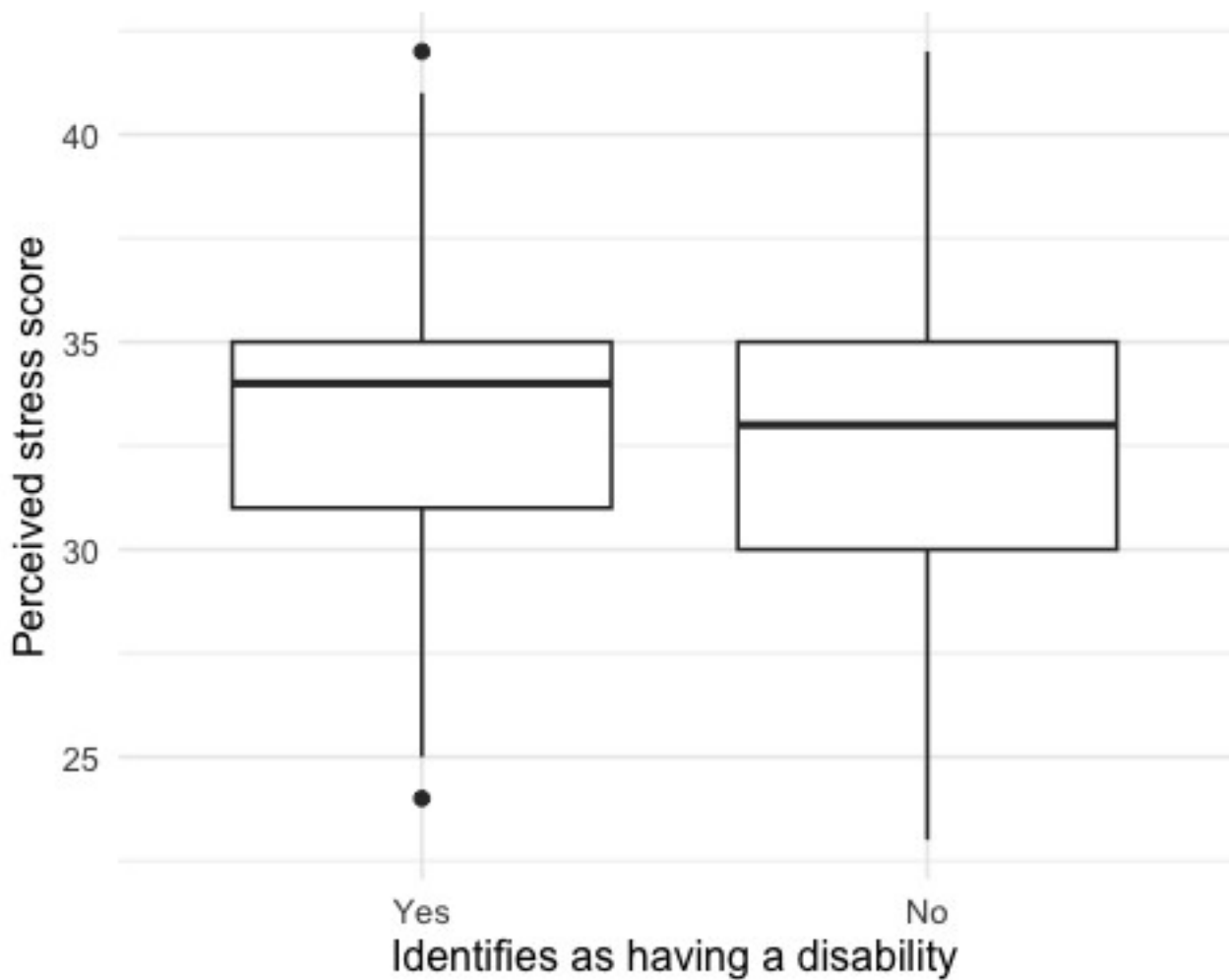


Figure 13: Filtered perceived stress scores for participants with and without a disability.

Calculate Cohen's d

Cohen's d describes the mean difference in standard deviation units:

$$d = \frac{\text{mean difference}}{\text{standard deviation}}$$

The setup file includes `independent_cohens_d()`, which gets each group's sample size, mean, and standard deviation, calculates the pooled standard deviation, and then calculates Cohen's d .

```
stress_disability_d <- independent_cohens_d(
  stress_filtered,
  Stress,
  Disability
)
```

```
stress_disability_d
```

```
# A tibble: 1 × 3
  MeanDifference PooledSD CohensD
```

	<dbl>	<dbl>	<dbl>
1	0.739	3.60	0.205

The three inputs are the data frame, the quantitative outcome, and the grouping variable. Focus on the result: Cohen's d expresses the difference between the group means relative to the variability in the scores.

Produce Your Interpretation

Before moving on to Research Question 2, write a summary that compares these two distributions of `Stress` and tells your audience about our approach to handling potential outliers. You can check your work at the end of this module.

Research Question 2: Produce An Outlier And Group-Comparison Analysis

Pick another variable in our data set that you suspect might not be normally distributed and use the IQR method to identify potential outliers. Filter out the potential outliers and then write a paragraph summarizing your final descriptive-statistics report. Then, compare two groups on this variable and write another descriptive-statistics report. You can check your work by comparing your paragraphs with the sample paragraphs at the end of this module.

`MOA_ACH`, `SocialSupport`, and `Symptoms` are good options. Use Research Question 1 as your model, and create descriptive object names for your own summaries and filtered data.

Check Your Work

Research Question 1

Summary For Perceived Stress

Using the EAMMi2 sample, we examined emerging adults' ratings of perceived stress over the last month, which had a possible score range of 10 to 50. An initial inspection of the data revealed potential outliers below 22.5 and above 42.5; these values were set aside for this demonstration analysis. Participants' ratings of perceived stress were normally distributed, with a mean of 32.70 and a median and mode of 33. The standard deviation was 3.60, with a range of 19 and an IQR of 5. The graph presents a boxplot of these data.

Summary For Perceived Stress Of Individuals With And Without A Disability

Using the EAMMi2 sample, we compared ratings of perceived stress over the last month for emerging adults with and without a disability. This scale had a possible score range of 10 to 50. An initial inspection of the data revealed potential outliers below 22.5 and above 42.5; these values were set aside for this demonstration analysis. The graph shows boxplots of perceived stress ratings for each group. The difference in means produced an effect size of $d = 0.21$.

The mean perceived stress rating for emerging adults with a disability was 33.35, and the median was 34. The mean for emerging adults without a disability was 32.61, and the median was 33. Thus, emerging adults with a disability reported slightly higher stress than emerging adults without a disability, although the mean difference of 0.74 points was small. Perceived stress ratings for those without a disability were slightly more variable ($SD = 3.63$, range = 19, IQR = 5) than ratings for those with a disability ($SD = 3.23$, range = 18, IQR = 4). Both distributions closely approximated a normal distribution.

Research Question 2 Sample

The following example uses MOA_ACH and compares participants who identified as Democrats or Republicans using PartyDichotomous.

Inspect The Original Variable

```
moa_original_summary <- eammi |>
  drop_na(MOA_ACH) |>
  summarise(
    N = n(),
    Mean = mean(MOA_ACH),
    Median = median(MOA_ACH),
    Mode = statistical_mode(MOA_ACH),
    SD = sd(MOA_ACH),
    Minimum = min(MOA_ACH),
    Maximum = max(MOA_ACH),
    Range = Maximum - Minimum,
    Q1 = quantile(MOA_ACH, 0.25),
    Q3 = quantile(MOA_ACH, 0.75),
    IQR = IQR(MOA_ACH)
  )

moa_original_summary
```

A tibble: 1 × 11

	N	Mean	Median	Mode	SD	Minimum	Maximum	Range	Q1	Q3	IQR
	<int>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	2073	37.8	37	36	6.56	0	60	60	34	41	7

```
eammi |>
  drop_na(MOA_ACH) |>
  ggplot(aes(x = MOA_ACH)) +
  geom_histogram(binwidth = 1, boundary = 20) +
  labs(
    x = "Markers of adulthood achieved",
    y = "Frequency"
  )
```

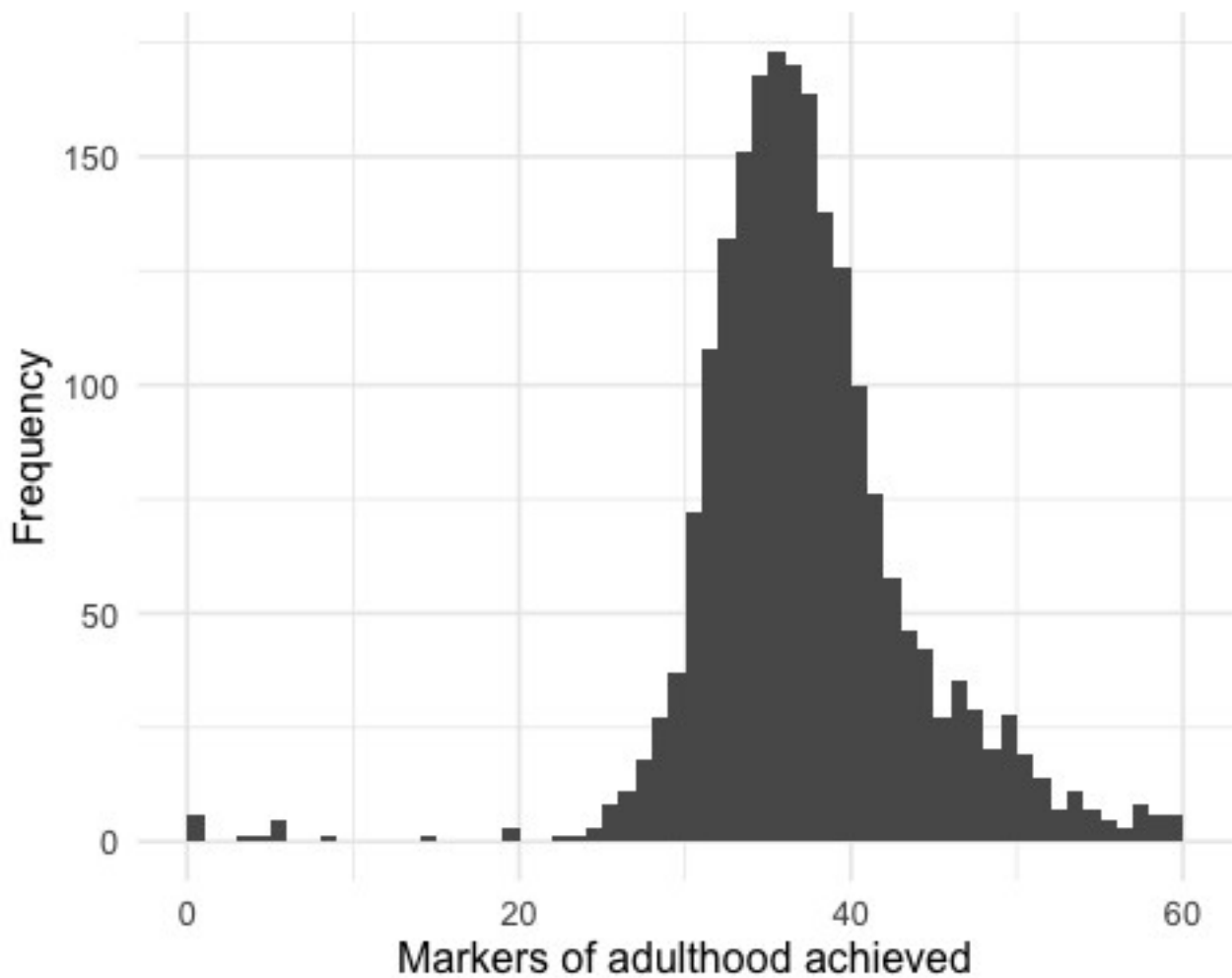


Figure 14: Original distribution of markers of adulthood achieved.

```
eammi |>
  drop_na(MOA_ACH) |>
  ggplot(aes(y = MOA_ACH)) +
  geom_boxplot() +
  labs(
    x = NULL,
    y = "Markers of adulthood achieved"
  )
```

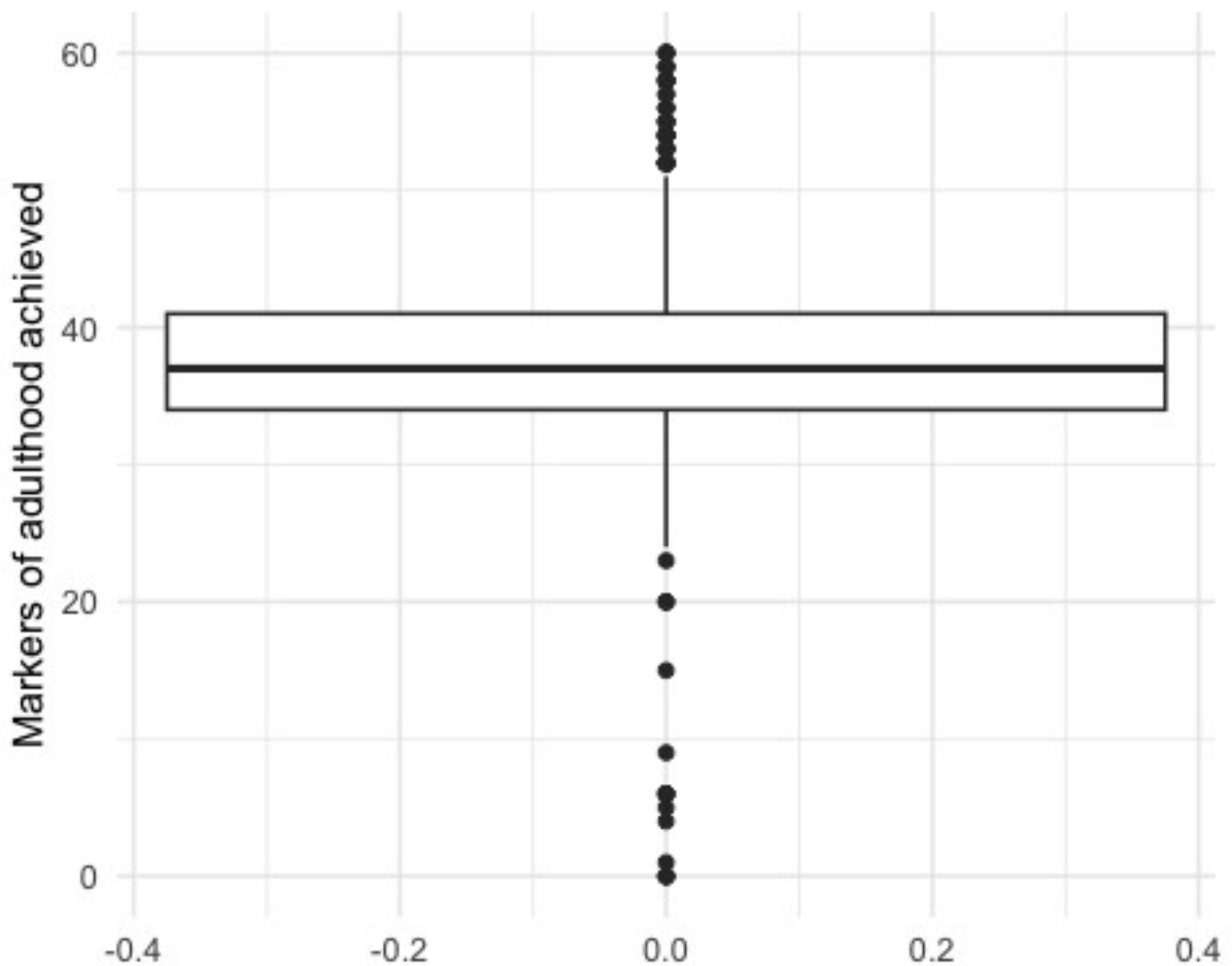


Figure 15: Original boxplot for markers of adulthood achieved.

The original distribution has a slight negative skew. Calculate the IQR boundaries.

```
moa_iqr <- eammi |>
  drop_na(MOA_ACH) |>
  summarise(
    Q1 = quantile(MOA_ACH, 0.25),
    Q3 = quantile(MOA_ACH, 0.75),
    IQR = IQR(MOA_ACH),
    LowerBoundary = Q1 - (1.5 * IQR),
    UpperBoundary = Q3 + (1.5 * IQR)
  )
```

```
moa_iqr
# A tibble: 1 × 5
  Q1    Q3    IQR LowerBoundary UpperBoundary
<dbl> <dbl> <dbl>         <dbl>         <dbl>
1    34    41     7           23.5           51.5
```

The lower boundary is 23.5, and the upper boundary is 51.5.

Filter And Describe The Variable

```

moa_filtered <- eammi |>
  filter(MOA_ACH > 23.5, MOA_ACH < 51.5)

moa_filtered_summary <- moa_filtered |>
  summarise(
    N = n(),
    Mean = mean(MOA_ACH),
    Median = median(MOA_ACH),
    Mode = statistical_mode(MOA_ACH),
    SD = sd(MOA_ACH),
    Minimum = min(MOA_ACH),
    Maximum = max(MOA_ACH),
    Range = Maximum - Minimum,
    Q1 = quantile(MOA_ACH, 0.25),
    Q3 = quantile(MOA_ACH, 0.75),
    IQR = IQR(MOA_ACH)
  )

moa_filtered_summary

# A tibble: 1 × 11
      N Mean Median Mode    SD Minimum Maximum Range  Q1  Q3  IQR
<int> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1  1987  37.5    37    36  5.00    24    51    27   34   40    6

moa_filtered |>
  ggplot(aes(x = MOA_ACH)) +
  geom_histogram(binwidth = 1, boundary = 20) +
  labs(
    x = "Markers of adulthood achieved",
    y = "Frequency"
  )

```

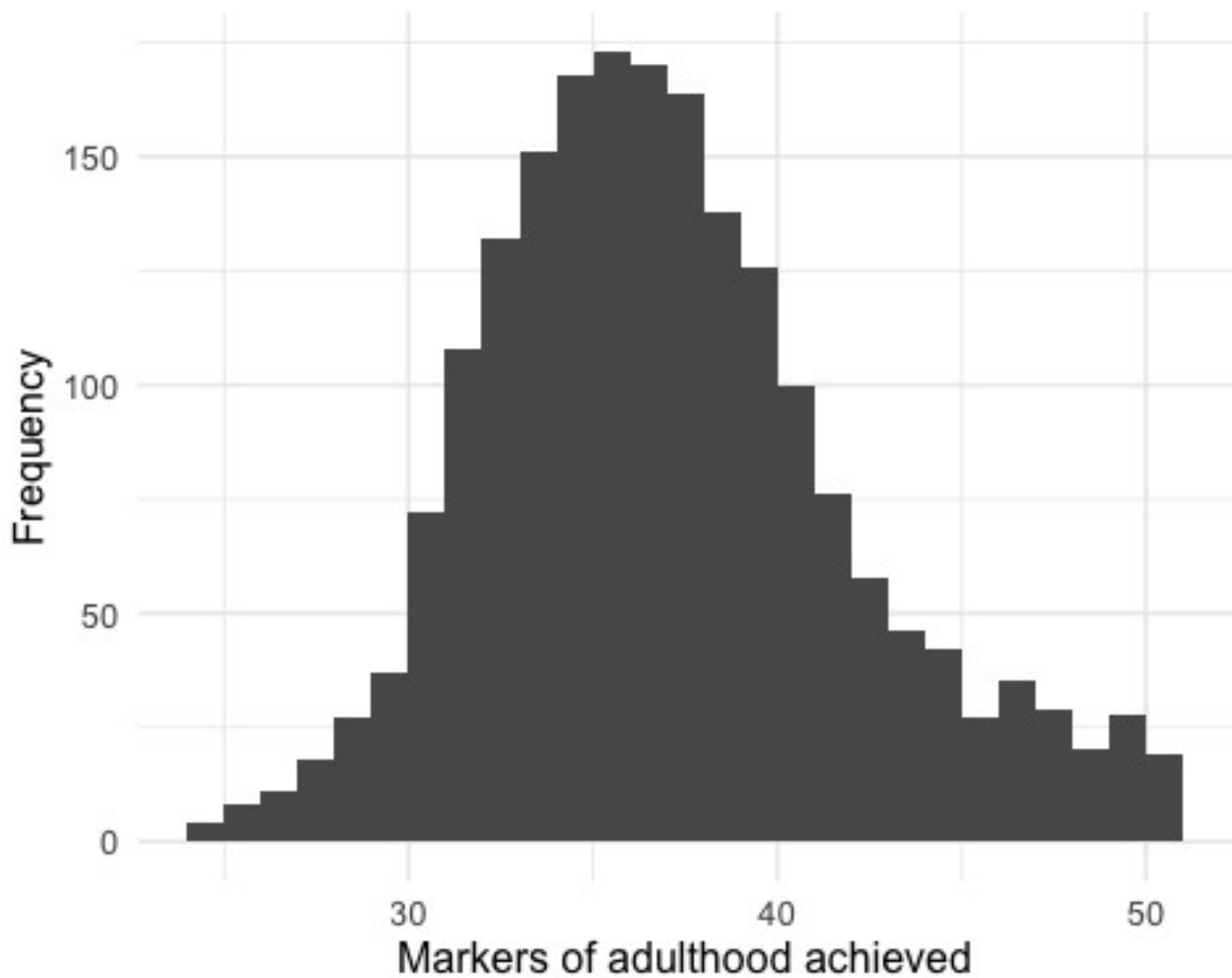


Figure 16: Filtered distribution of markers of adulthood achieved.

```
moa_filtered |>
  ggplot(aes(y = MOA_ACH)) +
  geom_boxplot() +
  labs(
    x = NULL,
    y = "Markers of adulthood achieved"
  )
```

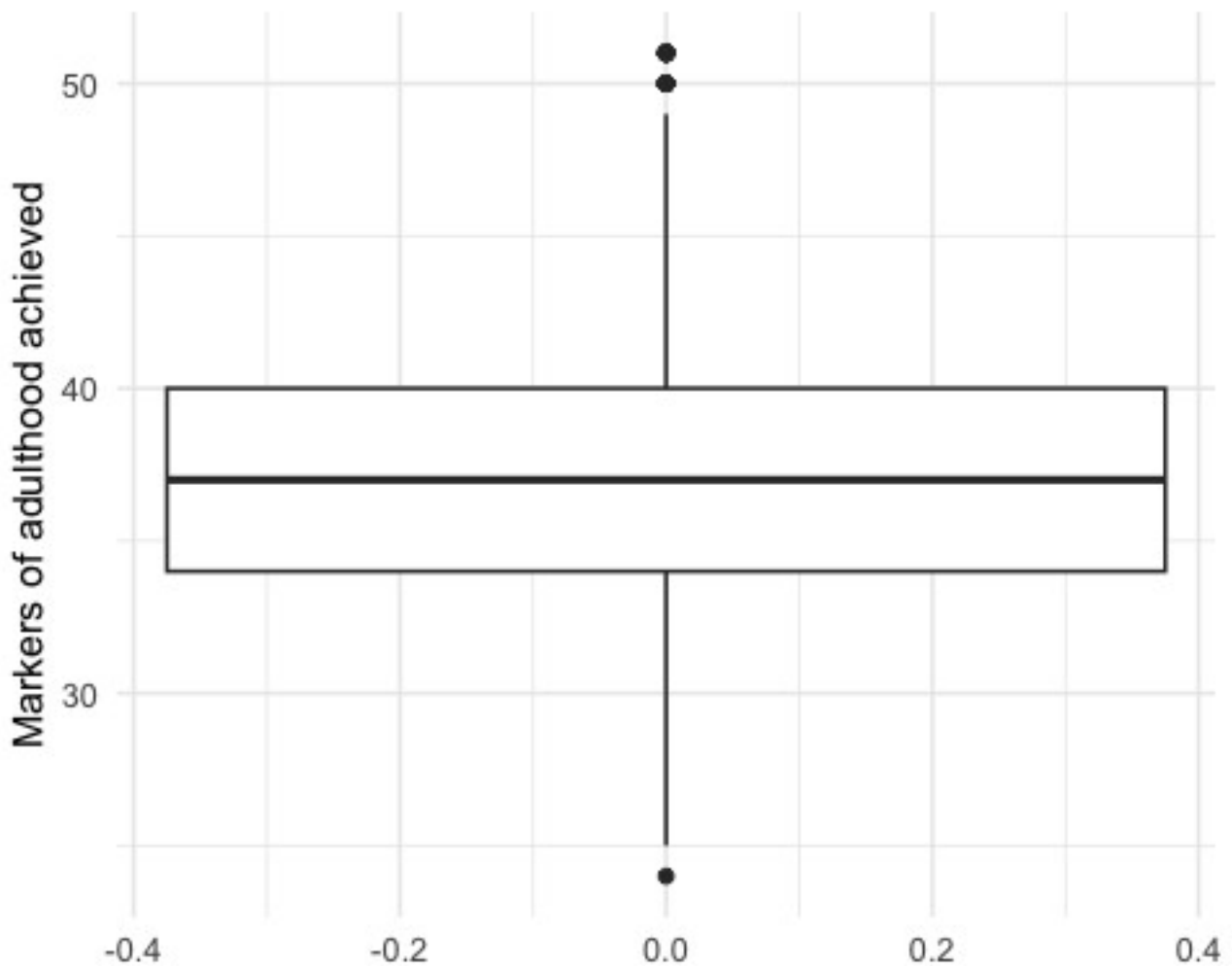


Figure 17: Filtered boxplot for markers of adulthood achieved.

Using the EAMMi2 sample, we examined the number of markers of adulthood achieved by emerging adults, which had a possible score range of 20, or none achieved, to 60, or all achieved. An initial inspection revealed potential outliers below 23.5 and above 51.5; these values were set aside for the analysis. Participants' achievement of markers of adulthood was close to normally distributed, with a mean of 37.50, a median of 37, and a mode of 36. The standard deviation was 5.00, with a range of 27 and an IQR of 6. The graph presents a boxplot of these data.

Compare Political-Party Groups

```
moa_filtered <- moa_filtered |>
  mutate(
    PartyDichotomous = factor(
      PartyDichotomous,
      levels = c(1, 2),
      labels = c("Democrat", "Republican")
    )
  )
moa_party_summary <- moa_filtered |>
  drop_na(PartyDichotomous) |>
  group_by(PartyDichotomous) |>
  summarise(
```

```

N = n(),
Mean = mean(MOA_ACH),
Median = median(MOA_ACH),
Mode = statistical_mode(MOA_ACH),
SD = sd(MOA_ACH),
Minimum = min(MOA_ACH),
Maximum = max(MOA_ACH),
Range = Maximum - Minimum,
Q1 = quantile(MOA_ACH, 0.25),
Q3 = quantile(MOA_ACH, 0.75),
IQR = IQR(MOA_ACH)
)

```

```
moa_party_summary
```

```
# A tibble: 2 × 12
```

```

PartyDichotomous      N Mean Median Mode   SD Minimum Maximum Range   Q1
<fct>               <int> <dbl> <dbl> <int> <dbl>   <dbl>   <dbl> <dbl> <dbl>
1 Democrat           1007  37.4    37   35  5.00    24    51    27    34
2 Republican         458  37.6    37   38  4.80    26    51    25    34
# i 2 more variables: Q3 <dbl>, IQR <dbl>

```

```

moa_filtered |>
  drop_na(PartyDichotomous) |>
  ggplot(aes(x = PartyDichotomous, y = MOA_ACH)) +
  geom_boxplot() +
  labs(
    x = "Political-party identification",
    y = "Markers of adulthood achieved"
  )

```

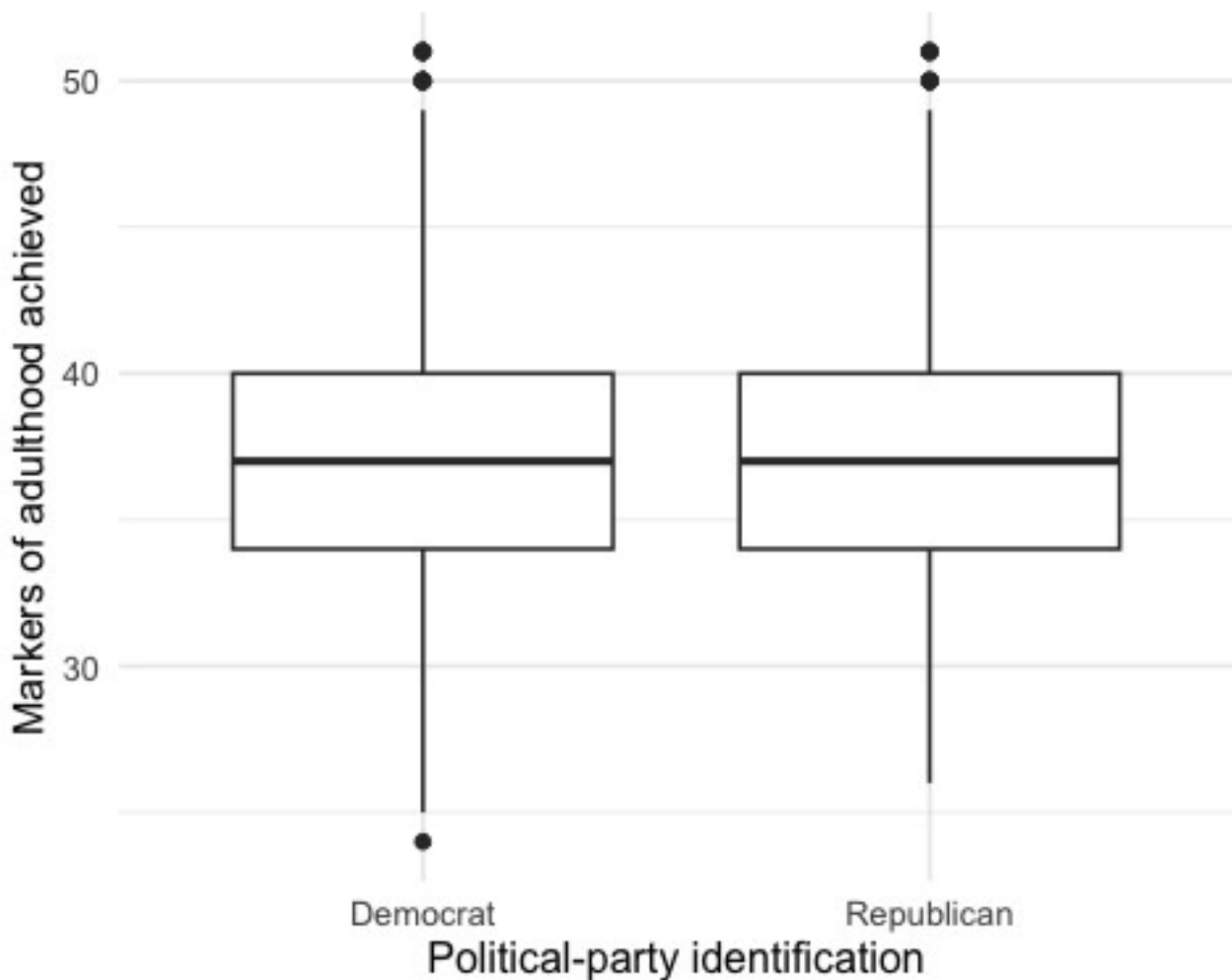


Figure 18: Filtered markers-of-adulthood scores for Democrats and Republicans.

```
moa_party_d <- independent_cohens_d(
  moa_filtered,
  MOA_ACH,
  PartyDichotomous
)
```

```
moa_party_d
# A tibble: 1 × 3
  MeanDifference PooledSD CohensD
      <dbl>      <dbl>   <dbl>
1      -0.230       4.94 -0.0466
```

Using the EAMMi2 sample, we compared achievement of markers of adulthood between participants who identified as Democrats and participants who identified as Republicans. This scale had a possible score range of 20, or none achieved, to 60, or all achieved. An initial inspection revealed potential outliers below 23.5 and above 51.5; these values were set aside for the analysis. The graph shows boxplots of achievement of markers of adulthood for each group. The difference in means produced an effect size of $d = -0.05$.

The mean achievement of markers of adulthood for emerging adults who identified as Democrats was 37.43, and the median was 37. The mean for participants who identified as Republicans was 37.66, and the median was 37.

The difference between these means was close to zero. Variability for Democrats ($SD = 5.00$, range = 27, IQR = 6) was very similar to variability for Republicans ($SD = 4.80$, range = 25, IQR = 6). Both distributions closely approximated a normal distribution and were very similar to one another.

Chapter Takeaway

Z scores describe how far a score is from the mean in standard deviation units. IQR boundaries and boxplots help identify potential outliers. When values are unusual but still possible, deciding whether to keep or set them aside is a research judgment that should be explained clearly.

In this chapter, we focused on these new R commands:

- `head()` displays the first observations or values in an object.
- `quantile()` calculates quartiles and other percentiles.
- `IQR()` calculates the interquartile range.
- `geom_boxplot()` creates a boxplot.
- `independent_cohens_d()` calculates Cohen's d for two independent groups.

Chapter 6: Correlation and Regression

Chapter 6 of *Exploring Statistics* introduces correlation and simple linear regression. We will ask whether perceived stress travels together with social support from family, friends, and a special someone, then use a regression line to describe the same relationships from a new angle.

Learning Goals

By the end of this chapter, you should be able to:

- calculate and interpret Pearson correlation coefficients;
- calculate and interpret shared variance with (r^2);
- fit a simple linear regression model;
- interpret the intercept and slope from a regression model;
- create a scatterplot with a line of best fit; and
- write a short interpretation of correlation and regression output.

Research Questions

1. A large research literature supports a negative correlation between social support and experiences of stress. Does this correlation exist across different sources of social support for emerging adults?
2. Choose two quantitative EAMMi2 variables that interest you. What do a correlation and a linear regression reveal about their relationship?

Research Question 1: Stress and Social Support

In the EAMMi2 study, perceived social support was measured across three sources: family, friends, and a “special someone.” We will correlate each source of social support with scores on the Perceived Stress Scale.

Get Ready

Open 06-correlation-regression.R from the scripts folder. Run the shared setup line first, and then run the script one section at a time as you work through this chapter.

```
source("scripts/_setup.R")
```

This research question uses four quantitative variables:

Variables used in Research Question 1

Variable	What it measures	Role in the analysis
Stress	Perceived stress over the last month	outcome
SupportFamily	Perceived social support from family	predictor
SupportFriends	Perceived social support from friends	predictor
SupportSpecial	Perceived social support from a	predictor

Variable	What it measures	Role in the analysis
	special someone	

Review these variables in Understanding EAMMi2 when you need their possible scores or survey descriptions.

Predict

Before running any code, write down your predictions:

- Should each relationship be positive, negative, or close to zero?
- Which source of support do you expect to have the strongest relationship with stress?
- What would a negative correlation mean for these variables?

These are predictions, not answers. The purpose is to make your expectations visible before you inspect the data.

Run

Begin with the relationship between stress and family support.

```
family_correlation <- cor(
  eammi$Stress,
  eammi$SupportFamily,
  use = "complete.obs"
)
```

```
family_correlation
[1] -0.03098761
```

`cor()` calculates a Pearson correlation coefficient. The two variables identify the scores to correlate. `use = "complete.obs"` tells R to use observations that have scores for both variables.

Now calculate several correlations at once.

```
support_correlations <- eammi |>
  select(Stress, SupportFamily, SupportFriends, SupportSpecial) |>
  cor(use = "pairwise.complete.obs")
```

```
support_correlations
```

	Stress	SupportFamily	SupportFriends	SupportSpecial
Stress	1.00000000	-0.03098761	0.01778431	0.08823717
SupportFamily	-0.03098761	1.00000000	0.59775855	0.36507167
SupportFriends	0.01778431	0.59775855	1.00000000	0.43018407
SupportSpecial	0.08823717	0.36507167	0.43018407	1.00000000

`select()` keeps the four variables needed for this analysis. `cor()` then creates a correlation matrix. Because each pair can have a different pattern of missing scores, `use = "pairwise.complete.obs"` uses the available complete pairs for each correlation.

Investigate

The same variables appear in the rows and columns of a correlation matrix. Find the point where the row for one variable meets the column for the other variable. The values above and below the diagonal repeat the same correlations.

Use the matrix to answer these questions:

- What is the r value for stress and family support?
- What is the r value for stress and friend support?
- What is the r value for stress and support from a special someone?
- What is the r value for family support and friend support?
- Which of your predictions were supported?

For each relationship with stress, identify its direction and strength. A correlation describes an association; it does not show that one variable causes the other.

Now calculate shared variance.

`support_correlations^2`

	Stress	SupportFamily	SupportFriends	SupportSpecial
Stress	1.0000000000	0.0009602322	0.0003162818	0.007785799
SupportFamily	0.0009602322	1.0000000000	0.3573152820	0.133277327
SupportFriends	0.0003162818	0.3573152820	1.0000000000	0.185058336
SupportSpecial	0.0077857990	0.1332773269	0.1850583360	1.0000000000

In R, 2 means “squared.” Squaring r gives (r^2), the proportion of variance shared by two variables. Multiply (r^2) by 100 to describe the percentage of shared variance.

Fit Regression Models

Next, examine the same relationships through simple linear regression. Start with family support.

```
family_model <- lm(Stress ~ SupportFamily, data = eammi)
```

```
coefficients(family_model)
```

```
(Intercept) SupportFamily
33.20246513 -0.09481975
```

`lm()` fits a linear model. R formulas follow the pattern `outcome ~ predictor`, which you can read as “outcome predicted by predictor.” Here, `Stress` is the outcome and `SupportFamily` is the predictor. `coefficients()` shows the intercept, (a), and slope, (b).

Interpret the coefficients before continuing:

- The intercept is the predicted stress score when family support equals 0.
- The slope is the predicted change in stress for a 1-unit increase in family support.

Now copy the model, give the new object a clear name, and replace only the predictor. Compare your code with these completed versions after you try it.

```
friends_model <- lm(Stress ~ SupportFriends, data = eammi)
special_model <- lm(Stress ~ SupportSpecial, data = eammi)
```

```
coefficients(friends_model)
      (Intercept) SupportFriends
      32.38134098      0.05486374
```

```
coefficients(special_model)
      (Intercept) SupportSpecial
      31.4298333      0.2238189
```

Create a scatterplot for the family-support model.

```
eammi |>
  drop_na(Stress, SupportFamily) |>
  ggplot(aes(x = SupportFamily, y = Stress)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE) +
  labs(
    x = "Perceived social support from family",
    y = "Perceived stress score"
  )
```

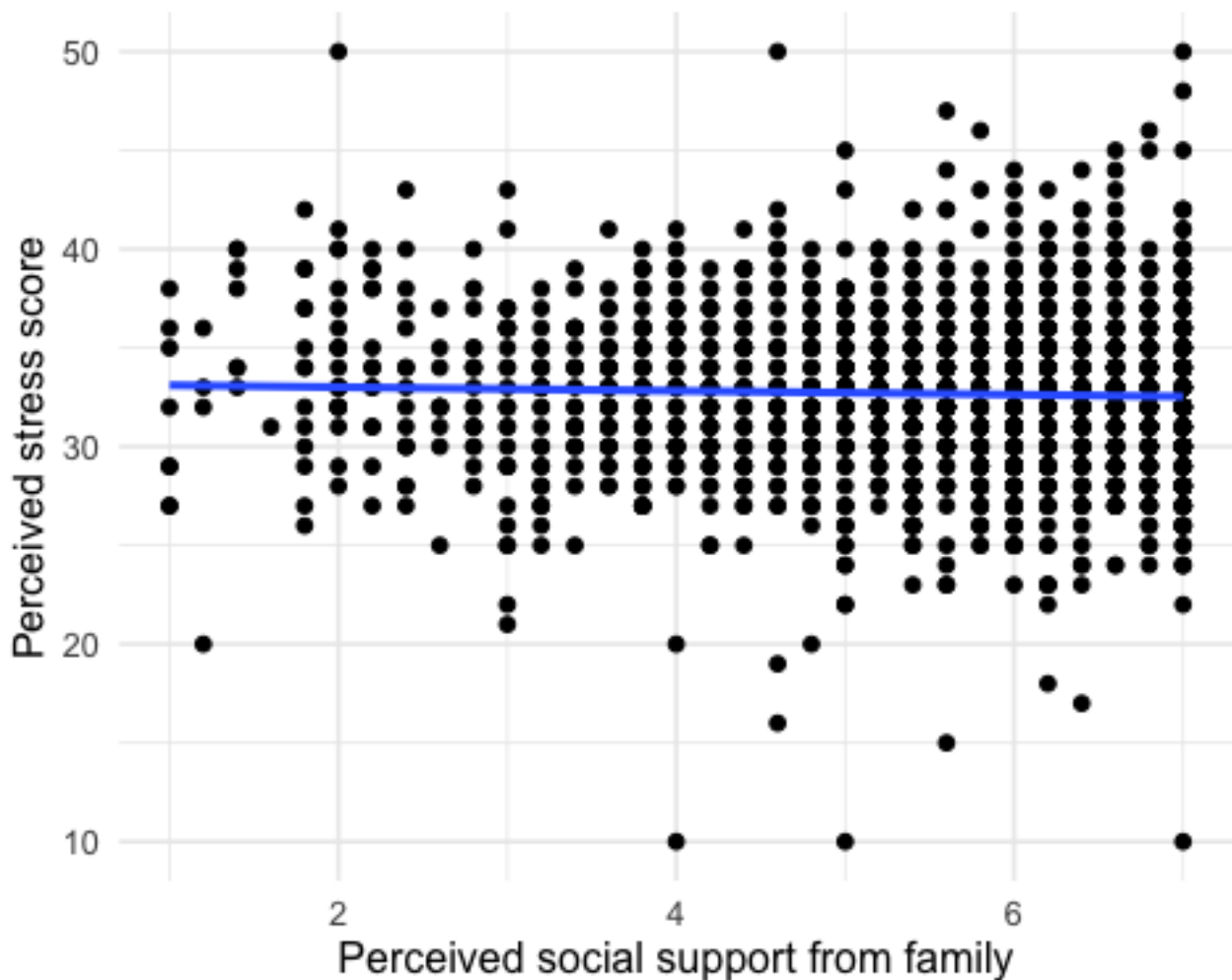


Figure 19: Perceived stress by perceived social support from family.

The + signs build one graph by adding layers in order. `geom_point()` adds the observed pairs of scores, and `geom_smooth(method = "lm", formula = y ~ x, se = FALSE)` places the fitted regression line over those points. `labs()` then adds readable labels. The plot places the predictor on the x-axis and the outcome on the y-axis, matching `Stress ~ SupportFamily`.

Modify this graph twice: first replace `SupportFamily` with `SupportFriends`, and then replace it with `SupportSpecial`. Update the axis label each time. Check whether each line rises, falls, or stays nearly flat.

Produce

Write a short paragraph that answers the research question. For each source of support:

- name both variables;
- report r and (r^2) ;
- describe the direction and strength of the relationship; and
- compare the result with your prediction.

Then write one sentence interpreting the intercept and one sentence interpreting the slope for each regression model. Use the actual variable names rather than only calling them X and Y.

Research Question 2: Choose a Relationship

Review Understanding EAMMi2 and choose two quantitative variables that interest you. Conduct both a correlation and a simple linear regression to learn about their relationship.

Predict

Write the direction and approximate strength you expect before running the analysis. Explain your reasoning in one sentence.

Run

Copy the code pattern from Research Question 1. Replace the variable names and create new object names that describe your analysis.

Investigate

Identify r , calculate (r^2) , and interpret the intercept and slope. Check that the predictor appears on the x-axis and the outcome appears on the y-axis of your scatterplot.

Check The Code And Labels

If your first object or axis names still refer to stress or support, revise them. Clear names and labels help you verify that the output belongs to the analysis you intended to run.

Produce

Write a short paragraph summarizing the relationship. Include the variables, direction, strength, r , (r^2) , and what the slope predicts. Do not use causal language.

Check Your Work

Research Question 1

Correlation coefficients:

- Stress and family support: $r = -.031$
- Stress and friend support: $r = .018$
- Stress and support from a special someone: $r = .088$
- Family support and friend support: $r = .598$

In our reduced EAMMi2 sample, the correlations between perceived stress and all three sources of social support were very small and close to zero. The relationship was strongest for support from a special someone, $r = .09$, followed by family support, $r = -.03$, and friend support, $r = .02$. The percentage of shared variance was less than 1% for each relationship: family ($r^2 = .001$), friends ($r^2 = .0003$), and special someone ($r^2 = .008$).

The regression coefficients can be interpreted as follows:

- Family support: if family support were 0, the predicted stress score would be 33.20. Each 1-unit increase in family support predicts a 0.09-point decrease in stress.
- Friend support: if friend support were 0, the predicted stress score would be 32.38. Each 1-unit increase in friend support predicts a 0.05-point increase in stress.
- Support from a special someone: if support from a special someone were 0, the predicted stress score would be 31.43. Each 1-unit increase in support from a special someone predicts a 0.22-point increase in stress.

Reporting note

A fully APA-formatted correlation paragraph would also include degrees of freedom and *p* values, which have not yet been emphasized in the course.

Research Question 2

Use these questions to check your work.

1. Did you select two quantitative variables?
2. For the correlation interpretation, did you reference:
 - the variables themselves;
 - the direction and strength of the relationship;
 - the *r* value; and
 - the (r^2) value and its interpretation?
3. For the regression interpretation, did you reference:
 - a correct interpretation of (a), the intercept, including its value? The intercept is the predicted value of the outcome when the predictor equals 0. R labels this coefficient (Intercept).
 - a correct interpretation of (b), the slope, including its value? The slope is the predicted change in the outcome for a 1-unit increase in the predictor. R labels this coefficient with the predictor's variable name.
 - the actual variables used in the analysis rather than only calling them X and Y?
4. Does the scatterplot use readable axis labels and a line of best fit?
5. Did you avoid saying that one measured variable caused the other?

Chapter Takeaway

Correlation describes the direction and strength of a linear relationship between quantitative variables. Squaring the correlation gives the proportion of shared variance. Simple linear regression uses an intercept and slope to predict an outcome from a predictor.

New R commands and patterns in this chapter:

- `select()` keeps specified variables.
- `cor()` calculates correlation coefficients.
- `outcome ~ predictor` identifies the outcome and predictor in an R formula.

- `lm()` fits a linear model.
- `coefficients()` shows the model's intercept and slope.
- `geom_point()` adds observed pairs of scores to a scatterplot.
- `geom_smooth()` adds a fitted line.

Chapter 9: One-Sample Designs

Chapter 9 of *Exploring Statistics* introduces one-sample designs. You will compare a sample mean with a known or expected value while practicing an important habit: pause and look at the data before trusting an inferential test.

Learning Goals

By the end of this chapter, you should be able to:

- write null and alternative hypotheses for a one-sample design;
- choose a directional or non-directional alternative hypothesis;
- conduct and interpret a one-sample t test.

Research Questions

1. The EAMMi2 consent form estimated that the study would take 30 minutes. Was that a reasonable estimate?
2. Are emerging adults significantly different in mindfulness from Zen practitioners? What about undergraduate students?

Research Question 1: Survey Duration

We want to know whether the average time participants took to complete the EAMMi2 survey is significantly different from 30 minutes.

Get Ready

Open 09-one-sample-designs.R from the scripts folder. Run the shared setup line first, and then run the script one section at a time.

```
source("scripts/_setup.R")
```

The analysis uses `Duration`, the number of minutes a participant took to finish the survey. Review it in Understanding EAMMi2.

Predict

Before you see the sample mean:

1. Write the null and alternative hypotheses in words and statistical notation.
2. Identify the comparison value.
3. Decide whether the alternative should be directional or non-directional.
4. Predict whether survey duration will be roughly symmetric or positively skewed.

The research question asks whether the mean is *different* from 30, without naming a direction. We will therefore use a two-tailed test with a comparison value of 30.

Run

Run the one-sample t test.

```
duration_first_test <- t.test(
  eammi$Duration,
  mu = 30,
  alternative = "two.sided"
)
```

```
duration_first_test
```

One Sample t-test

```
data: eammi$Duration
t = 7.1859, df = 2072, p-value = 9.274e-13
alternative hypothesis: true mean is not equal to 30
95 percent confidence interval:
 120.9337 189.1970
sample estimates:
mean of x
 155.0654
```

`t.test()` conducts the test. `eammi$Duration` selects the `Duration` variable from the `eammi` data frame. `mu = 30` supplies the comparison value, and `alternative = "two.sided"` represents a non-directional alternative hypothesis.

Do not interpret the test yet. First, examine the variable.

```
duration_first_summary <- eammi |>
  summarise(
    N = n(),
    Mean = mean(Duration),
    Median = median(Duration),
    SD = sd(Duration),
    Minimum = min(Duration),
    Q1 = quantile(Duration, 0.25),
    Q3 = quantile(Duration, 0.75),
    IQR = IQR(Duration),
    Maximum = max(Duration),
    Skewness = sample_skewness(Duration)
  )
```

```
duration_first_summary
```

```
# A tibble: 1 × 10
      N Mean Median   SD Minimum   Q1   Q3   IQR Maximum Skewness
<int> <dbl> <dbl> <dbl>   <dbl> <dbl> <dbl> <dbl>   <dbl>   <dbl>
1  2073  155.   27.3  792.    10.2  20.8  40.4  19.6   9887.    8.70
```

`sample_skewness()` is supplied by the setup file. It calculates a distribution's skewness.

```
eammi |>
  ggplot(aes(x = Duration)) +
  geom_histogram(binwidth = 100, boundary = 0) +
  labs(
    x = "Survey duration in minutes",
    y = "Frequency"
  )
```

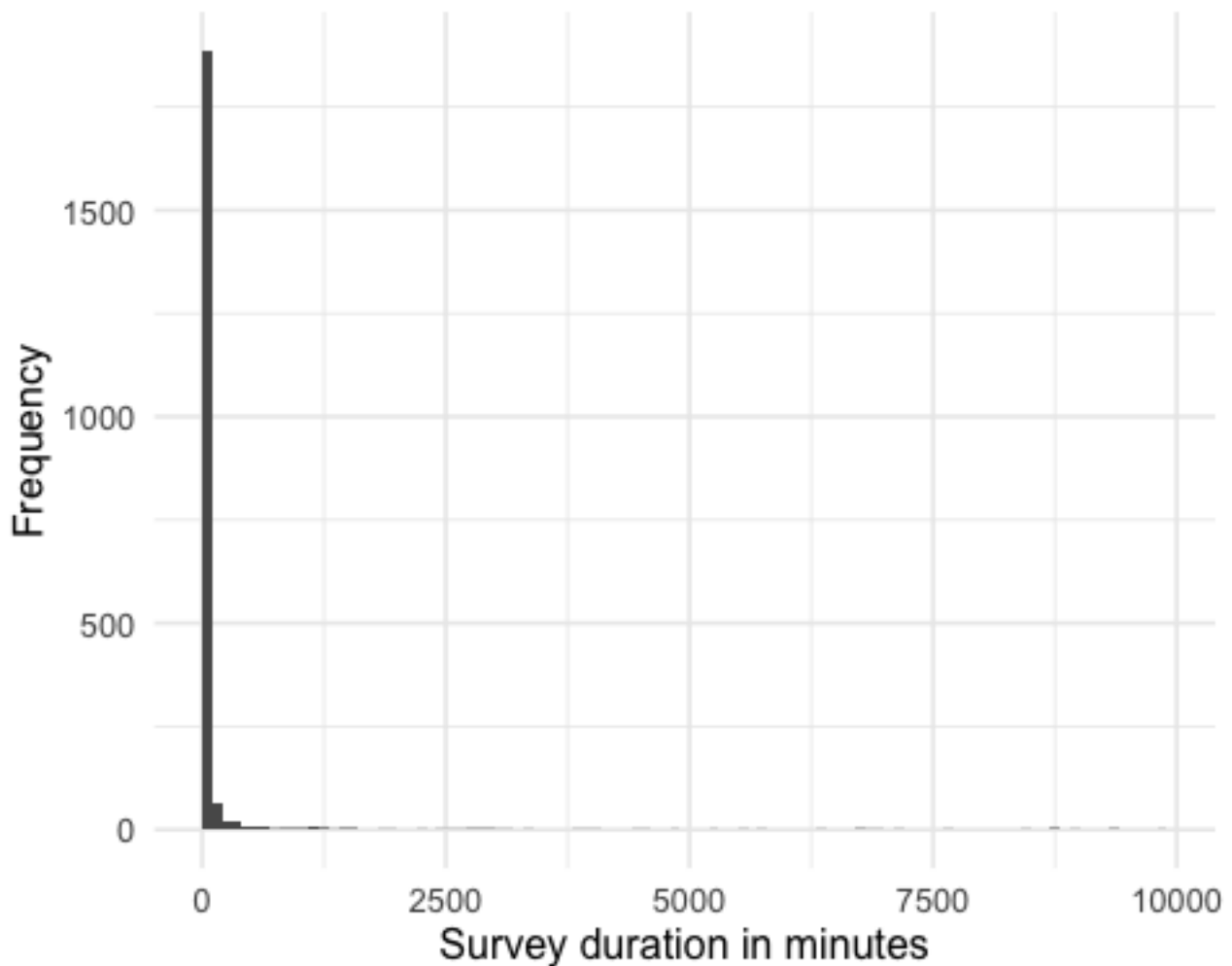


Figure 20: Survey duration before potential upper outliers are set aside.

Investigate

Does anything seem odd? Compare the mean with the median, inspect the standard deviation and maximum, and describe the histogram. A survey duration measured in the thousands of minutes deserves a second look.

The mean is much larger than the median, the standard deviation is extremely large, and the histogram has a long right tail. The initial distribution is strongly positively skewed. An important assumption of many statistical analyses, including t tests, is that the dependent variable is normally distributed. These data are not normally distributed, which means the results from our first t test will be misleading. We should not interpret or report that first test.

The next step is not to set unusual scores aside automatically. It is to identify potential outliers, decide what they represent, and state any decision we make.

Use the IQR boundaries:

$$\text{Lower boundary} = Q1 - (1.5 \times IQR)$$

$$\text{Upper boundary} = Q3 + (1.5 \times IQR)$$

```
duration_bounds <- eammi |>
  summarise(
    Q1 = quantile(Duration, 0.25),
    Q3 = quantile(Duration, 0.75),
    IQR = IQR(Duration),
    LowerBoundary = Q1 - (1.5 * IQR),
    UpperBoundary = Q3 + (1.5 * IQR)
  )
```

```
duration_bounds
```

```
# A tibble: 1 × 5
```

	Q1	Q3	IQR	LowerBoundary	UpperBoundary
1	20.8	40.4	19.6	-8.59	69.7

The lower boundary is negative, and survey duration cannot be negative. The upper boundary is a little under 70 minutes. For this demonstration, we will set aside durations above that boundary and state that decision in the final report.

Screen And Reanalyze

Store the upper boundary and use it to create a new data frame. The original eammi object remains unchanged.

```
duration_upper_boundary <- duration_bounds$UpperBoundary
```

```
duration_no_outliers <- eammi |>
  filter(Duration <= duration_upper_boundary)
```

filter() keeps observations that meet the condition. Reuse the descriptive code with duration_no_outliers as the data frame.

```
duration_filtered_summary <- duration_no_outliers |>
  summarise(
    N = n(),
    Mean = mean(Duration),
    Median = median(Duration),
    SD = sd(Duration),
    Minimum = min(Duration),
    Q1 = quantile(Duration, 0.25),
    Q3 = quantile(Duration, 0.75),
    IQR = IQR(Duration),
    Maximum = max(Duration),
    Skewness = sample_skewness(Duration)
  )
```

```
duration_filtered_summary
```

```
# A tibble: 1 × 10
```

	N	Mean	Median	SD	Minimum	Q1	Q3	IQR	Maximum	Skewness
1	1814	28.4	25.5	11.7	10.2	20.1	34.0	14.0	69.2	1.20

```
duration_no_outliers |>
  ggplot(aes(x = Duration)) +
  geom_histogram(binwidth = 5, boundary = 0) +
  labs(
    x = "Survey duration in minutes",
```

```
y = "Frequency"
)
```

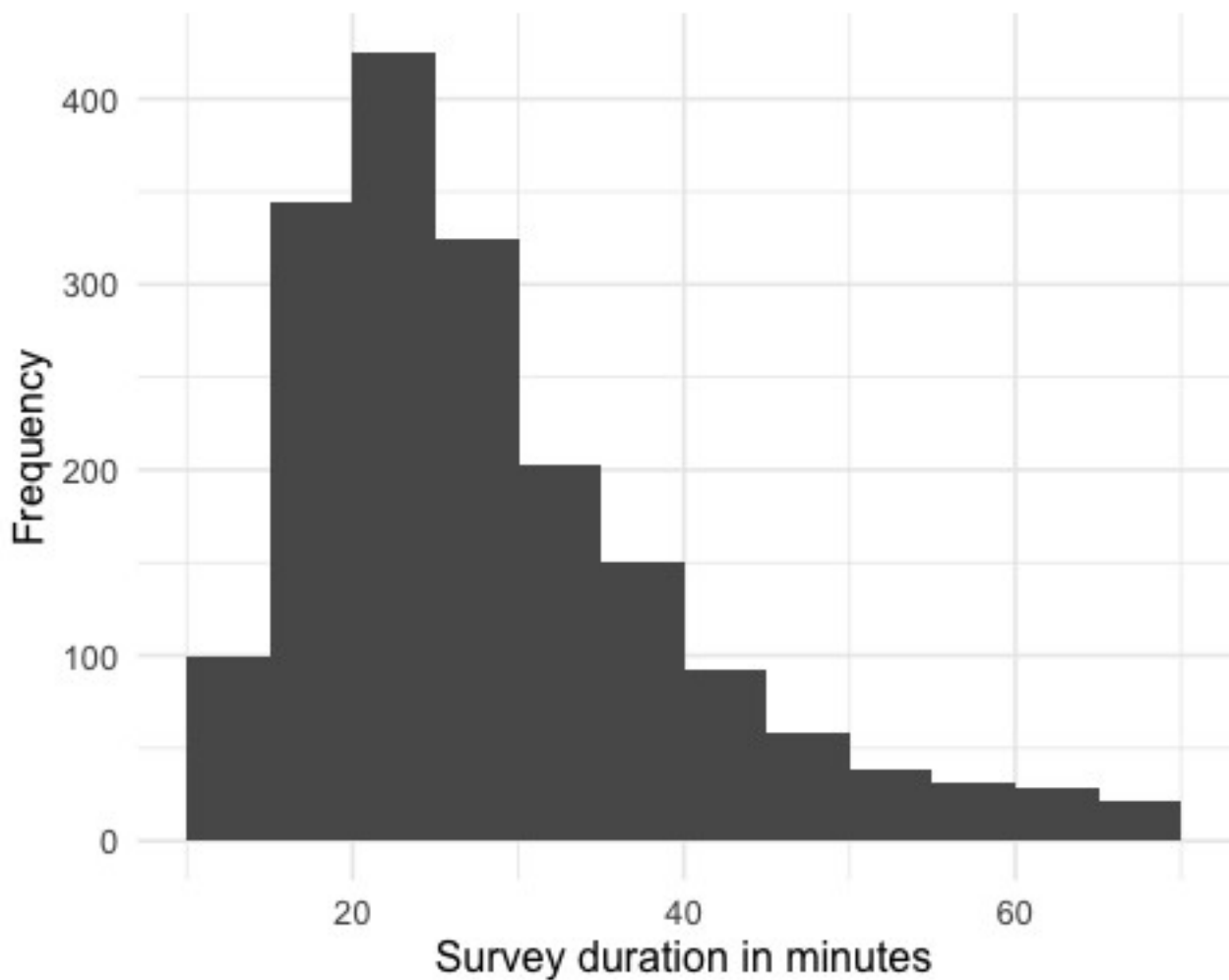


Figure 21: Survey duration after potential upper outliers are set aside.

The distribution still has a slight positive skew, but the skewness statistic is now within the guideline of -2.00 to +2.00. Run the final test on the modified data frame.

```
duration_final_test <- t.test(
  duration_no_outliers$Duration,
  mu = 30,
  alternative = "two.sided"
)
```

```
duration_final_test
```

One Sample t-test

```
data: duration_no_outliers$Duration
t = -5.8678, df = 1813, p-value = 5.24e-09
alternative hypothesis: true mean is not equal to 30
95 percent confidence interval:
```

```

27.85437 28.92939
sample estimates:
mean of x
28.39188

```

The confidence interval printed by `t.test()` is for the sample mean. The research question concerns the difference from 30, so subtract 30 from both confidence limits.

Calculate Cohen's *d*

`cohens_d()` is a workbook helper first introduced here and supplied by the setup file. For a one-sample design, it calculates Cohen's *d* by dividing the mean difference by the sample standard deviation.

```

duration_mean_difference <- mean(duration_no_outliers$Duration) - 30
duration_difference_ci <- duration_final_test$conf.int - 30

duration_d <- cohens_d(
  duration_mean_difference,
  sd(duration_no_outliers$Duration)
)

```

```

duration_mean_difference
[1] -1.608122
duration_difference_ci
[1] -2.145630 -1.070614
attr(,"conf.level")
[1] 0.95
duration_d
[1] -0.1377695

```

Produce

Write an interpretation that includes:

- the comparison value and the data decision;
- the sample mean and standard deviation;
- *t*, degrees of freedom, and *p*;
- Cohen's *d* and its magnitude;
- the confidence interval for the mean difference; and
- a conclusion in the context of survey duration.

Research Question 2: Mindfulness

Mindfulness is the tendency to focus on the present moment without judging one's experiences. Brown and Ryan (2003) reported a mean mindfulness score of 4.29 for Zen practitioners and 3.85 for undergraduate students using the Mindful Attention Awareness Scale. The same measure is named `Mindfulness` in the `EAMMI2` data.

Are emerging adults significantly different from Zen practitioners on mindfulness? What about undergraduate students?

Predict

Write a null and alternative hypothesis for each comparison in words and notation. Both questions ask whether the means are different, so identify the correct two-tailed alternative.

Run

Copy the descriptive summary and histogram pattern from Research Question 1. Replace Duration with Mindfulness, use an informative object name, and choose a bin width that shows the distribution clearly.

Investigate

Use the mean, median, skewness, and histogram to decide whether the distribution is close enough to normal for this analysis. Do not apply the duration boundary to a different variable.

Run The Comparisons

Copy the final `t.test()` pattern twice. In the first copy, use $\mu = 4.29$. In the second, use $\mu = 3.85$. Create distinct object names, mean differences, confidence intervals, and Cohen's d values.

Produce

Write a separate interpretation for each comparison. Explain which group had the higher mean and describe the magnitude of the difference.

Check Your Work

Research Question 1

$$H_0: \mu = 30$$

The average EAMMi2 survey duration does not differ from 30 minutes.

$$H_1: \mu \neq 30$$

The average EAMMi2 survey duration differs from 30 minutes.

A one-sample t test revealed a significant difference between the average time participants took to complete the EAMMi2 survey ($M = 28.39$ minutes, $SD = 11.67$) and the 30-minute estimate in the informed consent form, $t(1813) = -5.87$, $p < .001$, $d = -0.14$, 95% CI $[-2.15, -1.07]$. Potential upper outliers above the IQR boundary were set aside before this final analysis. On average, participants took less time than expected, although the magnitude of the difference was small.

Research Question 2

For the Zen-practitioner comparison:

$$H_0: \mu = 4.29$$

The mean mindfulness score for emerging adults does not differ from 4.29.

$$H_1: \mu \neq 4.29$$

The mean mindfulness score for emerging adults differs from 4.29.

For the undergraduate comparison:

$$H_0: \mu = 3.85$$

The mean mindfulness score for emerging adults does not differ from 3.85.

$$H_1: \mu \neq 3.85$$

The mean mindfulness score for emerging adults differs from 3.85.

Check the distribution first.

```
mindfulness_summary <- eammi |>
  summarise(
    N = n(),
    Mean = mean(Mindfulness),
    Median = median(Mindfulness),
    SD = sd(Mindfulness),
    Minimum = min(Mindfulness),
    Maximum = max(Mindfulness),
    Skewness = sample_skewness(Mindfulness)
  )

mindfulness_summary
# A tibble: 1 × 7
      N   Mean Median    SD Minimum Maximum Skewness
  <int> <dbl> <dbl> <dbl>   <dbl>   <dbl>   <dbl>
1  2073  3.69   3.73 0.839     0       6   -0.159

eammi |>
  ggplot(aes(x = Mindfulness)) +
  geom_histogram(binwidth = 0.25, boundary = 0) +
  labs(
    x = "Mindfulness score",
    y = "Frequency"
  )
```

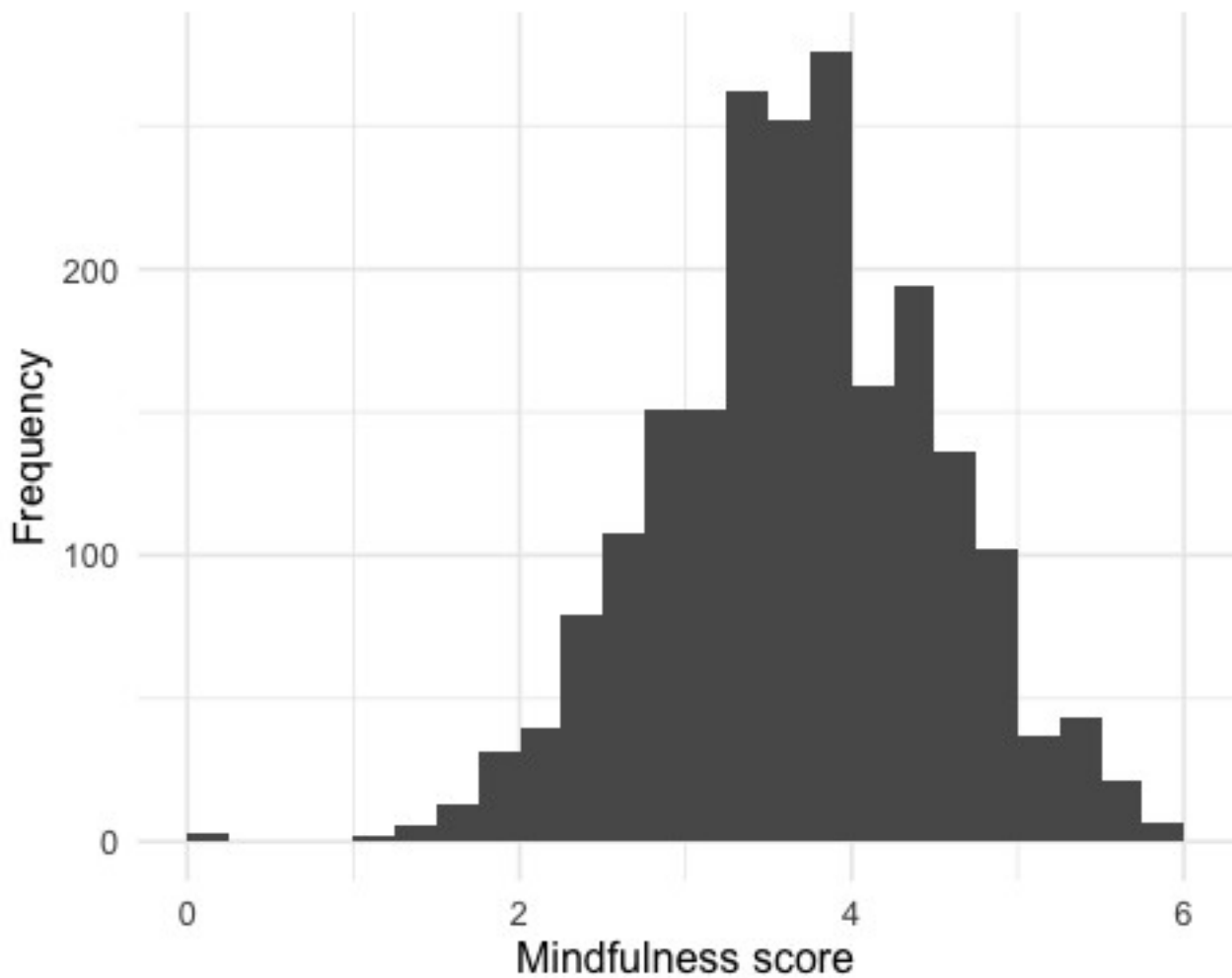


Figure 22: Mindfulness scores in the reduced EAMMi2 sample.

Mindfulness is close to normally distributed.

For the Zen-practitioner comparison:

```
mindfulness_zen_test <- t.test(
  eammi$Mindfulness,
  mu = 4.29,
  alternative = "two.sided"
)
```

```
mindfulness_zen_difference <- mean(eammi$Mindfulness) - 4.29
mindfulness_zen_ci <- mindfulness_zen_test$conf.int - 4.29
mindfulness_zen_d <- cohens_d(
  mindfulness_zen_difference,
  sd(eammi$Mindfulness)
)
```

```
mindfulness_zen_test
```

One Sample t-test

```

data: eammi$Mindfulness
t = -32.651, df = 2072, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 4.29
95 percent confidence interval:
 3.652204 3.724480
sample estimates:
mean of x
 3.688342

mindfulness_zen_ci
[1] -0.6377955 -0.5655201
attr(,"conf.level")
[1] 0.95

mindfulness_zen_d
[1] -0.7171186

```

Zen practitioners had higher mindfulness ($M = 4.29$) than emerging adults ($M = 3.69$, $SD = 0.84$), $t(2072) = -32.65$, $p < .001$, $d = -0.72$, 95% CI $[-0.64, -0.57]$. The difference was medium-to-large.

For the undergraduate comparison:

```

mindfulness_undergraduate_test <- t.test(
  eammi$Mindfulness,
  mu = 3.85,
  alternative = "two.sided"
)

mindfulness_undergraduate_difference <- mean(eammi$Mindfulness) - 3.85
mindfulness_undergraduate_ci <- mindfulness_undergraduate_test$conf.int - 3.85
mindfulness_undergraduate_d <- cohens_d(
  mindfulness_undergraduate_difference,
  sd(eammi$Mindfulness)
)

mindfulness_undergraduate_test

```

One Sample t-test

```

data: eammi$Mindfulness
t = -8.7728, df = 2072, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 3.85
95 percent confidence interval:
 3.652204 3.724480
sample estimates:
mean of x
 3.688342

mindfulness_undergraduate_ci
[1] -0.1977955 -0.1255201
attr(,"conf.level")
[1] 0.95

mindfulness_undergraduate_d
[1] -0.1926807

```

Undergraduate students had higher mindfulness ($M = 3.85$) than emerging adults ($M = 3.69$, $SD = 0.84$), $t(2072) = -8.77$, $p < .001$, $d = -0.19$, 95% CI $[-0.20, -0.13]$. The difference was small.

Chapter Takeaway

A one-sample t test compares a sample mean with a known or expected value. Examine the distribution before interpreting the test. When extreme values create severe skew, make a defensible data decision and report it.

New R commands and patterns in this chapter:

- `t.test()` conducts a one-sample t test when supplied with one variable and `mu`.
- `sample_skewness()` is supplied by the setup file and calculates sample skewness.
- `cohens_d()` is supplied by the setup file and divides a mean difference by a standard deviation.
- `mu` = supplies the comparison value.
- `alternative = "two.sided"` requests a non-directional test.
- `conf.int` extracts the confidence interval stored in a test object.

Chapter 10: Two-Sample Designs

Chapter 10 of *Exploring Statistics* introduces two-sample designs. Sometimes two means come from different groups of people; other times the same people supply both scores. That distinction determines which *t* test we use and what assumptions we check.

Learning Goals

By the end of this chapter, you should be able to:

- identify the independent and dependent variables in a two-sample question;
- distinguish independent-samples from paired-samples designs;
- examine normality and homogeneity of variance as appropriate;
- conduct and interpret both types of *t* test;
- calculate Cohen's *d* for both designs; and
- report the means, standard deviations, *t*, *p*, Cohen's *d*, and confidence interval.

Research Questions

1. Do students with conservative and liberal political views differ in feelings of belonging?
2. Do participants differ in how strongly they value the American Dream and how strongly they believe they can achieve it?
3. Do participants differ in their ratings of emerging adulthood as a time for exploration and identity formation?

Research Question 1: Political Views and Belonging

Belonging is measured from 1, not at all, to 5, very much. `PoliticsDichotomous` places participants into liberal and conservative groups. Because the scores come from different groups of people, we will conduct an independent-samples *t* test.

Get Ready

Open `10-two-sample-designs.R` from the `scripts` folder. Run the shared setup line first, and then run one section at a time.

```
source("scripts/_setup.R")
```

Review the variables in Understanding EAMMi2. Give the political-view codes readable labels.

```
eammi <- eammi |>
  mutate(
    PoliticsDichotomous = factor(
      PoliticsDichotomous,
      levels = c(1, 2),
      labels = c("Liberal", "Conservative")
    )
  )
```

Predict

Before running the analysis:

1. Identify the independent and dependent variables.
2. Write the null and alternative hypotheses in words and notation.
3. Predict which group will have the higher mean, or predict no meaningful difference.
4. State what you would need to inspect before using Student's independent-samples *t* test.

Run

Start with a table and a graph. These are familiar code patterns, so focus on what they reveal about the assumptions and the direction of any difference.

```

belonging_data <- eammi |>
  drop_na(PoliticsDichotomous, Belonging)

belonging_summary <- belonging_data |>
  group_by(PoliticsDichotomous) |>
  summarise(
    N = n(),
    Mean = mean(Belonging),
    Median = median(Belonging),
    SD = sd(Belonging),
    Skewness = sample_skewness(Belonging)
  )

belonging_summary
# A tibble: 2 × 6
  PoliticsDichotomous     N  Mean Median    SD Skewness
  <fct>              <int> <dbl>  <dbl> <dbl>   <dbl>
1 Liberal             897  3.47     4  1.00  -0.566
2 Conservative        438  3.75     4  1.04  -0.778

belonging_data |>
  ggplot(aes(x = PoliticsDichotomous, y = Belonging)) +
  geom_boxplot() +
  labs(
    x = "Political orientation",
    y = "Belonging score"
  )

```

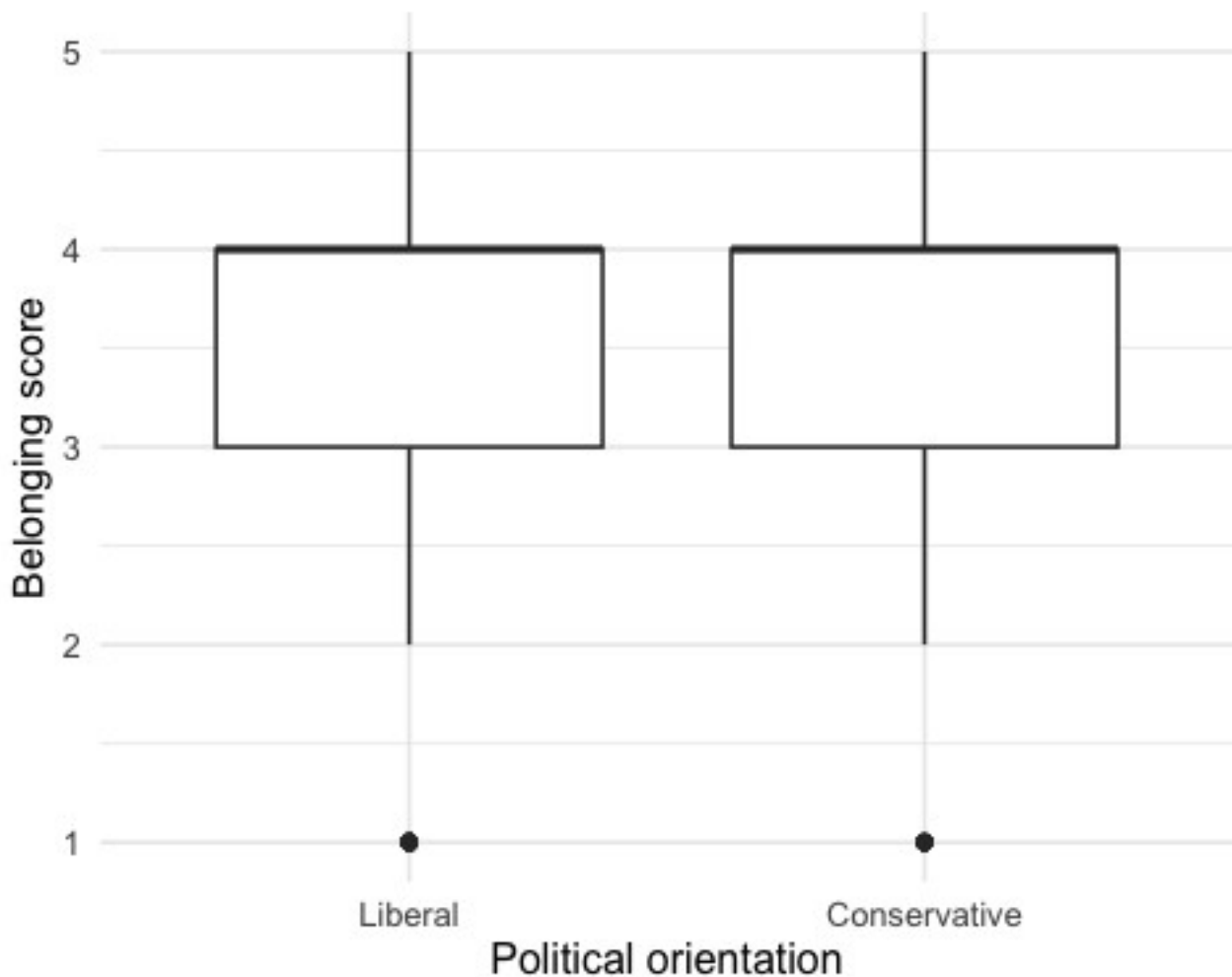


Figure 23: Belonging scores for liberal and conservative participants.

Two important assumptions must be considered before conducting Student's independent-samples t test. First, the dependent variable should be close enough to normally distributed within each group. Second, the groups should have similar variances.

Use the Shapiro-Wilk test to check normality within each group.

```
liberal_belonging <- belonging_data |>
  filter(PoliticsDichotomous == "Liberal") |>
  pull(Belonging)

conservative_belonging <- belonging_data |>
  filter(PoliticsDichotomous == "Conservative") |>
  pull(Belonging)

shapiro.test(liberal_belonging)
```

Shapiro-Wilk normality test

```
data: liberal_belonging
W = 0.88271, p-value < 2.2e-16
shapiro.test(conservative_belonging)
```

Shapiro-Wilk normality test

```
data: conservative_belonging
W = 0.86074, p-value < 2.2e-16
```

`pull()` takes one variable from a data frame so that a function can use its values. `shapiro.test()` tests the null hypothesis that the scores came from a normal distribution. A p value below .05 provides evidence that the distribution differs from normal.

Both Shapiro-Wilk tests are significant, so the normality assumption is not met. However, tests of normality are sensitive to small departures from normality in large samples. Belonging contains only valid scores from 1 to 5, and the skewness values are below 2.00 in both groups. Setting valid endpoint scores aside would be hard to justify. For this teaching analysis, we will keep all valid scores, proceed with the test, and report the assumption result.

Now examine whether the groups have similar variance.

```
belonging_variance_test <- var.test(
  Belonging ~ PoliticsDichotomous,
  data = belonging_data
)
```

```
belonging_variance_test
```

F test to compare two variances

```
data: Belonging by PoliticsDichotomous
F = 0.92717, num df = 896, denom df = 437, p-value = 0.3527
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.7867828 1.0876209
sample estimates:
ratio of variances
 0.9271695
```

`var.test()` conducts an F test comparing the two group variances. Focus on the p value. A value above .05 means the output does not provide evidence that the variances differ. Here, p is greater than .05, so we will treat the variances as similar enough for Student's t test.

The formula repeats the pattern introduced in Chapter 6: `outcome ~ predictor`. When the predictor identifies groups, you can read it as `outcome ~ group`.

Run the independent-samples t test.

```
belonging_t_test <- t.test(
  Belonging ~ PoliticsDichotomous,
  data = belonging_data,
  var.equal = TRUE
)
```

belonging_t_test

Two Sample t-test

```
data: Belonging by PoliticsDichotomous
t = -4.6974, df = 1333, p-value = 2.908e-06
alternative hypothesis: true difference in means between group Liberal and group Conservative is
not equal to 0
95 percent confidence interval:
 -0.3948206 -0.1621959
sample estimates:
    mean in group Liberal mean in group Conservative
           3.474916           3.753425

var.equal = TRUE requests Student's independent-samples t test.
```

Investigate

Read the output in this order:

1. Use the p value to decide whether the group means significantly differ.
2. Use the group means to identify the direction of the difference.
3. Use the confidence interval to identify a plausible range for the mean difference.
4. Check whether the result supports your prediction.

The test shows a significant difference, and conservative participants have the higher mean. The overlapping boxplots remind us that a statistically significant mean difference does not mean the two groups are completely separate.

Calculate Effect Size

Calculate Cohen's d for the magnitude of the difference.

```
belonging_group_stats <- eammi |>
  drop_na(PoliticsDichotomous, Belonging) |>
  group_by(PoliticsDichotomous) |>
  summarise(
    N = n(),
    Mean = mean(Belonging),
    SD = sd(Belonging)
  )

belonging_d <- independent_cohens_d(
  eammi,
  Belonging,
  PoliticsDichotomous
)

belonging_group_stats
# A tibble: 2 × 4
  PoliticsDichotomous     N Mean   SD
    <fct>          <int> <dbl> <dbl>
```

```

1 Liberal          897  3.47  1.00
2 Conservative    438  3.75  1.04

belonging_d
# A tibble: 1 × 3
  MeanDifference PooledSD CohensD
    <dbl>         <dbl>   <dbl>
1   -0.279       1.02   -0.274

```

This reuses `independent_cohens_d()` from Chapter 5. For independent samples, Cohen's d divides the mean difference by a pooled standard deviation. The function finds each group's sample size, mean, and standard deviation and then performs that calculation. To reuse it, change the data frame, outcome, and grouping variable.

Produce

Write an interpretation with the assumption results and decision to proceed, analysis name, both group means and standard deviations, t , degrees of freedom, p , Cohen's d , its magnitude, and the confidence interval. Describe a difference or association; political views were measured, not assigned.

Research Question 2: American Dream Ratings

Participants rated both the importance of achieving the American Dream (`USdream1`) and whether they believed they would achieve it (`USdream2`) on a 1-to-5 scale. Because both scores come from each participant, we will conduct a paired-samples t test.

Predict

Identify the independent and dependent variables, write the hypotheses, and predict which mean will be higher. Explain why the observations are paired rather than independent.

Run

Begin by creating a data frame that contains the paired scores and each participant's difference score. For a paired-samples t test, the normality assumption applies to these difference scores.

```

usdream_data <- eammi |>
  drop_na(USdream1, USdream2) |>
  mutate(
    Difference = USdream1 - USdream2
  )

usdream_summary <- usdream_data |>
  summarise(
    N = n(),
    USdream1_Mean = mean(USdream1),
    USdream1_SD = sd(USdream1),
    USdream2_Mean = mean(USdream2),
    USdream2_SD = sd(USdream2),
    DifferenceSkewness = sample_skewness(Difference)
  )

usdream_summary
# A tibble: 1 × 6
  N USdream1_Mean USdream1_SD USdream2_Mean USdream2_SD DifferenceSkewness

```

	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	2063	3.41	1.22	3.57	1.07	-0.282

Run the Shapiro-Wilk test on the difference scores.

```
usdream_normality_test <- shapiro.test(usdream_data$Difference)
```

```
usdream_normality_test
```

```
Shapiro-Wilk normality test
```

```
data: usdream_data$Difference
W = 0.86755, p-value < 2.2e-16
```

The Shapiro-Wilk test is significant, so the difference scores differ statistically from normal. Their skewness is about -0.28, however, which does not indicate severe skew. The two ratings also use a short 1-to-5 scale with no invalid values. For this teaching analysis, we will proceed and report the assumption result.

Run the paired test.

```
usdream_t_test <- t.test(
  usdream_data$USdream1,
  usdream_data$USdream2,
  paired = TRUE
)
```

```
usdream_t_test
```

```
Paired t-test
```

```
data: usdream_data$USdream1 and usdream_data$USdream2
t = -7.6047, df = 2062, p-value = 4.312e-14
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.2073099 -0.1223071
sample estimates:
mean difference
 -0.1648085
```

`paired = TRUE` tells R that the scores in the two variables belong to the same participants. R analyzes each participant's difference score rather than treating the variables as unrelated groups.

Investigate

Identify which measure has the higher mean. Then use t and p to decide whether the mean difference is statistically significant. Check the confidence interval and determine whether it contains 0.

Calculate Effect Size

Create the difference scores and calculate Cohen's d .

```
usdream_difference <- usdream_data$Difference
usdream_d <- cohens_d(
  mean(usdream_difference),
```

```
sd(usdream_difference)
)

usdream_d
[1] -0.1674288
```

For paired samples, Cohen's d is the mean of the difference scores divided by their standard deviation. The subtraction order affects the sign. Here, negative values mean that attainability ratings tend to be higher than importance ratings.

Produce

Write an interpretation that reports the normality result and decision to proceed, names both ratings, reports both means and standard deviations, and includes t , p , Cohen's d , its magnitude, and the confidence interval.

Research Question 3: Exploration and Identity Formation

Participants rated emerging adulthood as a time for exploration (Exploration) and identity formation (Identity) on a scale from 1, strongly disagree, to 4, strongly agree.

Predict

Identify the variables, select an independent- or paired-samples test, write the hypotheses, and predict which mean will be higher.

Run

Copy the Research Question 2 code. Replace the American Dream variables with Exploration and Identity, create the participant-level difference scores, and use new object names.

Investigate

Use `shapiro.test()` to check the difference scores. Then check the means, t , p , confidence interval, and the sign of the difference. Confirm that `paired = TRUE` remains in your test.

Calculate Effect Size

Create `exploration_identity_difference`, then use its mean and standard deviation with `cohens_d()`.

Produce

Write a paragraph that answers the research question and reports the normality result, decision to proceed, and required statistics.

Check Your Work

Research Question 1

Independent variable: political orientation. Dependent variable: feelings of belonging.

$$H_0: \mu_{\text{liberal}} = \mu_{\text{conservative}}$$

$$H_1: \mu_{\text{liberal}} \neq \mu_{\text{conservative}}$$

An independent-samples t test compared belonging among emerging adults who identified as liberal and conservative. Conservative emerging adults reported significantly greater belonging ($M = 3.75$, $SD = 1.04$) than liberal emerging adults ($M = 3.47$, $SD = 1.00$), $t(1333) = -4.70$, $p < .001$, $d = -0.27$, 95% CI $[-0.39, -0.16]$. The magnitude of the difference was small.

The Shapiro-Wilk tests were significant for both groups, so the normality assumption was not met. The skewness values were below 2.00, and all scores were valid values on the 1-to-5 scale, so we proceeded with the analysis. The F test was not significant, so the homogeneity-of-variance assumption was met.

Research Question 2

Independent variable: type of American Dream rating. Dependent variable: American Dream rating.

$$H_0: \mu_{\text{importance}} = \mu_{\text{attainability}}$$

$$H_1: \mu_{\text{importance}} \neq \mu_{\text{attainability}}$$

A paired-samples t test compared ratings of the importance and attainability of the American Dream. Emerging adults believed more strongly that they would achieve the American Dream ($M = 3.57$, $SD = 1.07$) than that the American Dream was important ($M = 3.41$, $SD = 1.22$), $t(2062) = -7.60$, $p < .001$, $d = -0.17$, 95% CI $[-0.21, -0.12]$. The magnitude of the difference was small.

The Shapiro-Wilk test of the paired difference scores was significant, so the normality assumption was not met. The difference scores were not severely skewed, so we proceeded with the analysis and reported the assumption result.

Research Question 3

The appropriate analysis is a paired-samples t test because each participant supplied both ratings.

```
exploration_identity_data <- eammi |>
  drop_na(Exploration, Identity) |>
  mutate(
    Difference = Exploration - Identity
  )

exploration_identity_summary <- exploration_identity_data |>
  summarise(
    N = n(),
    Exploration_Mean = mean(Exploration),
    Exploration_SD = sd(Exploration),
    Identity_Mean = mean(Identity),
    Identity_SD = sd(Identity),
    DifferenceSkewness = sample_skewness(Difference)
  )

exploration_identity_normality_test <- shapiro.test(
  exploration_identity_data$Difference
)

exploration_identity_t_test <- t.test(
```

```

exploration_identity_data$Exploration,
exploration_identity_data$Identity,
paired = TRUE
)

exploration_identity_difference <- exploration_identity_data$Difference
exploration_identity_d <- cohens_d(
  mean(exploration_identity_difference),
  sd(exploration_identity_difference)
)

exploration_identity_summary
# A tibble: 1 × 6
      N Exploration_Mean Exploration_SD Identity_Mean Identity_SD
<int>      <dbl>      <dbl>      <dbl>      <dbl>
1  2073      3.64      0.493      3.58      0.521
# i 1 more variable: DifferenceSkewness <dbl>

exploration_identity_normality_test

Shapiro-Wilk normality test

data:  exploration_identity_data$Difference
W = 0.89736, p-value < 2.2e-16

exploration_identity_t_test

```

Paired t-test

```

data:  exploration_identity_data$Exploration and exploration_identity_data$Identity
t = 4.3895, df = 2072, p-value = 1.193e-05
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 0.02815515 0.07362971
sample estimates:
mean difference
 0.05089243

exploration_identity_d
[1] 0.09640884

```

Respondents rated emerging adulthood significantly more strongly as a time for exploration ($M = 3.64$, $SD = 0.49$) than identity formation ($M = 3.58$, $SD = 0.52$), $t(2072) = 4.39$, $p < .001$, $d = 0.10$, 95% CI [0.03, 0.07]. The magnitude of the difference was quite small.

The Shapiro-Wilk test of the paired difference scores was significant, so the normality assumption was not met. The difference scores were approximately symmetric, and we proceeded while keeping the assumption result in mind.

Chapter Takeaway

Use an independent-samples t test when two means come from different groups of people. Use a paired-samples t test when two means come from the same or matched participants.

New R commands and patterns in this chapter:

- `pull()` takes one variable from a data frame for use by a function.
- `shapiro.test()` checks whether scores differ significantly from a normal distribution.
- `var.test()` conducts an F test comparing two group variances.
- `outcome ~ group` repeats the formula pattern from Chapter 6 for a group comparison.
- `var.equal = TRUE` requests Student's independent-samples *t* test.
- `paired = TRUE` identifies related scores in a paired-samples *t* test.

Chapter 11: Analysis of Variance

Chapter 11 of *Exploring Statistics* introduces one-way analysis of variance, usually called one-way ANOVA. A *t* test compares two means; ANOVA lets us bring three independent groups into the same analysis without running a collection of separate *t* tests.

Learning Goals

By the end of this chapter, you should be able to:

- identify the independent and dependent variables in a one-way ANOVA;
- explain when one-way ANOVA is appropriate;
- examine normality and homogeneity of variance;
- conduct and interpret an omnibus ANOVA;
- calculate eta squared for the overall effect;
- use Tukey HSD tests to identify which groups differ; and
- write a complete interpretation without making an unsupported causal claim.

Research Questions

1. Does achieved independence differ depending on whether participants identify as adults?
2. Does self-efficacy differ depending on whether participants identify as adults?
3. Does stress differ depending on the extent to which participants no longer live at home?

Research Question 1: Achieved Independence and Adult Identity

We will compare `MOA_ACH`, the extent to which participants have achieved markers of adulthood, across the three categories of `Adult`: no, maybe, and yes.

Get Ready

Open `11-anova-independent.R` from the `scripts` folder. Run the shared setup line first, and then run one section at a time.

```
source("scripts/_setup.R")
```

Review `Adult` and `MOA_ACH` in Understanding EAMMi2.

First, we will give `Adult` readable labels.

```
eammi <- eammi |>
  mutate(
    Adult = factor(
      Adult,
      levels = c(3, 2, 1),
      labels = c("No", "Maybe", "Yes")
    )
  )
```

The category order is meaningful and will remain No, Maybe, Yes in tables and graphs.

The earlier descriptive-statistics work identified potential MOA_ACH outliers with the IQR method. Recalculate those boundaries and create a separate data frame for this analysis.

```
moa_bounds <- eammi |>
  summarise(
    Q1 = quantile(MOA_ACH, 0.25),
    Q3 = quantile(MOA_ACH, 0.75),
    IQR = IQR(MOA_ACH),
    LowerBoundary = Q1 - (1.5 * IQR),
    UpperBoundary = Q3 + (1.5 * IQR)
  )

moa_bounds

# A tibble: 1 × 5
   Q1    Q3   IQR LowerBoundary UpperBoundary
<dbl> <dbl> <dbl>         <dbl>         <dbl>
1   34   41    7           23.5           51.5

moa_adult_data <- eammi |>
  filter(
    MOA_ACH >= moa_bounds$LowerBoundary,
    MOA_ACH <= moa_bounds$UpperBoundary
  ) |>
  drop_na(MOA_ACH, Adult)
```

This keeps eammi unchanged and makes the data decision visible in the object name.

Predict

Before running the ANOVA:

1. Identify the independent and dependent variables.
2. Write the null and alternative hypotheses in words and notation.
3. Predict the order of the three group means.
4. If the overall test is significant, predict which pairs of groups will differ.

Run

Begin by examining each group.

The graph later in this chapter will show 95% confidence intervals around the group means. `qt()` supplies the critical *t* value needed to calculate those intervals; the remaining pieces use the sample size and standard error you have already seen.

```
moa_adult_summary <- moa_adult_data |>
  group_by(Adult) |>
  summarise(
    N = n(),
    Mean = mean(MOA_ACH),
    SD = sd(MOA_ACH),
    SE = SD / sqrt(N),
    LowerCI = Mean - qt(0.975, df = N - 1) * SE,
    UpperCI = Mean + qt(0.975, df = N - 1) * SE,
    Skewness = sample_skewness(MOA_ACH)
  )
```

```
moa_adult_summary
# A tibble: 3 × 8
  Adult      N Mean    SD    SE LowerCI UpperCI Skewness
  <fct> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 No      141  34.7  4.50 0.379  34.0   35.5   0.899
2 Maybe   504  36.1  4.33 0.193  35.7   36.5   0.720
3 Yes    1341  38.3  5.06 0.138  38.1   38.6   0.367
```

One-way ANOVA assumes that the outcome is close enough to normally distributed within each group and that group variances are reasonably similar. Begin by fitting the model. Then reuse the Shapiro-Wilk test from Chapter 10 and add Bartlett's test.

```
moa_adult_anova <- aov(
  MOA_ACH ~ Adult,
  data = moa_adult_data
)

moa_variance_test <- bartlett.test(
  MOA_ACH ~ Adult,
  data = moa_adult_data
)

moa_normality_test <- shapiro.test(
  residuals(moa_adult_anova)
)
```

```
moa_variance_test
```

```
Bartlett test of homogeneity of variances
```

```
data: MOA_ACH by Adult
Bartlett's K-squared = 18.338, df = 2, p-value = 0.0001042
moa_normality_test
```

```
Shapiro-Wilk normality test
```

```
data: residuals(moa_adult_anova)
W = 0.98193, p-value = 3.676e-15
summary(moa_adult_anova)
      Df Sum Sq Mean Sq F value Pr(>F)
Adult      2   2976  1488.2    63.3 <2e-16 ***
Residuals 1983  46625    23.5
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

`aov()` conducts the ANOVA. The formula repeats the pattern from Chapters 6 and 10: `outcome ~ group`. Here, `MOA_ACH` is the outcome and `Adult` identifies the groups.

`bartlett.test()` checks whether the three group variances differ. The F test used in Chapter 10 cannot be used here because it compares only two variances. `residuals()` retrieves the differences between the observed scores and the scores predicted by the model. The Shapiro-Wilk test checks whether those residuals differ from normal.

Both assumption tests are significant, so this analysis does not meet either assumption. Large samples can make small departures statistically significant, but the group standard deviations also show a noticeable variance difference. Because this chapter demonstrates the one-way ANOVA process, we will continue while treating the violated assumptions as an important limitation.

`summary()` prints the ANOVA table stored in `moa_adult_anova`.

Investigate

The `Adult` row contains the between-group effect. The `Residuals` row represents variability remaining within groups. Read the table in this order:

1. Find the degrees of freedom for `Adult` and `Residuals`.
2. Find the F value.
3. Use the p value to decide whether at least one group mean differs.

The p value is less than .001, so the overall effect is significant. This omnibus test does not yet tell us which groups differ.

Calculate eta squared, the proportion of outcome variability associated with the grouping variable.

```
moa_eta_squared <- anova_eta_squared(
  moa_adult_anova,
  "Adult"
)
```

```
moa_eta_squared
[1] 0.06000782
```

`anova_eta_squared()` is supplied by the setup file. Its first argument is the ANOVA object. Its second argument is the grouping-variable name in quotation marks. The result is about .06, a medium effect.

Because the omnibus ANOVA is significant, use Tukey HSD comparisons.

```
moa_tukey <- TukeyHSD(moa_adult_anova)
```

```
moa_tukey
  Tukey multiple comparisons of means
    95% family-wise confidence level
```

```
Fit: aov(formula = MOA_ACH ~ Adult, data = moa_adult_data)
```

```
$Adult
      diff      lwr      upr    p adj
Maybe-No 1.383443 0.2999328 2.466953 0.0078278
Yes-No     3.600965 2.5940836 4.607846 0.0000000
Yes-Maybe  2.217522 1.6233026 2.811741 0.0000000
```

`TukeyHSD()` uses the fitted ANOVA object and compares every pair of adult-identity groups. The `diff` column gives the mean difference, `lwr` and `upr` give its confidence interval, and `p adj` gives a p value adjusted for making several comparisons.

Calculate Effect Sizes

Calculate Cohen's d for the pairwise comparisons.

```
moa_pairwise_d <- tukey_cohens_d(
  moa_adult_anova,
  moa_tukey,
  "Adult"
)
```

```
moa_pairwise_d

# A tibble: 3 × 2
  Comparison CohensD
  <chr>      <dbl>
1 Maybe-No   0.285
2 Yes-No     0.743
3 Yes-Maybe  0.457
```

`tukey_cohens_d()` uses three pieces that you have already created: the ANOVA object, the Tukey object, and the grouping-variable name. When you adapt this pattern, all three names must refer to the same analysis.

At first glance, the Tukey output can seem confusing. It provides three pairwise comparisons and uses adjusted p values to correct for the increased risk of Type I errors. R lists the groups being compared at the beginning of each row:

- Maybe-No compares participants who answered maybe with participants who answered no. The adjusted p value is .008, so the groups significantly differ with a small-to-moderate effect ($d = 0.29$).
- Yes-No compares participants who answered yes with participants who answered no. The adjusted p value is below .001, so the groups significantly differ with a moderate-to-large effect ($d = 0.74$).
- Yes-Maybe compares participants who answered yes with participants who answered maybe. The adjusted p value is below .001, so the groups significantly differ with a moderate effect ($d = 0.46$).

The Tukey table and `moa_pairwise_d` use the same comparison labels, so you can match each adjusted p value with its effect size.

Graph the group means.

```
moa_adult_summary |>
  ggplot(aes(x = Adult, y = Mean)) +
  geom_errorbar(
    aes(ymin = LowerCI, ymax = UpperCI),
    width = 0.12
  ) +
  geom_point(size = 3) +
  labs(
    x = "Adult identity",
    y = "Mean achieved independence"
  )
```

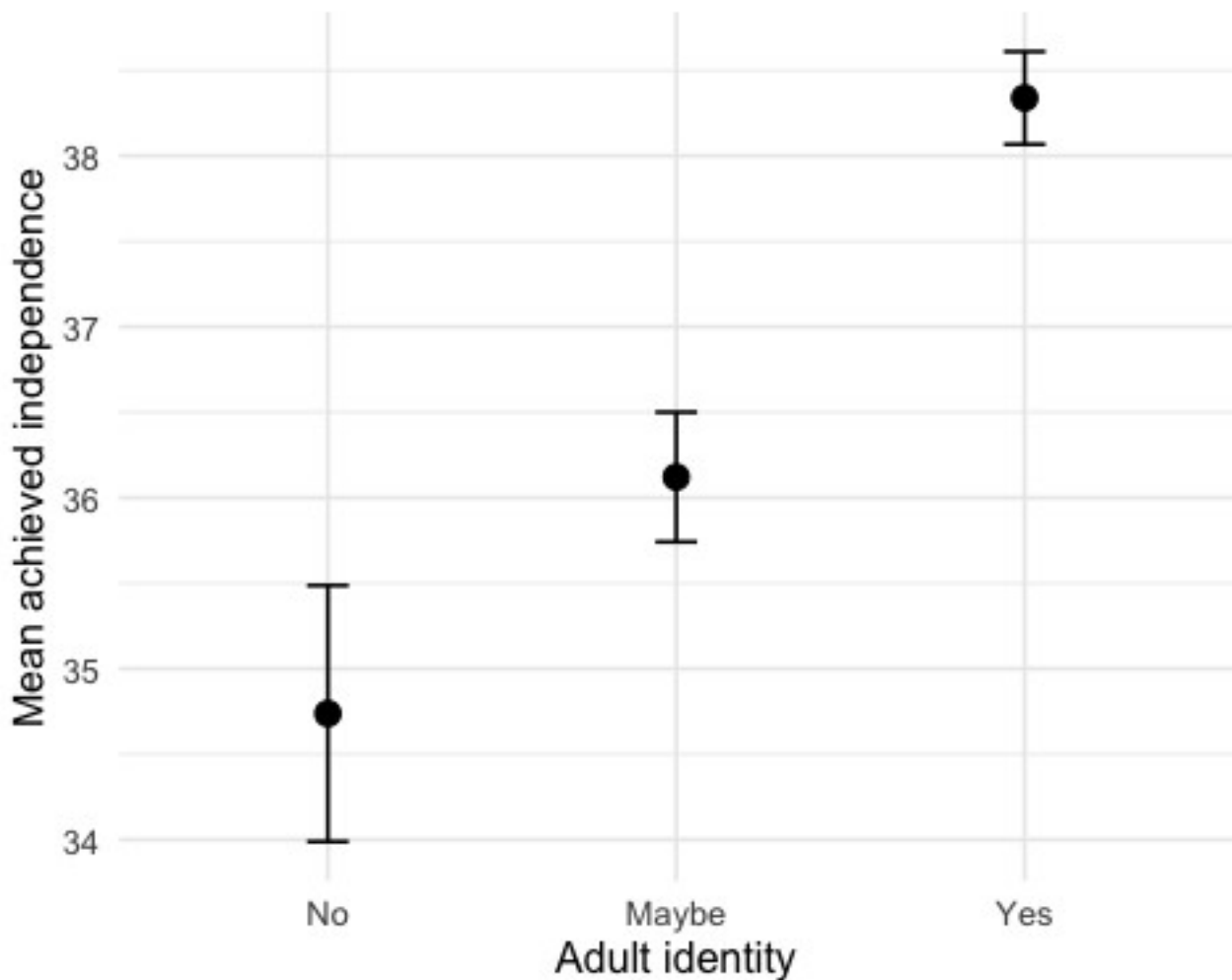


Figure 24: Mean achieved independence by adult identity.

This graph uses the layered `ggplot()` logic introduced in Chapter 6. Each `+` adds another component to the same graph:

- `ggplot()` starts with `moa_adult_summary` and maps adult status to the x-axis and the group mean to the y-axis;
- `geom_errorbar()` adds the 95% confidence interval around each mean;
- `geom_point()` adds a dot at each group mean; and
- `labs()` adds readable labels.

Earlier, `geom_point()` placed individual observations on a scatterplot. Here, each dot represents a group mean because the graph starts with the summary table. The graph uses points and confidence intervals rather than bars so both the estimate and its uncertainty are easy to see.

Use the Tukey table, effect sizes, means, and graph together. Each pair significantly differs: the Yes group has the highest mean, followed by Maybe, followed by No.

Produce

Write a paragraph that reports:

- the analysis and variables;
- the assumption results and decision to proceed;
- F , both degrees of freedom, p , and eta squared;
- the magnitude of the overall effect;
- each significant Tukey comparison with group means and pairwise d ; and
- a conclusion that answers the research question.

Both variables were measured. Say that achieved independence *differs across* adult-identity groups. Do not say that adult identity caused the difference.

Research Question 2: Self-Efficacy and Adult Identity

Does SelfEfficacy differ across the No, Maybe, and Yes adult-identity groups?

Predict

Write the null and alternative hypotheses and predict the order of the group means.

Run

Copy the complete Research Question 1 pattern, but do not reuse the MOA_ACH outlier filter. Create a data frame containing complete SelfEfficacy and Adult scores. Then create a group summary, fit `SelfEfficacy ~ Adult`, and run both assumption tests.

Investigate

Evaluate the assumptions from the descriptive output. Read the omnibus table before deciding whether Tukey comparisons are needed.

Calculate Effect Sizes

Create new eta-squared, Tukey, and pairwise- d objects. Check that every object name refers to self-efficacy rather than achieved independence.

Produce

Write a complete interpretation and note whether the data conditions affect your confidence in the result.

Research Question 3: Stress and Living at Home

Does Stress differ across the No, Somewhat, and Yes categories of NoLongerHome?

Predict

Write the null and alternative hypotheses and predict the pattern of group means.

Run

Give NoLongerHome readable labels in the order No, Somewhat, Yes. Create a complete-case data frame, group summary, the model `Stress ~ NoLongerHome`, and both assumption tests.

Investigate

Evaluate the assumptions and read the omnibus p value. Decide whether post hoc comparisons are needed before running them.

Calculate Effect Size

Calculate eta squared. If the omnibus test is not significant, stop there; post hoc tests are used to investigate a significant omnibus difference.

Produce

Write a conclusion that reports the nonsignificant result and the magnitude of the effect. Do not interpret group differences that the omnibus test did not support.

Check Your Work

Research Question 1

Independent variable: adult identity. Dependent variable: achieved independence.

$$H_0: \mu_{\text{no}} = \mu_{\text{maybe}} = \mu_{\text{yes}}$$

$$H_1: \text{at least one group mean differs}$$

Both formal assumption tests were significant, so normality and homogeneity of variance were not met. We proceeded because this was a demonstration analysis, but the results should be interpreted cautiously.

A one-way ANOVA revealed that achieved independence significantly differed across adult-identity groups, $F(2, 1983) = 63.30$, $p < .001$, $\eta^2 = .06$. The overall effect was medium. Tukey HSD tests showed that participants who identified as adults reported greater achieved independence ($M = 38.30$, $SE = 0.14$) than participants who answered maybe ($M = 36.09$, $SE = 0.19$), $p < .001$, $d = 0.46$, and participants who did not identify as adults ($M = 34.70$, $SE = 0.38$), $p < .001$, $d = 0.74$. Participants who answered maybe also reported greater achieved independence than participants who answered no, $p = .008$, $d = 0.29$.



Why this wording matters

The first sentence intentionally says that achieved independence *differed across* adult-identity groups. Both variables were measured, so there was no manipulated independent variable. Without a manipulated independent variable, we cannot say that adult identity caused differences in achieved independence.

This interpretation reports standard errors because they correspond with the 95% confidence intervals in the graph. To report group standard deviations instead, use the grouped descriptive-statistics pattern from Chapters 3 and 4. `moa_adult_summary` includes both SD and SE.

Research Question 2

$$H_0: \mu_{\text{no}} = \mu_{\text{maybe}} = \mu_{\text{yes}}$$

Self-efficacy does not differ across adult-identity groups.

$$H_1: \text{at least one group mean differs}$$

Self-efficacy differs across at least one pair of adult-identity groups.

```
self_efficacy_data <- eammi |>
  drop_na(SelfEfficacy, Adult)

self_efficacy_summary <- self_efficacy_data |>
  group_by(Adult) |>
  summarise(
    N = n(),
    Mean = mean(SelfEfficacy),
    SD = sd(SelfEfficacy),
    SE = SD / sqrt(N),
    LowerCI = Mean - qt(0.975, df = N - 1) * SE,
    UpperCI = Mean + qt(0.975, df = N - 1) * SE,
    Skewness = sample_skewness(SelfEfficacy)
  )

self_efficacy_anova <- aov(
  SelfEfficacy ~ Adult,
  data = self_efficacy_data
)

self_efficacy_variance_test <- bartlett.test(
  SelfEfficacy ~ Adult,
  data = self_efficacy_data
)

self_efficacy_normality_test <- shapiro.test(
  residuals(self_efficacy_anova)
)

self_efficacy_eta_squared <- anova_eta_squared(
  self_efficacy_anova,
```

```

"Adult"
)

self_efficacy_tukey <- TukeyHSD(self_efficacy_anova)
self_efficacy_pairwise_d <- tukey_cohens_d(
  self_efficacy_anova,
  self_efficacy_tukey,
  "Adult"
)

self_efficacy_summary
# A tibble: 3 × 8
  Adult      N Mean    SD    SE LowerCI UpperCI Skewness
  <fct> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 No      147  29.1  5.00  0.413   28.3   29.9 -0.568
2 Maybe   518  30.4  4.00  0.176   30.0   30.7 -0.0103
3 Yes    1407  31.5  4.42  0.118   31.2   31.7 -0.162

self_efficacy_variance_test

Bartlett test of homogeneity of variances

data: SelfEfficacy by Adult
Bartlett's K-squared = 13.925, df = 2, p-value = 0.0009466

self_efficacy_normality_test

Shapiro-Wilk normality test

data: residuals(self_efficacy_anova)
W = 0.98622, p-value = 3.045e-13

summary(self_efficacy_anova)
      Df Sum Sq Mean Sq F value    Pr(>F)
Adult     2   1034    516.8    27.12 2.37e-12 ***
Residuals 2069  39426     19.1
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

self_efficacy_eta_squared
[1] 0.0255454

self_efficacy_tukey
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = SelfEfficacy ~ Adult, data = self_efficacy_data)

$Adult
      diff      lwr      upr    p adj
Maybe-No  1.266685  0.3099061  2.223464  0.0054725
Yes-No     2.358514  1.4710631  3.245964  0.0000000
Yes-Maybe  1.091829  0.5656565  1.618001  0.0000036

self_efficacy_pairwise_d

```

```
# A tibble: 3 × 2
  Comparison CohensD
  <chr>      <dbl>
1 Maybe-No   0.290
2 Yes-No     0.540
3 Yes-Maybe  0.250

self_efficacy_summary |>
  ggplot(aes(x = Adult, y = Mean)) +
  geom_errorbar(
    aes(ymin = LowerCI, ymax = UpperCI),
    width = 0.12
  ) +
  geom_point(size = 3) +
  labs(
    x = "Adult identity",
    y = "Mean self-efficacy"
  )
)
```

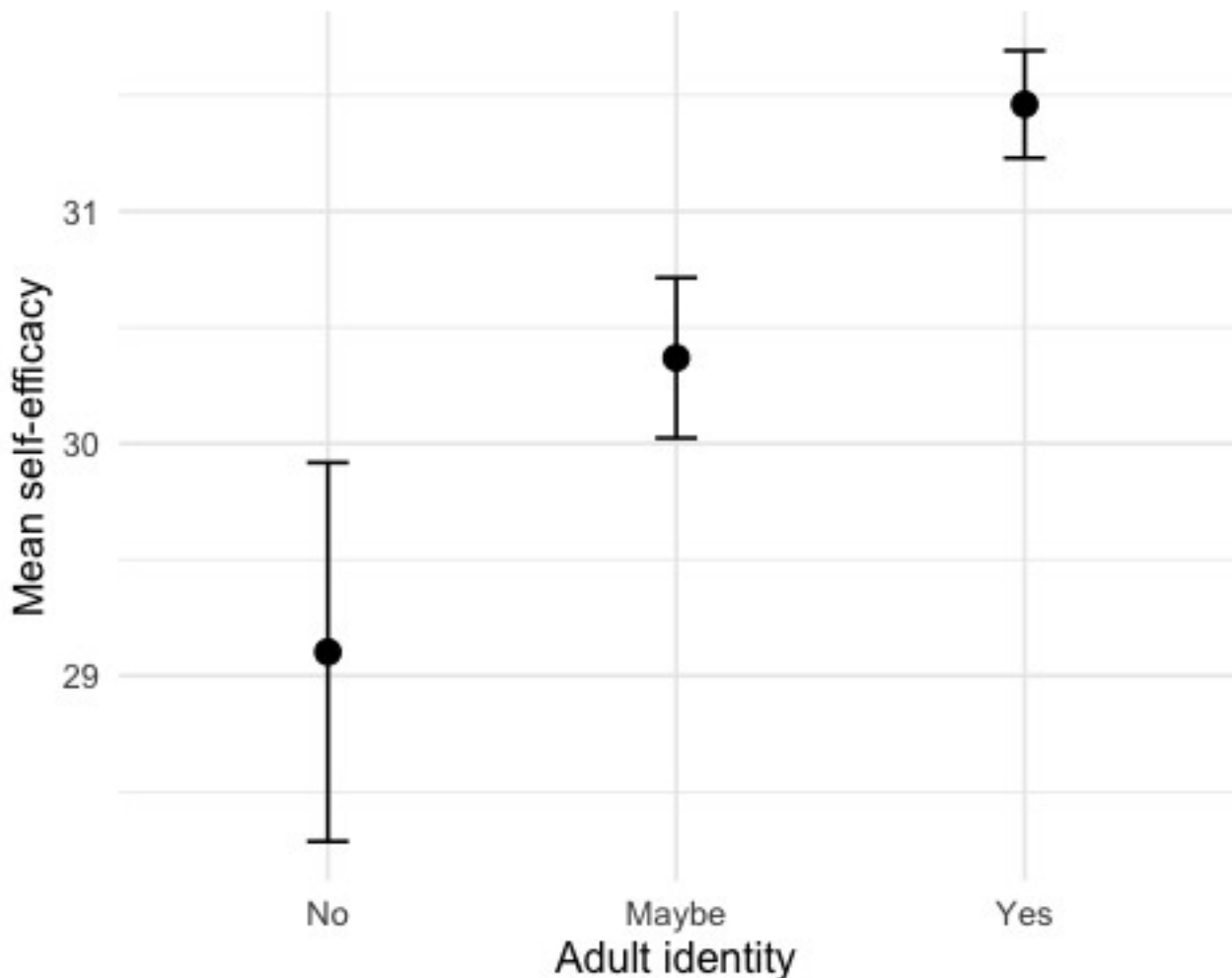


Figure 25: Mean self-efficacy by adult identity.

Both assumption tests were significant, so normality and homogeneity of variance were not met. Self-efficacy significantly differed across adult-identity groups, $F(2, 2069) = 27.12$, $p < .001$, $\eta^2 = .03$. The overall effect was small to medium. Participants who answered yes had higher self-efficacy ($M = 31.46$, $SE = 0.12$) than participants who

answered maybe ($M = 30.37$, $SE = 0.18$), $p < .001$, $d = 0.25$, and no ($M = 29.10$, $SE = 0.41$), $p < .001$, $d = 0.54$. The maybe group also had higher self-efficacy than the no group, $p = .005$, $d = 0.29$. Because both assumptions were violated, this demonstration result should be interpreted cautiously.

Research Question 3

$$H_0: \mu_{\text{no}} = \mu_{\text{somewhat}} = \mu_{\text{yes}}$$

Stress does not differ across the extent to which participants no longer live at home.

$$H_1: \text{at least one group mean differs}$$

Stress differs across at least one pair of living-at-home groups.

```
eammi <- eammi |>
  mutate(
    NoLongerHome = factor(
      NoLongerHome,
      levels = c(1, 2, 3),
      labels = c("No", "Somewhat", "Yes")
    )
  )

stress_home_data <- eammi |>
  drop_na(Stress, NoLongerHome)

stress_home_summary <- stress_home_data |>
  group_by(NoLongerHome) |>
  summarise(
    N = n(),
    Mean = mean(Stress),
    SD = sd(Stress),
    SE = SD / sqrt(N),
    Skewness = sample_skewness(Stress)
  )

stress_home_anova <- aov(
  Stress ~ NoLongerHome,
  data = stress_home_data
)

stress_home_variance_test <- bartlett.test(
  Stress ~ NoLongerHome,
  data = stress_home_data
)

stress_home_normality_test <- shapiro.test(
  residuals(stress_home_anova)
)

stress_home_eta_squared <- anova_eta_squared(
  stress_home_anova,
  "NoLongerHome"
)
```

```

stress_home_summary
# A tibble: 3 × 6
  NoLongerHome   N Mean   SD   SE Skewness
  <fct>         <int> <dbl> <dbl> <dbl>   <dbl>
1 No             829  32.7  4.04 0.140 -0.0929
2 Somewhat       656  32.8  3.91 0.153  0.397
3 Yes            565  32.6  4.22 0.178 -0.619

stress_home_variance_test

    Bartlett test of homogeneity of variances

data: Stress by NoLongerHome
Bartlett's K-squared = 3.5913, df = 2, p-value = 0.166

stress_home_normality_test

    Shapiro-Wilk normality test

data: residuals(stress_home_anova)
W = 0.97937, p-value < 2.2e-16

summary(stress_home_anova)
              Df Sum Sq Mean Sq F value Pr(>F)
NoLongerHome   2     16    7.807   0.476  0.621
Residuals    2047  33573   16.401
stress_home_eta_squared
[1] 0.0004648843

```

The Bartlett test was not significant, so the homogeneity-of-variance assumption was met. The Shapiro-Wilk test was significant, so the normality assumption was not met. Stress did not significantly differ across the extent to which participants no longer lived at home, $F(2, 2047) = 0.48$, $p = .62$, $\eta^2 < .001$. The overall effect was very small. Because the omnibus test was not significant, post hoc comparisons were not needed.

Chapter Takeaway

A one-way ANOVA tests whether at least one mean differs across three or more independent groups. A significant omnibus test is followed by comparisons such as Tukey HSD to identify the specific differences.

New R commands and patterns in this chapter:

- `aov()` fits an analysis-of-variance model.
- `summary()` prints the ANOVA table stored in that model.
- `residuals()` retrieves observed-minus-predicted differences from a model.
- `bartlett.test()` compares variances across three or more groups.
- `anova_eta_squared()` calculates eta squared from an ANOVA object and grouping-variable name.
- `TukeyHSD()` conducts adjusted pairwise comparisons.
- `tukey_cohens_d()` calculates Cohen's d for Tukey comparisons.
- `geom_errorbar()` adds confidence intervals to a graph.

Chapter 12: Analysis of Variance

Chapter 12 of *Exploring Statistics* introduces repeated-measures analysis of variance. A repeated-measures ANOVA compares three or more related means. You will use it when the same participants provide more than one score on the dependent variable, just as they do when rating several reasons for using social media.

Learning Goals

By the end of this chapter, you should be able to:

- identify when a repeated-measures ANOVA is appropriate;
- reshape data from wide format to long format;
- conduct and interpret a repeated-measures ANOVA;
- calculate partial eta squared;
- use paired post hoc tests to compare repeated-measures means; and
- write a short interpretation of repeated-measures ANOVA output.

Research Questions

1. Are emerging adults using social media more for some purposes than for others?
2. Do emerging adults report different amounts of social support from family, friends, and a special someone?

Research Question 1 is fully worked because reshaping repeated scores is new. Research Question 2 asks you to modify that pattern.

Before You Begin

You have already used `group_by()`, `summarise()`, `aov()`, and formula syntax. This chapter introduces `row_number()` and `pivot_longer()`, along with a model term that identifies repeated scores from the same participant.

Research Question 1: Reasons For Social Media Use

We want to determine if emerging adults are using social media more for some purposes than for others. Respondents rated the extent to which they use social media, from 1 = never to 5 = a lot, to maintain existing connections, make new connections, and get information.

Plan The Analysis

Variables used to answer Research Question 1

Variable	What it measures	Type	Role
SMmaintain	use to maintain existing connections	quantitative rating	repeated score
SMnewconnect	use to make new connections	quantitative rating	repeated score
SMinformation	use to get information	quantitative rating	repeated score

The repeated condition is the reason for using social media. The dependent variable is the social media use rating. Write the null and alternative hypotheses before continuing.

Get Ready

Open 12-anova-repeated.R from the scripts folder. Run the shared setup line, then run one section at a time.

```
source("scripts/_setup.R")
```

Repeated-measures ANOVA requires R to know which scores came from the same participant. Because the data do not include a participant ID, create one from the observation number and keep the three repeated scores.

```
social_media_wide <- eammi |>
  mutate(
    Participant = factor(row_number())
  ) |>
  select(
    Participant,
    SMmaintain,
    SMnewconnect,
    SMinformation
  ) |>
  drop_na(
    SMmaintain,
    SMnewconnect,
    SMinformation
  )
```

```
head(social_media_wide, n = 3)
```

```
# A tibble: 3 × 4
  Participant SMmaintain SMnewconnect SMinformation
  <fct>      <dbl>      <dbl>      <dbl>
1 1          3.8        4.75        4.5
2 2          2.4        1.25         3
3 3          3         3.25         3
```

`row_number()` gives each observation a number. `factor()` treats those numbers as participant labels rather than quantities.

In this wide version, each row is one participant. The participant's three social-media ratings appear in three separate variables. The first three rows therefore show three participants and one column for each repeated score.

Run And Investigate: Reshape The Data

The repeated-measures analysis needs the data in **long format**. In long data, each participant has one row for each rating instead of one row containing all three ratings. One variable identifies what the rating represents, and another variable contains the score. The code below reorganizes the wide data into that structure.

```
social_media_long <- social_media_wide |>
  pivot_longer(
    cols = c(SMmaintain, SMnewconnect, SMinformation),
    names_to = "Reason",
    values_to = "SMUse"
  ) |>
  mutate(
```

```

Reason = factor(
  Reason,
  levels = c("SMmaintain", "SMnewconnect", "SMinformation"),
  labels = c(
    "Maintain existing connections",
    "Make new connections",
    "Get information"
  )
)

head(social_media_long, n = 9)

# A tibble: 9 × 3
  Participant Reason          SMUse
  <fct>      <fct>          <dbl>
1 1          Maintain existing connections 3.8
2 1          Make new connections 4.75
3 1          Get information 4.5
4 2          Maintain existing connections 2.4
5 2          Make new connections 1.25
6 2          Get information 3
7 3          Maintain existing connections 3
8 3          Make new connections 3.25
9 3          Get information 3

```

`pivot_longer()` gathers the three score variables into two variables:

- Reason identifies which purpose the score represents; and
- SMUse contains the score itself.

We chose the names Reason and SMUse because they describe the information the new variables contain. These names did not exist in the original data, so different clear names could have been used.

Inside `factor()`, `levels` contains the original variable names exactly as they appear in `social_media_wide`: `SMmaintain`, `SMnewconnect`, and `SMinformation`. The codebook tells us what those variables measure. We used those meanings to write the readable labels. The first level receives the first label, the second level receives the second label, and so on.

Each participant now contributes three rows. The first nine rows show three participants with three rows apiece. For each participant, the Participant label repeats, Reason identifies the social-media purpose, and SMUse contains the score that previously appeared in a separate column. No scores were added or removed; R reorganized the same information into the structure needed for this analysis. Reshaping can look more dramatic than it is.

Compare the two tables produced by `head()`. Choose one participant in `social_media_wide`, note the three rating values, and find those same three numbers in that participant's rows in `social_media_long`. Only their organization has changed.

Run The Analysis

First summarize each reason.

```

social_media_summary <- social_media_long |>
  group_by(Reason) |>

```

```

summarise(
  N = n(),
  Mean = mean(SMUse),
  SD = sd(SMUse),
  SE = SD / sqrt(N),
  LowerCI = Mean - qt(0.975, df = N - 1) * SE,
  UpperCI = Mean + qt(0.975, df = N - 1) * SE
)

social_media_summary
# A tibble: 3 × 7
  Reason          N Mean   SD   SE LowerCI UpperCI
  <fct>      <int> <dbl> <dbl> <dbl>   <dbl>   <dbl>
1 Maintain existing connections 2072  3.09 0.858 0.0189    3.06    3.13
2 Make new connections          2072  3.08 0.964 0.0212    3.03    3.12
3 Get information                2072  3.36 1.04  0.0228    3.32    3.41

social_media_anova <- aov(
  SMUse ~ Reason + factor(Participant),
  data = social_media_long
)

summary(social_media_anova)
              Df Sum Sq Mean Sq F value Pr(>F)
Reason          2    106   53.14 115.444 <2e-16 ***
factor(Participant) 2071  3776    1.82   3.961 <2e-16 ***
Residuals      4142   1907    0.46
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

SMUse ~ Reason identifies the outcome and repeated condition. Adding factor(Participant) accounts for stable differences among participants who contributed multiple scores. Fitting this model may take about 30 seconds because R must account for more than 2,000 participant labels. `summary(social_media_anova)` only displays the completed model and should appear quickly. In the ANOVA table, the Reason row tests whether at least one mean differs.

```

social_media_eta_squared <- repeated_eta_squared(
  social_media_anova,
  "Reason"
)

```

```

social_media_eta_squared
[1] 0.05280006

```

`repeated_eta_squared()` is supplied by the setup file. Give it the fitted ANOVA object and the name of the repeated condition in quotation marks. The function calculates partial eta squared, which describes the proportion of variability associated with the repeated condition after participant differences are accounted for.

Because the overall ANOVA is significant, compare each pair of repeated means.

```

social_media_pairwise <- repeated_pairwise_tests(
  social_media_long,
  SMUse,
  Reason,
  Participant,

```

```
anova_model = social_media_anova
)
```

```
social_media_pairwise
```

```
# A tibble: 3 × 4
```

Comparison <chr>	MeanDifference <dbl>	p <dbl>	CohensD <dbl>
1 Maintain existing connections - Make new conn...	0.0165	6.41e- 1	0.0244
2 Maintain existing connections - Get informati...	-0.269	5.76e-11	0.396
3 Make new connections - Get information	-0.285	5.76e-11	0.420

`repeated_pairwise_tests()` is supplied by the setup file. It compares each pair of scores from the same participants and reports the mean difference, Tukey-adjusted p value, and Cohen's d . These are paired comparisons because each comparison uses two scores from the same participant.

Inspect And Interpret

Check the output in this order:

1. Confirm that the three condition means are present.
2. Read the Reason row of the ANOVA table.
3. Inspect partial eta squared.
4. Use the pairwise p values to identify which pairs differ.
5. Use Cohen's d to describe the pairwise effects.

The mean was highest for getting information. Getting information was significantly higher than maintaining existing connections or making new connections. Maintaining existing connections and making new connections did not significantly differ.

```
social_media_summary |>
  ggplot(aes(x = Reason, y = Mean)) +
  geom_errorbar(
    aes(ymin = LowerCI, ymax = UpperCI),
    width = 0.12
  ) +
  geom_point(size = 3) +
  coord_flip() +
  labs(
    x = "Reason for using social media",
    y = "Mean social media use"
  )
```

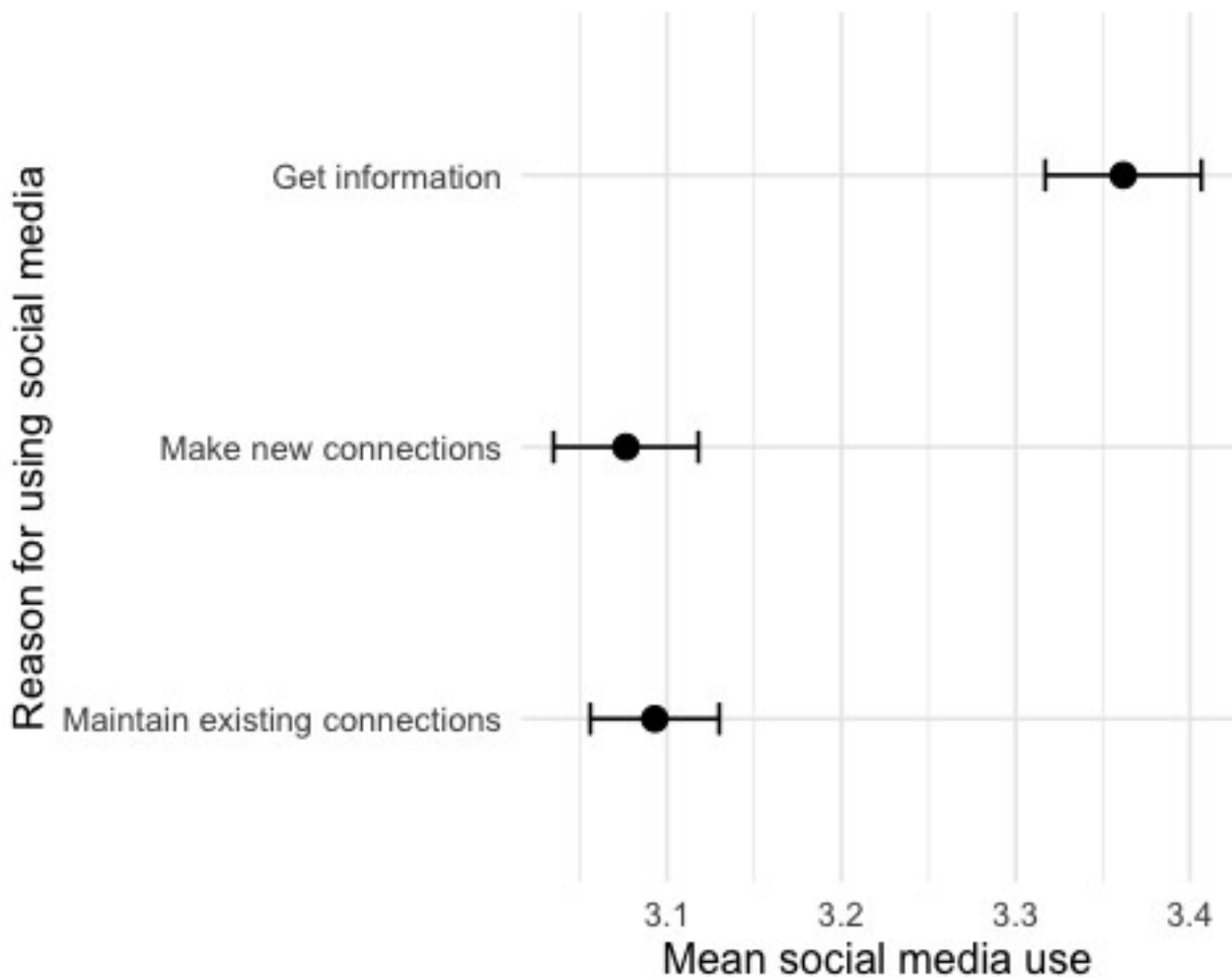


Figure 26: Mean social media use by reason for use.

Write an interpretation that answers the research question and reports the ANOVA, effect size, means, and pairwise comparisons.

Research Question 2: Sources Of Social Support

Do respondents report different amounts of social support from family, friends, and a special someone? Each source was rated from 1 = very strongly disagree to 7 = very strongly agree.

Complete The Analysis

Use Research Question 1 as your model. Replace the three social media variables with `SupportFamily`, `SupportFriends`, and `SupportSpecial`. Name the repeated condition `SupportSource` and the score `Support`.

Before running copied code, mark each place where you must change:

- the wide and long object names;
- the three repeated variables;
- the condition and score names;

- the model formula; and
- the data, score, condition, and participant names supplied to the pairwise function.

Write the independent variable, dependent variable, null hypothesis, and alternative hypothesis. Then produce the descriptive statistics, repeated-measures ANOVA, partial eta squared, paired comparisons, Cohen's d values, and a written interpretation.

Check Your Work

Research Question 1

Independent variable: reason for using social media, with three repeated conditions. Dependent variable: social media use score.

$$H_0: \mu_{\text{maintain}} = \mu_{\text{new connections}} = \mu_{\text{information}}$$

The alternative hypothesis states that at least one repeated-measures mean differs.

A repeated-measures ANOVA revealed that participants significantly differed in their reasons for using social media, $F(2, 4142) = 115.44, p < .001$, partial $\eta^2 = .05$. Use was highest for getting information ($M = 3.36, SE = 0.02$), followed by maintaining existing connections ($M = 3.09, SE = 0.02$), and making new connections ($M = 3.08, SE = 0.02$). Getting information was significantly higher than maintaining existing connections, $p < .001, d = 0.40$, and making new connections, $p < .001, d = 0.42$. Maintaining existing connections and making new connections did not significantly differ, $p = .64, d = 0.02$.

Research Question 2

Independent variable: source of social support, with family, friends, and special someone as repeated conditions. Dependent variable: social support score.

$$H_0: \mu_{\text{family}} = \mu_{\text{friends}} = \mu_{\text{special someone}}$$

The alternative hypothesis states that at least one source-of-support mean differs.

```
support_wide <- eammi |>
  mutate(Participant = factor(row_number())) |>
  select(Participant, SupportFamily, SupportFriends, SupportSpecial) |>
  drop_na(SupportFamily, SupportFriends, SupportSpecial)

support_long <- support_wide |>
  pivot_longer(
    cols = c(SupportFamily, SupportFriends, SupportSpecial),
    names_to = "SupportSource",
    values_to = "Support"
  ) |>
  mutate(
    SupportSource = factor(
      SupportSource,
      levels = c("SupportFamily", "SupportFriends", "SupportSpecial"),
      labels = c("Family", "Friends", "Special someone")
    )
  )
```

```

support_summary <- support_long |>
  group_by(SupportSource) |>
  summarise(
    N = n(),
    Mean = mean(Support),
    SD = sd(Support),
    SE = SD / sqrt(N),
    LowerCI = Mean - qt(0.975, df = N - 1) * SE,
    UpperCI = Mean + qt(0.975, df = N - 1) * SE
  )

support_summary
# A tibble: 3 × 7
  SupportSource      N Mean   SD   SE LowerCI UpperCI
  <fct>          <int> <dbl> <dbl> <dbl>   <dbl>   <dbl>
1 Family          2071  5.46  1.32 0.0291    5.40    5.51
2 Friends          2071  5.53  1.31 0.0289    5.48    5.59
3 Special someone  2071  5.61  1.60 0.0351    5.54    5.68

support_anova <- aov(
  Support ~ SupportSource + factor(Participant),
  data = support_long
)

summary(support_anova)

              Df Sum Sq Mean Sq F value    Pr(>F)
SupportSource      2      25  12.671   11.471 1.08e-05 ***
factor(Participant) 2070   7904   3.818    3.457 < 2e-16 ***
Residuals         4140   4573   1.105
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

support_eta_squared <- repeated_eta_squared(support_anova, "SupportSource")
support_pairwise <- repeated_pairwise_tests(
  support_long,
  Support,
  SupportSource,
  Participant,
  anova_model = support_anova
)

support_eta_squared
[1] 0.005511009

support_pairwise
# A tibble: 3 × 4
  Comparison      MeanDifference      p CohensD
  <chr>          <dbl>      <dbl>   <dbl>
1 Family - Friends -0.0777 0.00798 0.0739
2 Family - Special someone -0.156 0.0000563 0.149
3 Friends - Special someone -0.0787 0.0588 0.0749

support_summary |>
  ggplot(aes(x = SupportSource, y = Mean)) +
  geom_errorbar(
    aes(ymin = LowerCI, ymax = UpperCI),
    width = 0.12
  )

```

```

) +
geom_point(size = 3) +
coord_flip() +
labs(
  x = "Source of social support",
  y = "Mean social support"
)

```

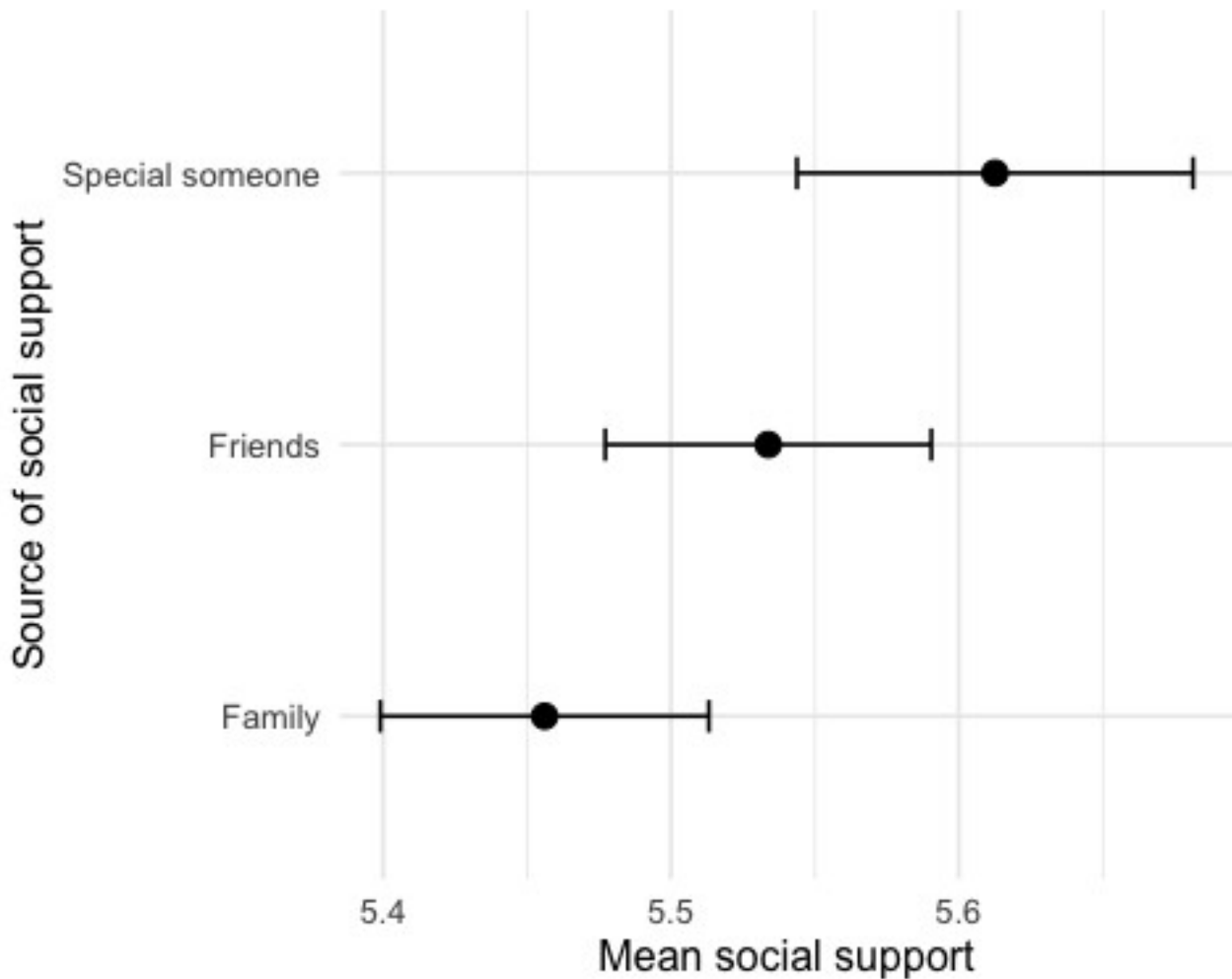


Figure 27: Mean social support by source.

A repeated-measures ANOVA revealed significant differences in reported support, $F(2, 4140) = 11.47$, $p < .001$, partial $\eta^2 = .006$. Support was highest from a special someone ($M = 5.61$, $SE = 0.04$), followed by friends ($M = 5.53$, $SE = 0.03$), and family ($M = 5.46$, $SE = 0.03$). Friends were significantly higher than family, $p = .008$, $d = 0.07$. A special someone was higher than family, $p < .001$, $d = 0.15$. Support from a special someone and friends did not significantly differ, $p = .059$, $d = 0.07$.

**Careful interpretation**

These are measured responses from the same participants. We can say reported support differs across sources, but not that the source caused the difference.

Chapter Takeaway

A repeated-measures ANOVA compares three or more related means and accounts for the fact that each participant contributes multiple scores. Wide data store repeated scores in separate variables; long data identify the condition and score in separate columns.

R Skills Practiced

- `row_number()` creates a participant number for each observation.
- `pivot_longer()` reshapes repeated scores from wide to long format.
- `factor(Participant)` identifies participant labels in the model.
- `repeated_eta_squared()` calculates partial eta squared.
- `repeated_pairwise_tests()` is supplied by the setup file and compares each pair of repeated scores.

Chapter 13: Two-Way Independent Analysis of Variance

Chapter 13 of *Exploring Statistics* introduces two-way independent analysis of variance. A two-way ANOVA fits three questions into one model: one main effect for each independent variable and an interaction asking whether the pattern for one independent variable depends on the level of the other.

Learning Goals

By the end of this chapter, you should be able to:

- identify the dependent variable and both independent variables;
- distinguish main effects from interactions;
- check the assumptions of a two-way ANOVA;
- conduct and interpret a two-way ANOVA;
- use post hoc tests when they are needed; and
- write a cautious conclusion.

Research Questions

1. Does satisfaction with life differ by political views, adult status, or the combination of political views and adult status?
2. Does social support differ by political views, adult status, or the combination of political views and adult status?

Before You Begin

You will reuse `aov()`, `TukeyHSD()`, `bartlett.test()`, `shapiro.test()`, grouped summaries, and formula syntax. New patterns include `predictor_1 * predictor_2`, combining groups for an assumption check, and an interaction plot.

Research Question 1: Satisfaction With Life, Political Views, And Adult Status

Previous research has found that conservatives tend to report greater satisfaction with life than liberals. We will examine whether that pattern appears in EAMMi2 and whether it depends on adult status.

Plan The Analysis

Variables used to answer Research Question 1

Variable	What it measures	Type	Role
SWLS	satisfaction with life	quantitative	dependent variable
PoliticsDichotomous	liberal or conservative views	categorical	independent variable
Adult	adult status	categorical	independent variable

Before running the analysis, write a null and alternative hypothesis for each of the three tests:

1. **Political-views main effect:** Do liberals and conservatives differ in satisfaction with life?
2. **Adult-status main effect:** Does satisfaction with life differ among people who answered no, maybe, or yes?
3. **Interaction:** Does the difference between liberals and conservatives depend on adult status?

The interaction question asks whether the political-views difference is the same across the three adult-status groups.

Get Ready

Open `13-two-way-anova.R`, run the setup line, and then run one section at a time.

```
source("scripts/_setup.R")

eammi <- eammi |>
  mutate(
    PoliticsDichotomous = factor(
      PoliticsDichotomous,
      levels = c(1, 2),
      labels = c("Liberal", "Conservative")
    ),
    Adult = factor(
      Adult,
      levels = c(3, 2, 1),
      labels = c("No", "Maybe", "Yes")
    )
  )

swls_data <- eammi |>
  drop_na(SWLS, PoliticsDichotomous, Adult)
```

Run The Model

The formula for a two-way ANOVA shows the dependent variable on the left and both independent variables on the right.

```
swls_anova <- aov(
  SWLS ~ PoliticsDichotomous * Adult,
  data = swls_data,
  contrasts = list(
    PoliticsDichotomous = contr.sum,
    Adult = contr.sum
  )
)
```

In `SWLS ~ PoliticsDichotomous * Adult`:

- `SWLS` is the dependent variable;
- `PoliticsDichotomous` and `Adult` are the two independent variables; and
- `*` tells R to test both main effects and the interaction between them.

The model is stored as `swls_anova`. The `contrasts` lines tell R how to compare the categories when the six groups contain different numbers of participants.

Check The Assumptions

```
bartlett.test(
  SWLS ~ interaction(PoliticsDichotomous, Adult),
  data = swls_data
)
```

Bartlett test of homogeneity of variances

```
data: SWLS by interaction(PoliticsDichotomous, Adult)
Bartlett's K-squared = 4.12, df = 5, p-value = 0.5323
```

`interaction()` combines the independent variables into six groups. Here, Bartlett's p value is greater than .05, so the homogeneity assumption is met.

```
shapiro.test(residuals(swls_anova))
```

Shapiro-Wilk normality test

```
data: residuals(swls_anova)
W = 0.97518, p-value = 2.192e-14
```

Residuals are differences between observed and model-predicted scores. `residuals(swls_anova)` retrieves those differences from the model fitted above. They differ significantly from normal. Because the sample is large and SWLS is not severely skewed, we proceed but remember this result.

Run The Analysis

```
swls_cell_summary <- swls_data |>
  group_by(Adult, PoliticsDichotomous) |>
  summarise(
    N = n(),
    Mean = mean(SWLS),
    SD = sd(SWLS),
    SE = SD / sqrt(N),
    .groups = "drop"
  )
```

```
swls_cell_summary
# A tibble: 6 × 6
  Adult PoliticsDichotomous      N Mean   SD   SE
  <fct> <fct>          <int> <dbl> <dbl> <dbl>
1 No    Liberal           81  19.2  7.20 0.800
2 No    Conservative       33  22.6  6.42 1.12
3 Maybe Liberal        268  21.8  6.59 0.402
4 Maybe Conservative    93  24.2  6.77 0.702
5 Yes   Liberal          547  21.9  7.20 0.308
6 Yes   Conservative     312  24.2  6.73 0.381
```

Crossing two political categories with three adult-status categories creates six groups. The summary table gives the sample size, mean, standard deviation, and standard error for each group.

The ANOVA model was already fitted and stored as `swls_anova`. The next code does not specify the model again. Instead, `two_way_anova_table()` takes that fitted model and organizes its results into a table containing the two main effects, the interaction, and partial eta squared. This function is supplied by the setup file.

```
swls_anova_table <- two_way_anova_table(swls_anova)
swls_anova_table
```

	Effect	Df	SumOfSquares	F	p
1	PoliticsDichotomous	1	1041.49049	21.7381907	3.439891e-06
2	Adult	2	390.25401	4.0727286	1.724403e-02
3	PoliticsDichotomous:Adult	2	22.93445	0.2393462	7.871763e-01

```
PartialEtaSquared
1      0.0161054869
2      0.0060962353
3      0.0003603312
```

At first glance, the output table can seem confusing. It provides the results of three separate tests:

- The PoliticsDichotomous row tests whether satisfaction with life differs between liberals and conservatives.
- The Adult row tests whether satisfaction with life differs across the no, maybe, and yes groups.
- The PoliticsDichotomous:Adult row tests whether the political-views difference depends on adult status. The colon (:) identifies the interaction in the output.

Inspect And Interpret

Read a two-way table in this order:

1. Inspect the interaction.
2. If the interaction is not significant, interpret each main effect.
3. Use post hoc comparisons for a significant factor with more than two levels.
4. Compare the table with the cell means and interaction plot.

The interaction is not significant, so the political difference does not appear to depend on adult status. Both main effects are significant. Conservatives reported higher satisfaction than liberals. Adult status has three levels, so compare its groups.

```
swls_adult_tukey <- TukeyHSD(swls_anova, "Adult")
swls_adult_tukey
```

```
Tukey multiple comparisons of means
95% family-wise confidence level
```

```
Fit: aov(formula = SWLS ~ PoliticsDichotomous * Adult, data = swls_data, contrasts =
list(PoliticsDichotomous = contr.sum, Adult = contr.sum))
```

```
$Adult
      diff      lwr      upr    p adj
Maybe-No 2.32061155 0.5758061 4.065417 0.0052359
Yes-No    2.34367025 0.7247940 3.962546 0.0020269
Yes-Maybe 0.02305871 -0.9956170 1.041734 0.9984460
```

```
swls_adult_pairwise_d <- tukey_cohens_d(
  swls_anova,
  swls_adult_tukey,
  "Adult"
)
swls_adult_pairwise_d
```

```
# A tibble: 3 × 2
  Comparison CohensD
  <chr>      <dbl>
1 Maybe-No   0.335
2 Yes-No     0.339
3 Yes-Maybe  0.00333
```

Participants answering yes or maybe reported higher satisfaction than those answering no. The yes and maybe groups did not differ.

```
swls_cell_summary |>
  ggplot(aes(
    x = Adult,
    y = Mean,
    group = PoliticsDichotomous,
    color = PoliticsDichotomous,
    linetype = PoliticsDichotomous,
    shape = PoliticsDichotomous
  )) +
  geom_point() +
  geom_line() +
  labs(
    x = "Adult status",
    y = "Mean satisfaction with life",
    color = "Political views",
    linetype = "Political views",
    shape = "Political views"
  )
```

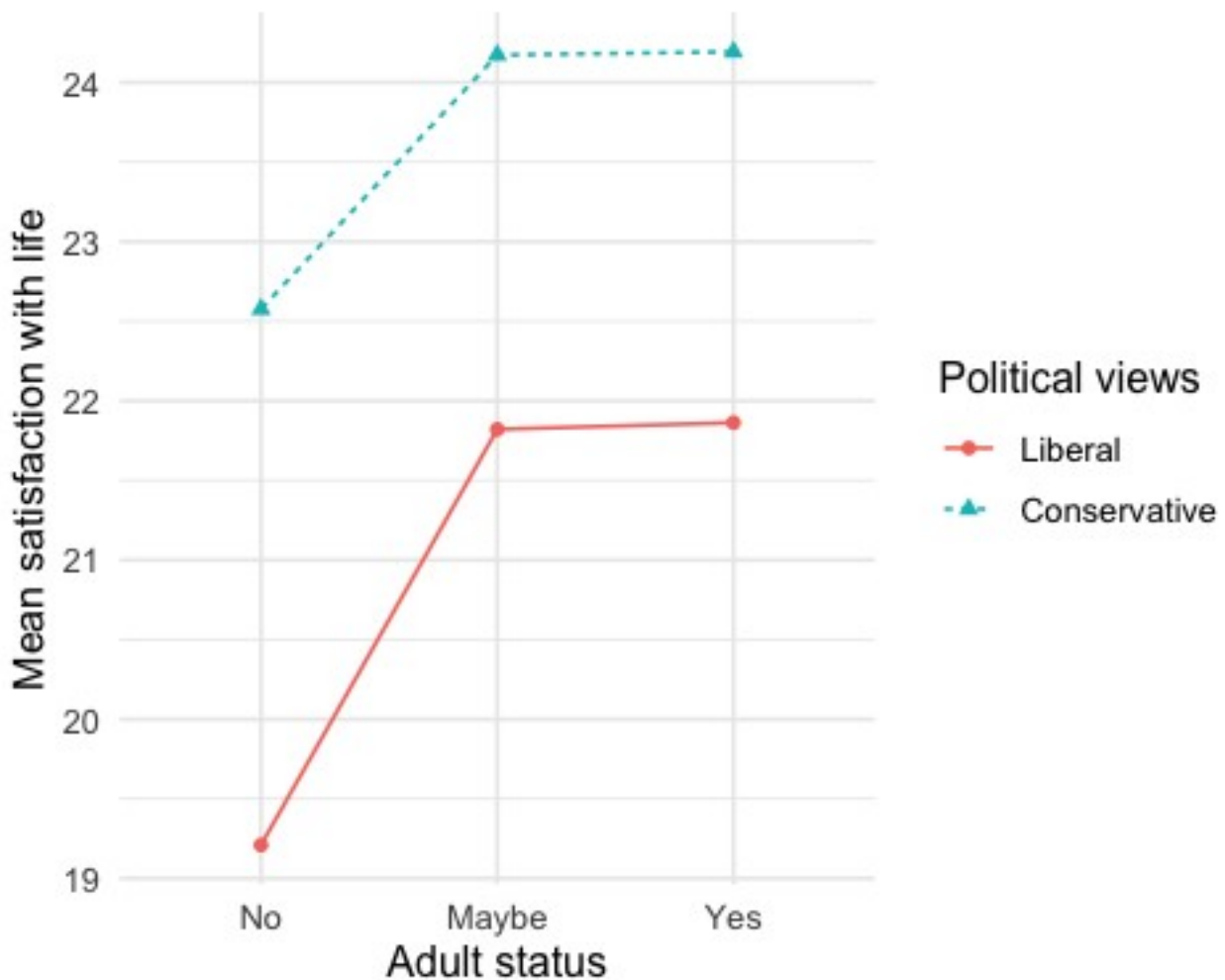


Figure 28: Mean satisfaction with life by political views and adult status.

Produce a conclusion that reports all three tests and the necessary post hoc comparisons.

Research Question 2: Social Support, Political Views, And Adult Status

Use the same independent variables and change the dependent variable to SocialSupport.

Complete The Analysis

Copy the Research Question 1 pattern and replace every SWLS-specific object and variable name. Check both assumptions, calculate the six cell summaries, run the model and two-way table, decide whether post hoc tests are needed, and write an interpretation explaining how the assumptions affect confidence in the conclusions.

Before running the model, write separate null and alternative hypotheses for the political-views main effect, adult-status main effect, and interaction. Afterward, investigate the table in the same interaction-first order.

Check Your Work

Research Question 1

The independent variables are political views and adult status; the dependent variable is satisfaction with life.

The null hypotheses state that conservatives and liberals do not differ, the three adult-status means do not differ, and political views and adult status do not interact:

$$H_0: \mu_{\text{conservative}} = \mu_{\text{liberal}}$$

$$H_0: \mu_{\text{yes}} = \mu_{\text{maybe}} = \mu_{\text{no}}$$

The alternative hypotheses state that political groups differ, at least one adult-status mean differs, and political views and adult status interact.

A 2 x 3 two-way independent ANOVA found a significant political main effect: conservatives reported greater satisfaction than liberals, $F(1, 1328) = 21.74$, $p < .001$, partial $\eta^2 = .02$. Adult status also had a significant main effect, $F(2, 1328) = 4.07$, $p = .017$, partial $\eta^2 = .006$. Yes exceeded no, $p = .002$, $d = 0.34$; maybe exceeded no, $p = .005$, $d = 0.34$; and yes did not differ from maybe, $p = .998$, $d = 0.003$. The interaction was not significant, $F(2, 1328) = 0.24$, $p = .79$, partial $\eta^2 < .001$.

Research Question 2

```
social_support_data <- eammi |>
  drop_na(SocialSupport, PoliticsDichotomous, Adult)

bartlett.test(
  SocialSupport ~ interaction(PoliticsDichotomous, Adult),
  data = social_support_data
)
```

Bartlett test of homogeneity of variances

```
data: SocialSupport by interaction(PoliticsDichotomous, Adult)
Bartlett's K-squared = 16.588, df = 5, p-value = 0.005352
```

```
social_support_anova <- aov(
  SocialSupport ~ PoliticsDichotomous * Adult,
  data = social_support_data,
  contrasts = list(
    PoliticsDichotomous = contr.sum,
    Adult = contr.sum
  )
)
```

```
shapiro.test(residuals(social_support_anova))
```

Shapiro-Wilk normality test

```
data: residuals(social_support_anova)
W = 0.92614, p-value < 2.2e-16
```

```

social_support_cell_summary <- social_support_data |>
  group_by(Adult, PoliticsDichotomous) |>
  summarise(
    N = n(), Mean = mean(SocialSupport), SD = sd(SocialSupport),
    SE = SD / sqrt(N), .groups = "drop"
  )

social_support_anova_table <- two_way_anova_table(social_support_anova)
social_support_cell_summary

# A tibble: 6 × 6
  Adult PoliticsDichotomous      N Mean    SD    SE
  <fct> <fct>                <int> <dbl> <dbl> <dbl>
1 No    Liberal              81  62.7  14.4  1.60
2 No    Conservative         33   68    11.6  2.02
3 Maybe Liberal          268  67.5  11.5  0.704
4 Maybe Conservative      93  66.8  13.9  1.44
5 Yes   Liberal           547  66.4  13.1  0.559
6 Yes   Conservative       312  68.1  14.4  0.814

social_support_anova_table

      Effect Df SumOfSquares      F      p
1 PoliticsDichotomous 1 650.2719 3.724754 0.0538243
2 Adult 2 308.9036 0.884699 0.4130816
3 PoliticsDichotomous:Adult 2 663.0691 1.899028 0.1501204
PartialEtaSquared
1 0.002796940
2 0.001330605
3 0.002851826

```

The homogeneity assumption was not met and residuals differed significantly from normal, so interpret cautiously. The interaction was not significant, $F(2, 1328) = 1.90$, $p = .15$, partial $\eta^2 = .003$. Neither adult status, $F(2, 1328) = 0.88$, $p = .41$, partial $\eta^2 = .001$, nor political views, $F(1, 1328) = 3.72$, $p = .054$, partial $\eta^2 = .003$, had a significant main effect. Post hoc tests are unnecessary.

```

social_support_cell_summary |>
  ggplot(aes(
    x = Adult,
    y = Mean,
    group = PoliticsDichotomous,
    color = PoliticsDichotomous,
    linetype = PoliticsDichotomous,
    shape = PoliticsDichotomous
  )) +
  geom_point() +
  geom_line() +
  labs(
    x = "Adult status",
    y = "Mean social support",
    color = "Political views",
    linetype = "Political views",
    shape = "Political views"
  )

```

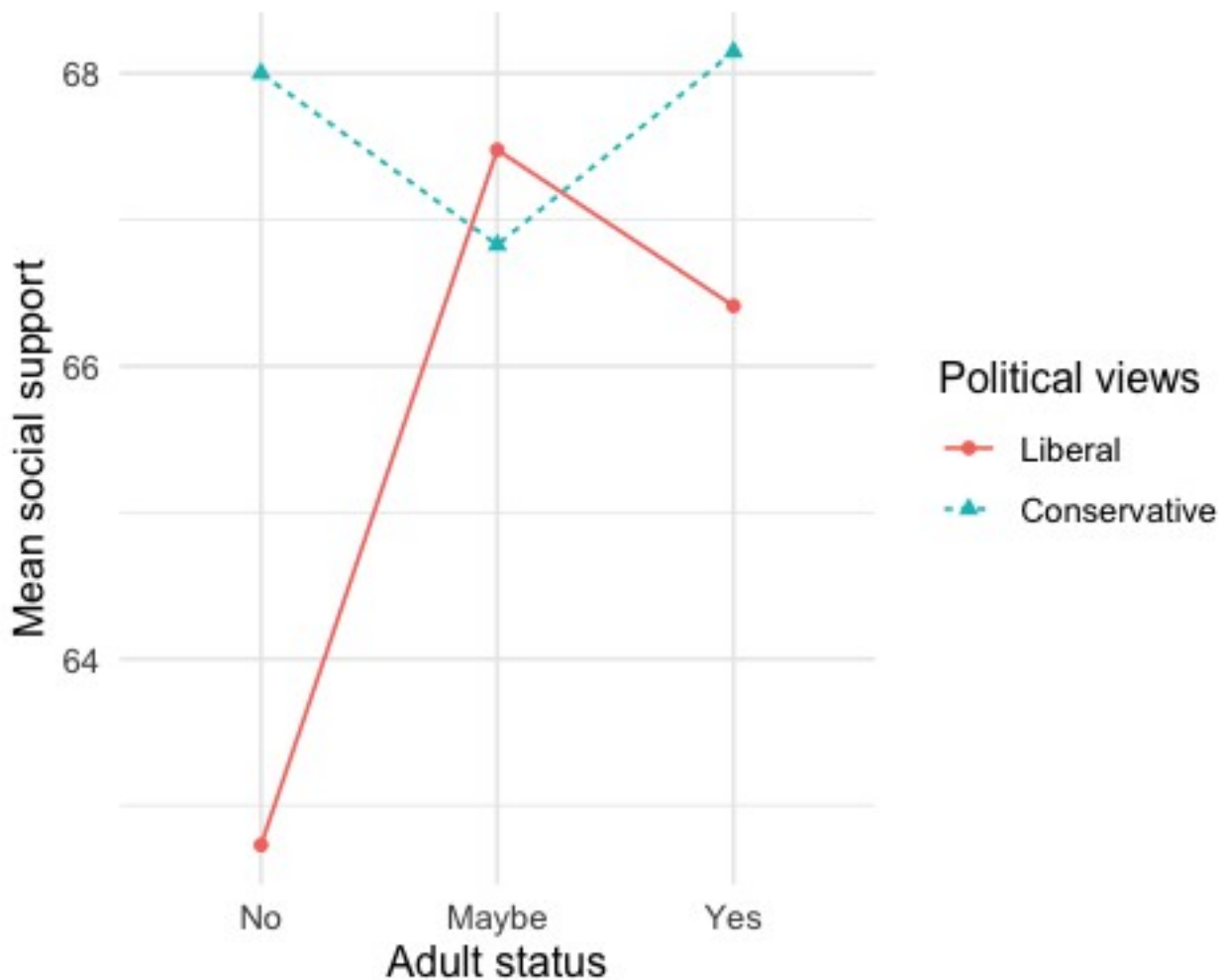


Figure 29: Mean social support by political views and adult status.

Chapter Takeaway

A two-way independent ANOVA tests two main effects and one interaction. Interpret the interaction first because a meaningful interaction changes how the main effects should be understood.

R Skills Practiced

- `bartlett.test()` and `shapiro.test()` repeat the assumption checks introduced in Chapters 10 and 11.
- `interaction()` combines grouping variables for an assumption check.
- `predictor_1 * predictor_2` requests both main effects and the interaction.
- `two_way_anova_table()` is supplied by the setup file and reports the two-way tests and partial eta squared from a fitted model.
- `geom_line()` connects means in an interaction plot.

Chapter 14: Chi-Square Tests

Chapter 14 of *Exploring Statistics* introduces two tests for categorical variables. A chi-square test of independence asks whether two categorical variables are related. A goodness-of-fit test compares the observed counts for one categorical variable with expected proportions.

Learning Goals

By the end of this chapter, you should be able to:

- choose between a chi-square test of independence and a goodness-of-fit test;
- create and compare observed and expected frequencies;
- conduct and interpret both types of chi-square test in R;
- calculate Cramer's V for a test of independence; and
- write a conclusion that answers the research question.

Research Questions

1. Is the extent to which an emerging adult is still living at home related to the extent to which they categorize themselves as an adult?
2. Does gender representation in the EAMMi2 sample differ from U.S. adult population values?
3. Is the extent to which an emerging adult considers themselves financially independent related to the extent to which they categorize themselves as an adult?
4. Are participants in the EAMMi2 sample equally likely to identify as Republican or Democrat? What about liberal or conservative?

Research Questions 1 and 2 provide one worked example of each test. Research Questions 3 and 4 ask you to modify the matching pattern.

Before You Begin

Use this decision:

- two categorical variables and a question about their relationship: test of independence;
- one categorical variable compared with stated proportions: goodness-of-fit test.

Research Question 1: Living At Home And Adult Status

Is the extent to which an emerging adult is still living at home related to the extent to which they categorize themselves as an adult?

Plan The Analysis

Variables used to answer Research Question 1

Variable	What it measures	Type	Role
NoLongerHome	extent moved out of parents' home	categorical	first variable
Adult	adult status	categorical	second variable

Variable	What it measures	Type	Role
----------	------------------	------	------

Because the question asks whether two categorical variables are related, use a chi-square test of independence.

Get Ready

Open 14-chi-square.R, run the setup line, and then run one section at a time.

```
source("scripts/_setup.R")
eammi <- eammi |>
  mutate(
    NoLongerHome = factor(
      NoLongerHome,
      levels = c(1, 2, 3),
      labels = c("No", "Somewhat", "Yes")
    ),
    Adult = factor(
      Adult,
      levels = c(3, 2, 1),
      labels = c("No", "Maybe", "Yes")
    )
  )

home_adult_data <- eammi |>
  drop_na(NoLongerHome, Adult)
```

NoLongerHome uses a double negative: No means still living at home, while Yes means no longer living at home.

Predict

If moving out and adult status are related, which combination should occur more often than independence would predict: no longer living at home and saying yes to being an adult, or no longer living at home and saying no?

Record your prediction.

Run The Analysis

```
home_adult_table <- table(
  home_adult_data$NoLongerHome,
  home_adult_data$Adult
)
home_adult_table
```

	No	Maybe	Yes
No	70	223	538
Somewhat	47	198	411
Yes	28	92	444

table() creates the observed contingency table.

```
home_adult_chi <- chisq.test(home_adult_table)
home_adult_chi
```

Pearson's Chi-squared test

```
data: home_adult_table
X-squared = 44.491, df = 4, p-value = 5.073e-09
home_adult_chi$expected
```

	No	Maybe	Yes
No	58.74939	207.8513	564.3993
Somewhat	46.37738	164.0800	445.5427
Yes	39.87323	141.0687	383.0580

`chisq.test()` compares observed counts with the counts expected if the variables were independent. `$expected` retrieves those expected counts from the saved test.

```
home_adult_v <- cramers_v(home_adult_chi)
home_adult_v
[1] 0.1041449
```

Cramer's V is appropriate for this 3 x 3 table. Guidelines are about .07 for small, .21 for medium, and .35 for large.

Investigate And Interpret

First decide whether the test is significant and describe the size of Cramer's V. Then compare each observed cell with the matching expected cell. Look for a consistent pattern rather than listing numbers without interpretation.

The relationship is significant and small. Those still living at home were more likely than expected to say no to being an adult. Those no longer living at home were more likely than expected to say yes and less likely to say maybe or no. Overall, it appears that participants who were less likely to be living at home were also more likely to describe themselves as adults.

Produce a paragraph that reports the test, effect size, and observed-versus-expected pattern.

Research Question 2: Gender Representation

According to the U.S. Census Bureau, in 2019 (close to the time of EAMMi2 data collection) the U.S. population was 51% female, 47% male, and 2% non-binary. Here, we are going to use our EAMMi2 data file to determine if our sample represents a similar population. Before continuing, decide whether the appropriate chi-square test should be a test of independence or goodness of fit.

Plan And Predict

Because we are comparing proportions in our data set with population-level proportions, use a goodness-of-fit test. Before running it, predict which category appears most over-represented in the sample.

```
eammi <- eammi |>
  mutate(
    Gender = factor(
      Gender,
      levels = c(3, 2, 1),
      labels = c("Another identity", "Female", "Male")
    )
  )

gender_data <- eammi |>
```

```
drop_na(Gender)
```

```
gender_counts <- table(gender_data$Gender)
gender_counts
```

```
Another identity      Female      Male
           30           1524           519
```

```
prop.table(gender_counts)
```

```
Another identity      Female      Male
    0.01447178    0.73516643    0.25036179
```

`prop.table()` converts the observed counts to proportions.

Run And Investigate

```
gender_chi <- chisq.test(
  gender_counts,
  p = c(.02, .51, .47)
)
gender_chi
```

Chi-squared test for given probabilities

```
data:  gender_counts
X-squared = 422.02, df = 2, p-value < 2.2e-16
```

The expected proportions must match the alphabetical category order in `gender_counts`: another identity, female, male. The significant result shows that representation differs from the comparison values. Females are over-represented; males and participants selecting another identity are under-represented.

Research Question 3: Financial Independence And Adult Status

Is financial independence related to whether participants categorize themselves as adults?

Complete The Analysis

Use Research Question 1 as the model. Replace `NoLongerHome` with `FinanciallyInd`, create new object names, and retain `Adult`. Produce the observed table, test, expected counts, Cramer's V, and written conclusion.

Before running copied code, identify every place the old variable or object name appears. Predict which financial-independence category should be more likely than expected to say yes to being an adult.

Research Question 4: Party Identification And Political Views

Are participants equally likely to identify as Republican or Democrat? What about liberal or conservative?

Complete The Analysis

Use Research Question 2 as the model. Conduct separate goodness-of-fit tests for `PartyDichotomous` and `PoliticsDichotomous`, each with expected proportions of .50 and .50. Create separate object names and interpret both results.

Check Your Work

Research Question 1

A 3 x 3 chi-square test found a small, significant relationship between living at home and adult status, ($\chi^2(4) = 44.49, p < .001$, Cramer's $V = .10$). Those still living at home were less likely than expected to say yes or maybe and more likely to say no. Those somewhat living at home were less likely to say yes, more likely to say maybe, and about as likely to say no. Those no longer living at home were more likely to say yes and less likely to say maybe or no. In short, it appears that participants who were less likely to be living at home were also more likely to describe themselves as adults.

Research Question 2

A goodness-of-fit test showed that gender representation differed from the comparison values, ($\chi^2(2) = 422.02, p < .001$). Females were over-represented (73.5% versus 51%), while males (25.0% versus 47%) and participants selecting another identity (1.4% versus 2%) were under-represented.

Research Question 3

```
eammi <- eammi |>
  mutate(
    FinanciallyInd = factor(
      FinanciallyInd,
      levels = c(1, 2, 3),
      labels = c("No", "Somewhat", "Yes")
    )
  )

financial_adult_data <- eammi |>
  drop_na(FinanciallyInd, Adult)

financial_adult_table <- table(
  financial_adult_data$FinanciallyInd,
  financial_adult_data$Adult
)

financial_adult_chi <- chisq.test(financial_adult_table)
financial_adult_v <- cramers_v(financial_adult_chi)

financial_adult_table
```

	No	Maybe	Yes
No	83	245	449
Somewhat	50	225	688
Yes	13	43	250

```
financial_adult_chi

Pearson's Chi-squared test

data: financial_adult_table
X-squared = 73.337, df = 4, p-value = 4.478e-15

financial_adult_chi$expected
```

	No	Maybe	Yes
No	55.44575	194.81965	526.7346
Somewhat	68.71848	241.45601	652.8255
Yes	21.83578	76.72434	207.4399

```
financial_adult_v
```

```
[1] 0.1338731
```

Financial independence and adult status had a small-to-moderate significant relationship, $\chi^2(4) = 73.34$, $p < .001$, Cramer's $V = .13$. Participants who were not financially independent were less likely than expected to say yes and more likely than expected to say maybe or no. Participants who were somewhat financially independent were more likely than expected to say yes and less likely than expected to say maybe or no. Participants who were financially independent were also more likely than expected to say yes and less likely than expected to say maybe or no. In short, it appears that participants who reported greater financial independence were also more likely to describe themselves as adults.

Research Question 4

```
eammi <- eammi |>
  mutate(
    PartyDichotomous = factor(
      PartyDichotomous,
      levels = c(1, 2),
      labels = c("Democrat", "Republican")
    ),
    PoliticsDichotomous = factor(
      PoliticsDichotomous,
      levels = c(1, 2),
      labels = c("Liberal", "Conservative")
    )
  )

party_data <- eammi |>
  drop_na(PartyDichotomous)

party_counts <- table(party_data$PartyDichotomous)
party_chi <- chisq.test(party_counts, p = c(.50, .50))

politics_data <- eammi |>
  drop_na(PoliticsDichotomous)

politics_counts <- table(politics_data$PoliticsDichotomous)
politics_chi <- chisq.test(politics_counts, p = c(.50, .50))

party_counts

Democrat Republican
1051          484

prop.table(party_counts)

Democrat Republican
0.6846906  0.3153094
```

```
party_chi
```

```
Chi-squared test for given probabilities
```

```
data: party_counts
X-squared = 209.44, df = 1, p-value < 2.2e-16
politics_counts
```

```
   Liberal Conservative
      897           438
```

```
prop.table(politics_counts)
```

```
   Liberal Conservative
0.6719101  0.3280899
```

```
politics_chi
```

```
Chi-squared test for given probabilities
```

```
data: politics_counts
X-squared = 157.81, df = 1, p-value < 2.2e-16
```

Participants were more likely than expected to identify as Democrat (68.5%) and less likely as Republican (31.5%), ($\chi^2(1) = 209.44, p < .001$). They were also more likely to identify as liberal (67.2%) and less likely as conservative (32.8%), ($\chi^2(1) = 157.81, p < .001$).

Chapter Takeaway

Use a test of independence for a relationship between two categorical variables. Use a goodness-of-fit test when one categorical distribution is compared with stated expected proportions.

R Skills Practiced

- `table()` creates observed frequency or contingency tables.
- `chisq.test()` conducts both chi-square tests.
- `test_object$expected` retrieves expected counts for a test of independence.
- `cramers_v()` calculates Cramer's V.
- `prop.table()` converts counts into proportions.
- `p = c(...)` supplies expected goodness-of-fit proportions.

Chapter 15: Nonparametric Statistics

Chapter 15 of *Exploring Statistics* introduces nonparametric tests. These tests convert scores to ranks before comparing groups or examining an association. You will use them with the importance variables, which are strongly skewed and offer one final reminder that real data do not always behave neatly.

Learning Goals

By the end of this chapter, you should be able to:

- choose a nonparametric counterpart for four familiar parametric tests;
- screen the importance variables before conducting an analysis;
- conduct and interpret Mann-Whitney U, Wilcoxon signed-rank, Kruskal-Wallis, and Spearman correlation tests;
- use descriptive statistics and effect sizes to explain significant results; and
- write conclusions that answer each research question.

Research Questions

1. Does the importance placed on hobbies and one other aspect of life differ between Democrats and Republicans?
2. Do emerging adults value careers or hobbies more? What about a second pair of importance ratings?
3. Does career importance differ by adult status? Does hobby importance differ by adult status?
4. Is hobby importance related to satisfaction with life? What about another importance rating?

Before You Begin

Each analysis in this chapter is a rank-based counterpart to a test used earlier:

Parametric tests and their nonparametric counterparts

Research design	Familiar parametric test	Nonparametric test
two independent groups	independent-samples <i>t</i> test	Mann-Whitney U test
two scores from the same participants	paired-samples <i>t</i> test	Wilcoxon signed-rank test
three or more independent groups	one-way independent ANOVA	Kruskal-Wallis test
association between two quantitative variables	Pearson correlation	Spearman correlation

The nonparametric tests answer familiar questions but use the relative ranks of scores. Because the importance distributions are skewed, pay particular attention to medians when describing them.

Research Question 1: Importance Ratings And Political Party

In the EAMMi2 study, participants divided 100 points among marriage, parenting, career, and hobbies to show the relative importance of these aspects of life. We will first examine whether Democrats and Republicans differ in the importance they place on hobbies.

Plan The Analysis

Variables used to answer the guided portion of Research Question 1

Variable	What it measures	Type	Role
IMP hobbies	importance assigned to hobbies	quantitative	dependent variable
PartyDichotomous	Democrat or Republican identification	categorical	grouping variable

The parametric test for two independent groups is an independent-samples t test. Because the importance ratings are strongly skewed, use its nonparametric counterpart: the Mann-Whitney U test.

The null hypothesis states that the distributions of hobby-importance ratings are the same for Democrats and Republicans. The alternative hypothesis states that the distributions differ.

Get Ready

Open 15-nonparametric-statistics.R, run the setup line, and then run one section at a time.

```
source("scripts/_setup.R")
```

One thing you should know about conducting research is that people do not always follow instructions exactly. Before analyzing the ratings, check whether participants assigned exactly 100 points across the four variables.

```
eammi <- eammi |>
  mutate(
    IMPtotal = IMPmarriage + IMPparenting + IMPcareer + IMPhobbies,
    PartyDichotomous = factor(
      PartyDichotomous,
      levels = c(1, 2),
      labels = c("Democrat", "Republican")
    ),
    Adult = factor(
      Adult,
      levels = c(3, 2, 1),
      labels = c("No", "Maybe", "Yes")
    )
  )
```

```
importance_check <- eammi |>
  summarise(
    FollowedDirections = sum(IMPtotal == 100, na.rm = TRUE),
    DidNotFollowDirections = sum(IMPtotal != 100, na.rm = TRUE),
    MissingTotal = sum(is.na(IMPtotal))
  )
```

```
importance_check
```

```
# A tibble: 1 × 3
```

```
  FollowedDirections DidNotFollowDirections MissingTotal
      <int>             <int>             <int>
1         1886               183                4
```

IMPtotal is a new variable containing the sum of the four ratings. The check shows that 1,886 participants assigned exactly 100 points, 183 assigned a different total, and four do not have enough information to calculate a total.

Keep the participants whose total is exactly 100. The other observations are set aside for every analysis in this chapter.

```
importance_data <- eammi |>
  filter(IMPtotal == 100)
```

Run The Analysis

First calculate descriptive statistics for each political group.

```
hobbies_party_data <- importance_data |>
  drop_na(IMP hobbies, PartyDichotomous)

hobbies_party_summary <- hobbies_party_data |>
  group_by(PartyDichotomous) |>
  summarise(
    N = n(),
    Mean = mean(IMP hobbies),
    Median = median(IMP hobbies),
    SD = sd(IMP hobbies),
    .groups = "drop"
  )
```

```
hobbies_party_summary
# A tibble: 2 × 5
  PartyDichotomous      N Mean Median    SD
  <fct>          <int> <dbl> <dbl> <dbl>
1 Democrat         965  21.8    20  13.3
2 Republican        445  16.3    15   9.96
```

Now conduct the Mann-Whitney U test.

```
hobbies_party_test <- wilcox.test(
  IMP hobbies ~ PartyDichotomous,
  data = hobbies_party_data,
  exact = FALSE
)
```

```
hobbies_party_test
```

Wilcoxon rank sum test with continuity correction

```
data: IMP hobbies by PartyDichotomous
W = 269788, p-value = 3.775e-15
alternative hypothesis: true location shift is not equal to 0
```

For two independent groups, `wilcox.test()` conducts the Mann-Whitney U test. The test statistic is *U*. `exact = FALSE` tells R to use a large-sample calculation because many participants gave equal ratings.

Use rank-biserial correlation to describe the size of the group difference.

```
hobbies_party_effect <- rank_biserial_independent(
  hobbies_party_test,
  hobbies_party_data$PartyDichotomous
)
```

```
hobbies_party_effect
```

```
[1] 0.2565081
```

`rank_biserial_independent()` is supplied by the setup file. Give it the saved test object and the grouping variable used in that test. Values around .10, .30, and .50 can be described as small, medium, and large.

```
hobbies_party_data |>
  ggplot(aes(x = PartyDichotomous, y = IMPhobbies)) +
  geom_boxplot() +
  labs(
    x = "Political party",
    y = "Importance assigned to hobbies"
  )
```

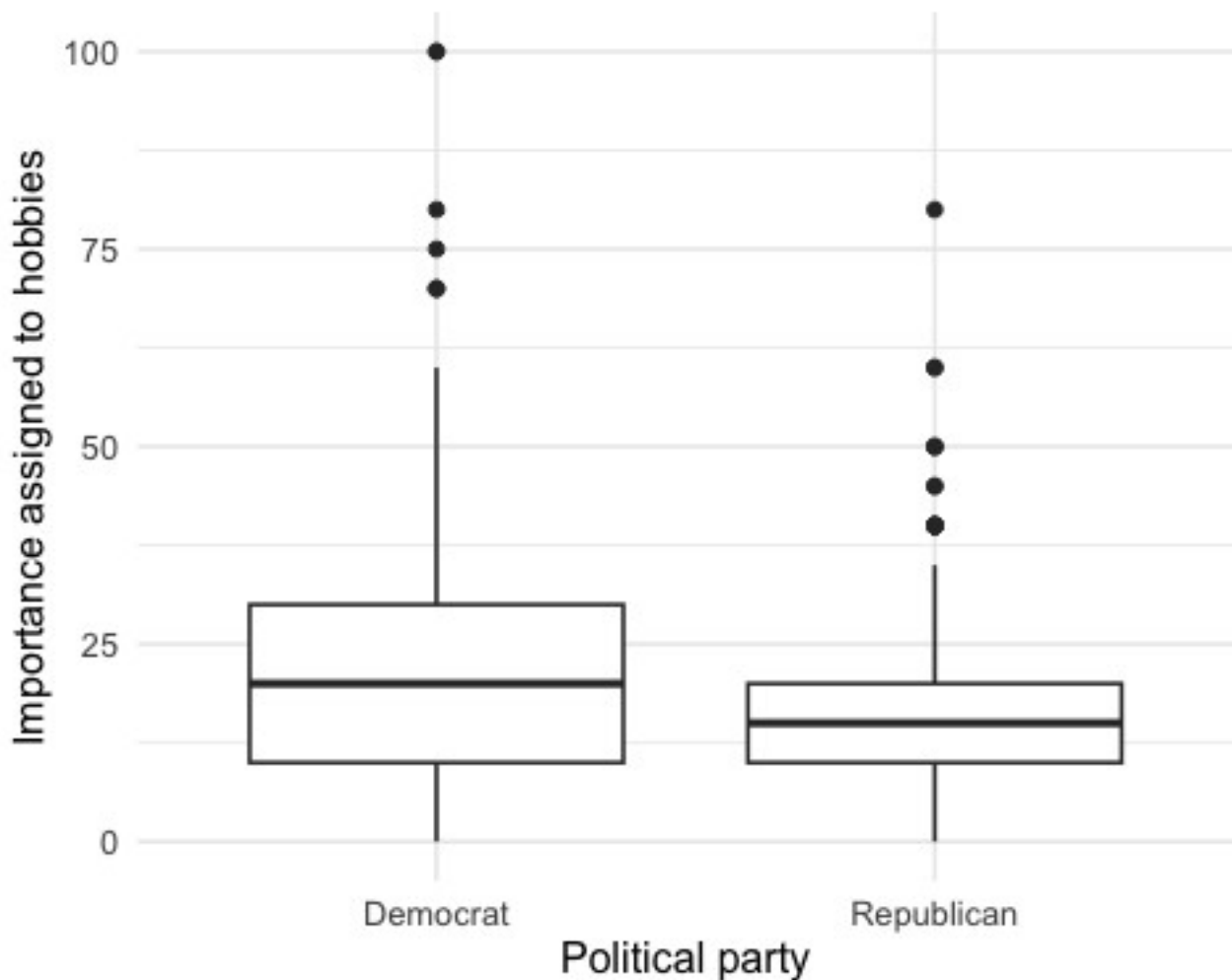


Figure 30: Hobby-importance ratings by political party.

Read The Output

The test is significant, so use the descriptive statistics and graph to understand the difference. Democrats assigned a higher median importance to hobbies than Republicans. The rank-biserial correlation falls between the small and medium guidelines.

Before writing your interpretation, choose `IMPmarriage`, `IMPparenting`, or `IMPcareer` and repeat the analysis. Create new object names rather than replacing the hobby objects. Then write one paragraph summarizing both analyses.

Research Question 2: Work Or Play

Do emerging adults value work or play more? Compare `IMPcareer` with `IMPhobbies`. The same participants provided both ratings, so the parametric test would be a paired-samples *t* test. Because the ratings are skewed, use the Wilcoxon signed-rank test.

The null hypothesis states that the distribution of paired differences is centered at zero. The alternative hypothesis states that the ratings differ.

Get Ready

The variables are already available in `importance_data`. Summarize each one before conducting the test.

```
career_summary <- importance_data |>
  summarise(
    N = n(),
    Mean = mean(IMPcareer),
    Median = median(IMPcareer),
    SD = sd(IMPcareer)
  )
```

```
hobby_summary <- importance_data |>
  summarise(
    N = n(),
    Mean = mean(IMPhobbies),
    Median = median(IMPhobbies),
    SD = sd(IMPhobbies)
  )
```

```
career_summary
# A tibble: 1 × 4
      N Mean Median   SD
<int> <dbl> <dbl> <dbl>
1  1886  33.3    30  15.2

hobby_summary
# A tibble: 1 × 4
      N Mean Median   SD
<int> <dbl> <dbl> <dbl>
1  1886  20.5    20  12.7
```

Run The Analysis

```
career_hobby_test <- wilcox.test(
  importance_data$IMPcareer,
```

```
importance_data$IMP hobbies,
paired = TRUE,
exact = FALSE
)
```

```
career_hobby_test
```

Wilcoxon signed rank test with continuity correction

```
data: importance_data$IMPcareer and importance_data$IMP hobbies
V = 1015505, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0
```

`paired = TRUE` tells R that each career rating is paired with the hobby rating in the same observation. The test statistic is `V`. `exact = FALSE` uses a large-sample calculation because many participants gave equal ratings.

```
career_hobby_effect <- rank_biserial_paired(
  importance_data$IMPcareer,
  importance_data$IMP hobbies
)
```

```
career_hobby_effect
```

```
[1] 0.7756254
```

`rank_biserial_paired()` is supplied by the setup file. Give it the two variables in the same order used in `wilcox.test()`.

Read The Output

The difference is significant. The descriptive statistics show that emerging adults placed more importance on their careers than their hobbies. The rank-biserial correlation indicates a large difference.

Choose a different pair from `IMP marriage`, `IMP parenting`, `IMP career`, and `IMP hobbies`. Repeat the summaries, test, and effect-size calculation using new object names. Then write one paragraph summarizing both comparisons.

Research Question 3: Importance Ratings And Adult Status

Does career importance differ among participants who answered no, maybe, or yes when asked whether they consider themselves adults? The parametric test would be a one-way independent ANOVA. Because career importance is skewed, use the Kruskal-Wallis test.

The null hypothesis states that the career-importance distributions are the same across the three adult-status groups. The alternative hypothesis states that at least one group differs.

Run The Analysis

```
career_adult_data <- importance_data |>
  drop_na(IMPcareer, Adult)

career_adult_summary <- career_adult_data |>
  group_by(Adult) |>
  summarise(
    N = n(),
```

```

    Mean = mean(IMPcareer),
    Median = median(IMPcareer),
    SD = sd(IMPcareer),
    .groups = "drop"
  )

career_adult_summary
# A tibble: 3 × 5
  Adult      N Mean Median   SD
  <fct> <int> <dbl> <dbl> <dbl>
1 No     141  33.9    30  14.9
2 Maybe  488  33.6    30  14.8
3 Yes   1257  33.1    30  15.4

career_adult_test <- kruskal.test(
  IMPcareer ~ Adult,
  data = career_adult_data
)

career_adult_test

```

Kruskal-Wallis rank sum test

```

data: IMPcareer by Adult
Kruskal-Wallis chi-squared = 1.0025, df = 2, p-value = 0.6058

```

`kruskal.test()` uses the familiar outcome ~ group formula. The test statistic is H .

```

career_adult_effect <- kruskal_epsilon_squared(
  career_adult_test,
  career_adult_data$Adult
)

```

```

career_adult_effect
[1] 0

```

`kruskal_epsilon_squared()` is supplied by the setup file. Give it the saved test and grouping variable. Epsilon-squared values around .01, .08, and .26 can be described as small, medium, and large.

Read The Output

The p value is greater than .05, so career importance does not significantly differ by adult status. Because the overall test is not significant, do not conduct post hoc comparisons.

Now repeat the analysis with `IMP_hobbies`. If the Kruskal-Wallis test is significant, conduct DSCF post hoc comparisons using the following pattern:

```
dscfAllPairsTest(outcome ~ group, data = data_object)
```

`dscfAllPairsTest()` uses the same formula pattern and compares every pair while accounting for the number of comparisons. Write one paragraph summarizing the career and hobby analyses.

Research Question 4: Hobby Importance And Satisfaction With Life

Is the importance placed on hobbies related to satisfaction with life? The parametric analysis would be a Pearson correlation. Because hobby importance is skewed, use Spearman correlation.

The null hypothesis states that the population Spearman correlation is zero. The alternative hypothesis states that it is not zero.

Run The Analysis

```
hobbies_swls_data <- importance_data |>
  drop_na(IMPhobbies, SWLS)

hobbies_swls_test <- cor.test(
  hobbies_swls_data$IMPhobbies,
  hobbies_swls_data$SWLS,
  method = "spearman",
  exact = FALSE
)

hobbies_swls_test

Spearman's rank correlation rho

data: hobbies_swls_data$IMPhobbies and hobbies_swls_data$SWLS
S = 1222239667, p-value = 5.083e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.09315736
nrow(hobbies_swls_data)
[1] 1886
```

`cor.test()` was introduced with Pearson correlation. `method = "spearman"` tells R to correlate ranks instead of the original scores. The coefficient is Spearman's r_s . To report degrees of freedom, subtract 2 from the number of complete observations.

Read The Output

Hobby importance has a small, significant negative association with satisfaction with life. Higher satisfaction-with-life scores are associated with slightly lower importance placed on hobbies. The relationship is close to zero and might not be significant in a smaller sample.

Repeat the Spearman correlation using SWLS and a different importance variable. Create new data and test objects. Then write one paragraph summarizing both correlations.

Check Your Work

Research Question 1

For hobbies, Democrats reported a mean of 21.8, median of 20.0, and standard deviation of 13.3. Republicans reported a mean of 16.3, median of 15.0, and standard deviation of 10.0.

A Mann-Whitney U test found that hobby importance significantly differed between Democrats and Republicans, $U = 269,788$, $p < .001$, $r_{rb} = .26$. The effect was small-to-medium, with Democrats assigning greater importance to hobbies.

The sample extension below uses IMPcareer.

```
career_party_data <- importance_data |>
  drop_na(IMPcareer, PartyDichotomous)

career_party_summary <- career_party_data |>
  group_by(PartyDichotomous) |>
  summarise(
    N = n(),
    Mean = mean(IMPcareer),
    Median = median(IMPcareer),
    SD = sd(IMPcareer),
    .groups = "drop"
  )

career_party_test <- wilcox.test(
  IMPcareer ~ PartyDichotomous,
  data = career_party_data,
  exact = FALSE
)

career_party_effect <- rank_biserial_independent(
  career_party_test,
  career_party_data$PartyDichotomous
)

career_party_summary
# A tibble: 2 × 5
  PartyDichotomous      N Mean Median    SD
  <fct>          <int> <dbl> <dbl> <dbl>
1 Democrat        965  35.3    30  15.9
2 Republican      445  29.1    25  13.7

career_party_test
```

Wilcoxon rank sum test with continuity correction

```
data: IMPcareer by PartyDichotomous
W = 265782, p-value = 4.368e-13
alternative hypothesis: true location shift is not equal to 0
career_party_effect
[1] 0.2378529
```

Career importance also significantly differed, $U = 265,782.5$, $p < .001$, $r_{rb} = .24$. Democrats reported a mean of 35.3, median of 30.0, and standard deviation of 15.9. Republicans reported a mean of 29.1, median of 25.0, and standard deviation of 13.7. Democrats therefore assigned greater importance to both hobbies and careers.

Research Question 2

Career importance had a mean of 33.3, median of 30.0, and standard deviation of 15.2. Hobby importance had a mean of 20.5, median of 20.0, and standard deviation of 12.7. Participants rated careers as more important than hobbies, $V = 1,015,505$, $p < .001$, $r_{rb} = .78$, a large difference.

The sample extension below compares marriage with hobbies.

```
marriage_hobby_test <- wilcox.test(
  importance_data$IMPmarriage,
  importance_data$IMPhobbies,
  paired = TRUE,
  exact = FALSE
)

marriage_hobby_effect <- rank_biserial_paired(
  importance_data$IMPmarriage,
  importance_data$IMPhobbies
)
```

marriage_hobby_test

Wilcoxon signed rank test with continuity correction

```
data: importance_data$IMPmarriage and importance_data$IMPhobbies
V = 772860, p-value = 6.661e-15
alternative hypothesis: true location shift is not equal to 0
marriage_hobby_effect
[1] 0.2251412
```

Marriage importance had a mean of 23.4, median of 25.0, and standard deviation of 12.4. Participants rated marriage as more important than hobbies, $V = 772,860$, $p < .001$, $r_{rb} = .23$, a small-to-medium difference.

Research Question 3

Career importance did not significantly differ by adult status, $H(2) = 1.00$, $p = .606$, $\epsilon^2 < .001$.

```
hobbies_adult_data <- importance_data |>
  drop_na(IMPhobbies, Adult)

hobbies_adult_summary <- hobbies_adult_data |>
  group_by(Adult) |>
  summarise(
    N = n(),
    Mean = mean(IMPhobbies),
    Median = median(IMPhobbies),
    SD = sd(IMPhobbies),
    .groups = "drop"
  )

hobbies_adult_test <- kruskal.test(
  IMPhobbies ~ Adult,
  data = hobbies_adult_data
)
```

```

hobbies_adult_effect <- kruskal_epsilon_squared(
  hobbies_adult_test,
  hobbies_adult_data$Adult
)

hobbies_adult_post_hoc <- dscfAllPairsTest(
  IMPhobbies ~ Adult,
  data = hobbies_adult_data
)

hobbies_adult_summary
# A tibble: 3 × 5
  Adult      N Mean Median    SD
  <fct> <int> <dbl>  <dbl> <dbl>
1 No      141  20.6    15  14.2
2 Maybe   488  21.9    20  12.8
3 Yes    1257  20.0    20  12.4

hobbies_adult_test

      Kruskal-Wallis rank sum test

data:  IMPhobbies by Adult
Kruskal-Wallis chi-squared = 8.88, df = 2, p-value = 0.0118

hobbies_adult_effect
[1] 0.003653725

hobbies_adult_post_hoc
      No      Maybe
Maybe 0.2254 -
Yes    0.9928 0.0094

```

Hobby importance significantly differed by adult status, $H(2) = 8.88$, $p = .012$, $\epsilon^2 = .004$, although the effect was very small. Participants answering maybe reported greater hobby importance than participants answering yes, $p = .009$. The no group did not significantly differ from the maybe group, $p = .225$, or the yes group, $p = .993$.

Research Question 4

There was a small, significant negative correlation between satisfaction with life and hobby importance, $r_s(1884) = -.09$, $p < .001$.

The sample extension below uses IMPcareer.

```

career_swls_data <- importance_data |>
  drop_na(IMPcareer, SWLS)

career_swls_test <- cor.test(
  career_swls_data$IMPcareer,
  career_swls_data$SWLS,
  method = "spearman",
  exact = FALSE
)

```

```
career_swls_test
```

```
Spearman's rank correlation rho
```

```
data: career_swls_data$IMPcareer and career_swls_data$WLS
S = 1269129046, p-value = 3.861e-09
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.1350947
nrow(career_swls_data)
[1] 1886
```

There was also a small, significant negative correlation between satisfaction with life and career importance, $r_s(1884) = -.14, p < .001$. Higher satisfaction-with-life scores were associated with lower importance placed on both hobbies and careers.

Chapter Takeaway

Nonparametric tests use ranks to answer familiar questions when a parametric analysis is not appropriate. Match the test to the research design: independent groups, paired scores, three or more independent groups, or an association between quantitative variables.

R Skills Practiced

- `wilcox.test(outcome ~ group)` conducts a Mann-Whitney U test for two independent groups.
- `wilcox.test(x, y, paired = TRUE)` conducts a Wilcoxon signed-rank test for paired scores.
- `rank_biserial_independent()` and `rank_biserial_paired()` calculate rank-biserial correlations. These helpers are supplied by the setup file.
- `kruskal.test()` conducts a Kruskal-Wallis test.
- `kruskal_epsilon_squared()` calculates epsilon squared. This helper is supplied by the setup file.
- `dscfAllPairsTest()` conducts DSCF post hoc comparisons.
- `cor.test(..., method = "spearman")` conducts a Spearman correlation.

R Command Index

Use this page when you recognize a command but need a reminder about what it does. The index records where each command is first explained. Later chapters may reuse a command without teaching it again.

Start Here Commands

Command or pattern	What it does	First explained
<code><-</code>	Stores a result in a named object.	Run Your First Script
<code>#</code>	Adds a comment for readers. R ignores the rest of that line.	Run Your First Script
<code> ></code>	Passes the result of one step into the next step.	Run Your First Script
<code>nrow()</code>	Counts observations in a data frame.	Run Your First Script
<code>ncol()</code>	Counts variables in a data frame.	Run Your First Script
<code>names()</code>	Lists variable names in a data frame.	Run Your First Script
<code>count()</code>	Counts how often values or categories occur.	Run Your First Script

How To Read The Examples

R functions use parentheses:

```
function_name(object, option = value)
```

Pipes organize a sequence from left to right:

```
data_object |>
  first_step() |>
  next_step()
```

Assignment saves a result so it can be inspected or reused:

```
result_name <- data_object |>
  analysis_step()
```

Chapter Commands

Chapters 1 And 2: Inspect, Label, Tabulate, And Graph

Command or pattern	What it does	First explained
<code>source()</code>	Runs code saved in another script.	Chapter 1
<code>dim()</code>	Reports observations and variables together.	Chapter 1
<code>factor()</code>	Gives category codes readable labels and an order.	Chapter 1

Command or pattern	What it does	First explained
<code>\$</code>	Selects one variable inside a data frame.	Chapter 1
<code>levels()</code>	Shows the labels and order of a factor.	Chapter 1
<code>mutate()</code>	Changes or creates variables.	Chapter 1
<code>ggplot() + aes()</code>	Starts a graph and maps variables to visual positions.	Chapter 1
<code>geom_col()</code>	Creates bars from values already calculated in a table.	Chapter 1
<code>labs()</code>	Adds readable graph labels.	Chapter 1
<code>class()</code>	Shows how R represents an object or variable.	Chapter 2
<code>cumsum()</code>	Creates a running total.	Chapter 2
<code>drop_na()</code>	Omits observations missing specified values from an analysis.	Chapter 2
<code>geom_histogram()</code>	Groups quantitative scores into intervals and graphs their frequencies.	Chapter 2
<code>coord_flip()</code>	Turns a graph so long category labels are easier to read.	Chapter 2

Chapters 3-5: Describe Distributions

Command or pattern	What it does	First explained
<code>filter()</code>	Keeps observations that meet a condition.	Chapters 3 and 4
<code>group_by()</code>	Requests later calculations separately for each group.	Chapters 3 and 4
<code>summarise()</code>	Reduces observations to named summary values.	Chapters 3 and 4
<code>n()</code>	Counts observations inside a summary.	Chapters 3 and 4
<code>mean(), median(), sd()</code>	Calculate center and variability.	Chapters 3 and 4
<code>min(), max()</code>	Find the observed endpoints.	Chapters 3 and 4
<code>sum(), is.na()</code>	Count missing values when they matter to a summary.	Chapters 3 and 4
<code>facet_wrap()</code>	Repeats a graph in separate group panels.	Chapters 3 and 4
<code>statistical_mode()</code>	Returns the most common observed value. This helper is	Chapters 3 and 4

Command or pattern	What it does	First explained
	supplied by the setup file.	
<code>head()</code>	Displays the first observations or values in an object.	Chapter 5
<code>quantile()</code>	Calculates quartiles or other percentiles.	Chapter 5
<code>IQR()</code>	Calculates the interquartile range.	Chapter 5
<code>geom_boxplot()</code>	Creates a boxplot.	Chapter 5
<code>independent_cohens_d()</code>	Calculates Cohen's d for two independent groups. This helper is supplied by the setup file.	Chapter 5

Chapters 6, 9, And 10: Correlation And t Tests

Command or pattern	What it does	First explained
<code>select()</code>	Keeps specified variables.	Chapter 6
<code>cor()</code>	Calculates Pearson correlation coefficients.	Chapter 6
<code>outcome ~ predictor</code>	Identifies the outcome and predictor in an R model formula.	Chapter 6
<code>lm()</code>	Fits a linear model.	Chapter 6
<code>coefficients()</code>	Shows a model's intercept and slope.	Chapter 6
<code>geom_point()</code> , <code>geom_smooth()</code>	Add observed score pairs and a fitted line.	Chapter 6
<code>t.test()</code>	Conducts a t test.	Chapter 9
<code>sample_skewness()</code>	Calculates sample skewness. This helper is supplied by the setup file.	Chapter 9
<code>cohens_d()</code>	Divides a mean difference by a standard deviation. This helper is supplied by the setup file.	Chapter 9
<code>mu =</code>	Supplies the comparison value for a one-sample test.	Chapter 9
<code>alternative = "two.sided"</code>	Requests a non-directional test.	Chapter 9
<code>\$conf.int</code>	Retrieves the confidence interval stored in a test object.	Chapter 9
<code>pull()</code>	Takes one variable from a data frame for use by a function.	Chapter 10
<code>shapiro.test()</code>	Checks whether scores differ significantly from a normal distribution.	Chapter 10

Command or pattern	What it does	First explained
<code>var.test()</code>	Conducts an F test comparing two group variances.	Chapter 10
<code>var.equal = TRUE</code>	Requests Student's independent-samples t test.	Chapter 10
<code>paired = TRUE</code>	Identifies two related sets of scores.	Chapter 10

Chapters 11-14: ANOVA And Chi-Square

Command or pattern	What it does	First explained
<code>aov()</code>	Fits an analysis-of-variance model.	Chapter 11
<code>summary(model)</code>	Prints the ANOVA table stored in a model object.	Chapter 11
<code>residuals()</code>	Retrieves observed-minus-predicted differences from a model.	Chapter 11
<code>bartlett.test()</code>	Checks whether three or more group variances differ.	Chapter 11
<code>qt()</code>	Supplies the critical t value used to calculate a confidence interval.	Chapter 11
<code>anova_eta_squared()</code>	Calculates eta squared for a one-way ANOVA. This helper is supplied by the setup file.	Chapter 11
<code>TukeyHSD()</code>	Conducts adjusted pairwise comparisons.	Chapter 11
<code>tukey_cohens_d()</code>	Calculates Cohen's <i>d</i> for Tukey comparisons. This helper is supplied by the setup file.	Chapter 11
<code>geom_errorbar()</code>	Adds confidence intervals to a graph.	Chapter 11
<code>row_number()</code>	Creates a participant label from observation order.	Chapter 12
<code>pivot_longer()</code>	Reshapes repeated scores from wide to long format.	Chapter 12
<code>repeated_eta_squared()</code>	Calculates partial eta squared for the repeated condition. This helper is supplied by the setup file.	Chapter 12
<code>repeated_pairwise_tests()</code>	Compares each pair of repeated scores and reports mean differences, Tukey-adjusted <i>p</i> values, and Cohen's <i>d</i> . This helper is supplied by the setup file.	Chapter 12

Command or pattern	What it does	First explained
<code>interaction()</code>	Combines grouping variables for an assumption check.	Chapter 13
<code>predictor_1 * predictor_2</code>	Requests two main effects and their interaction.	Chapter 13
<code>two_way_anova_table()</code>	Reports two-way tests with partial eta squared. This helper is supplied by the setup file.	Chapter 13
<code>geom_line()</code>	Connects means in an interaction plot.	Chapter 13
<code>table()</code>	Creates observed frequency or contingency tables.	Chapter 14
<code>chisq.test()</code>	Conducts chi-square tests.	Chapter 14
<code>test_object\$expected</code>	Retrieves expected counts.	Chapter 14
<code>cramers_v()</code>	Calculates Cramer's V. This helper is supplied by the setup file.	Chapter 14
<code>prop.table()</code>	Converts counts to proportions.	Chapter 14
<code>p = c(...)</code>	Supplies the expected proportions for a goodness-of-fit test.	Chapter 14

Chapter 15: Nonparametric Statistics

Command or pattern	What it does	First explained
<code>wilcox.test(outcome ~ group)</code>	Conducts a Mann-Whitney U test for two independent groups.	Chapter 15
<code>wilcox.test(x, y, paired = TRUE)</code>	Conducts a Wilcoxon signed-rank test for paired scores.	Chapter 15
<code>rank_biserial_independent()</code>	Calculates rank-biserial correlation for two independent groups. This helper is supplied by the setup file.	Chapter 15
<code>rank_biserial_paired()</code>	Calculates rank-biserial correlation for paired scores. This helper is supplied by the setup file.	Chapter 15
<code>kruskal.test()</code>	Conducts a Kruskal-Wallis test.	Chapter 15
<code>kruskal_epsilon_squared()</code>	Calculates epsilon squared for a Kruskal-Wallis test. This helper is supplied by the setup file.	Chapter 15
<code>dscfAllPairsTest()</code>	Conducts DSCF post hoc comparisons after a significant Kruskal-Wallis test.	Chapter 15
<code>cor.test(..., method = "spearman")</code>	Conducts a Spearman rank	Chapter 15

Command or pattern	What it does	First explained
	correlation.	

Troubleshooting

Errors are information about where R became unable to continue. Begin with the first error message, check the code immediately above it, and change only one thing at a time.

A Quick Recovery Routine

1. Stop and read the first message in the Console.
2. Check whether the `>` prompt has returned.
3. Compare the code with the workbook example.
4. Check spelling, capitalization, parentheses, commas, and quotation marks.
5. Run the setup line again if an expected object is missing.
6. Run the smallest complete section that could produce the result.

Common Problems

What you see	What it usually means	What to try
<code>object 'eammi' not found</code>	The shared setup has not created the data object in this R session.	Open the project and run <code>source("scripts/_setup.R")</code> .
R cannot find <code>scripts/_setup.R</code> or the data file	The RStudio project may not be open, or part of the starter folder was moved.	Open <code>exploring-statistics-r.Rproj</code> and keep the starter files together.
A <code>+</code> prompt	R is waiting for the rest of an unfinished command.	Press Esc, check for a missing parenthesis or quotation mark, and run the complete command.
<code>could not find function</code>	The setup section may not have run, or the function name may be misspelled.	Run the setup line and compare the function spelling with the example.
<code>object 'VariableName' not found</code>	The variable name may be misspelled or used outside the data frame.	Check names (<code>eammi</code>) and remember that capitalization matters.
A graph does not appear	The plotting code may be incomplete or the wrong Output tab may be selected.	Run the entire graph expression and open the Plots tab.
The output differs from the workbook	An earlier recode, filter, or object may differ.	Restart R, run setup, and repeat the chapter code in order.

Read An Error Message

You do not need to understand every word. Look first for:

- the object or function named in the message;
- a line number, if one is shown;
- wording such as `not found`, `unexpected`, or `missing`; and
- the first error, rather than messages that followed it.

Copying only the final line of a long message can remove useful context. When asking for help, include the code you ran and the first complete error message.

Restart When The Sequence Is Unclear

If you have run sections out of order and no longer know which objects have been changed:

1. choose **Session > Restart R**;
2. confirm that the project is still open;
3. run `source("scripts/_setup.R");` and
4. repeat the chapter code from the beginning.

Restarting R clears objects from memory. It does not delete your script or data files.

Before Asking For Help

Be ready to describe:

- which script you opened;
- which line or section you ran;
- what you expected;
- what appeared instead; and
- the complete first error message.

This information helps another person reproduce the problem and identify the relevant step.

Statistical Language

This page is a reminder, not a replacement for the textbook. Return to the relevant textbook chapter when you need a fuller explanation or example.

Variables And Roles

Term	Meaning
Categorical variable	Places observations into categories.
Nominal variable	Uses categories without a meaningful rank order.
Ordinal variable	Uses categories with a meaningful order.
Quantitative variable	Records numerical amounts or scores for which numerical differences are meaningful.
Outcome or dependent variable	The response an analysis describes, predicts, or compares.
Predictor or independent variable	The variable used to predict an outcome or define comparison groups.

The role of a variable depends on the research question. A variable can be an outcome in one analysis and a predictor in another.

Describing A Distribution

When describing one variable, consider:

- frequency and percent for categories;
- center, including the mean, median, or mode;
- variability, including the range, standard deviation, or IQR;
- shape, including symmetry or skew; and
- missing, impossible, or unusual values.

Report what the output shows before offering a broader explanation for the pattern.

Comparing And Relating Variables

- A difference describes how groups or conditions compare.
- An association describes how two variables vary together.
- A prediction uses one or more variables to estimate an outcome.
- An interaction means the relationship between one predictor and the outcome differs across values of another predictor.

An observed association does not by itself establish that one variable caused the other.

Formula Syntax In R

Many R models use a formula:

```
outcome ~ predictor
```

Read this as “model the outcome using the predictor.” The variable on the left is the outcome. Variables on the right are predictors or grouping variables.

A two-predictor model may use:

```
outcome ~ predictor_1 * predictor_2
```

The * asks R to estimate both main effects and their interaction. The chapter that introduces this model will explain how to read the resulting output.

Writing A Conclusion

A complete short interpretation usually states:

1. which variables or groups were analyzed;
2. the important pattern in the output;
3. the relevant statistics;
4. whether the result supports the stated hypothesis; and
5. any limitation needed to avoid an unsupported conclusion.

Use the wording and level of detail requested in each chapter. Statistical significance, practical importance, and causation are different claims and should not be treated as interchangeable.